

Lecture-5: Deep Learning

Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD,
Stochastic GD, AdaGrad, RMSProp, Adam

Acknowledgements

- CS6910: Deep Learning Course by Prof. Mitesh M. Khapra, IIT Madras, India

vector of parameters,
say, randomly initialized


$$\theta = [w, b]$$

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change in the
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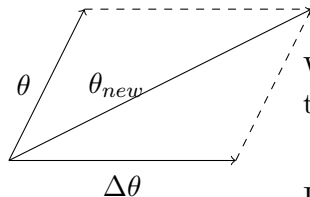
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We moved in the direc-
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Let us be a bit conservat-
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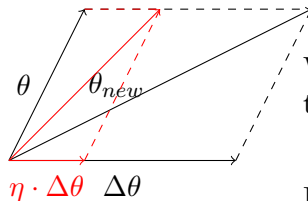
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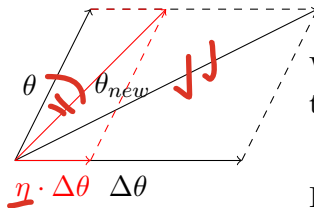
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$\theta_{new} = \theta + \eta \cdot \Delta\theta$

Question: What is the right $\Delta\theta$ to use?

The answer comes from Taylor series

For ease of notation, let $\Delta\theta = u$, then from Taylor series, we have,

$$\begin{aligned} * \left\{ \begin{aligned} \mathcal{L}(\theta + \eta u) &= \mathcal{L}(\theta) + \eta * u^T \nabla \mathcal{L}(\theta) + \frac{\eta^2}{2!} * u^T \nabla^2 \mathcal{L}(\theta) u + \frac{\eta^3}{3!} * \dots + \frac{\eta^4}{4!} * \dots \\ &= \mathcal{L}(\theta) + \eta * \underline{u^T \nabla \mathcal{L}(\theta)} \quad [\eta \text{ is typically small, so } \eta^2, \eta^3, \dots \rightarrow 0] \end{aligned} \right. \end{aligned}$$

Note that the move (ηu) would be favorable only if,

$$\mathcal{L}(\theta + \eta u) - \mathcal{L}(\theta) < 0 \text{ [i.e., if the new loss is less than the previous loss]}$$

This implies,

$$u^T \nabla \mathcal{L}(\theta) < 0$$

Okay, so we have,

$$u^T \nabla \mathcal{L}(\theta) < 0$$

But, what is the range of $u^T \nabla \mathcal{L}(\theta)$? Let's see....

Let β be the angle between u^T and $\nabla \mathcal{L}(\theta)$, then we know that,

$$-1 \leq \cos(\beta) = \frac{u^T \nabla \mathcal{L}(\theta)}{\|u\| * \|\nabla \mathcal{L}(\theta)\|} \leq 1$$

Multiply throughout by $k = \|u\| * \|\nabla \mathcal{L}(\theta)\|$

$$-k \leq k * \cos(\beta) = u^T \nabla \mathcal{L}(\theta) \leq k$$

Thus, $\mathcal{L}(\theta + \eta u) - \mathcal{L}(\theta) = u^T \nabla \mathcal{L}(\theta) = k * \cos(\beta)$ will be most negative when $\cos(\beta) = -1$ i.e., when β is 180°

- Let's create an algorithm from this rule ...

Algorithm 1: `gradient_descent()`

$t \leftarrow 0;$

$max_iterations \leftarrow 1000;$

while $t < max_iterations$ **do**

| ✓ $w_{t+1} \leftarrow w_t - \eta \nabla w_t;$

| ✓ $b_{t+1} \leftarrow b_t - \eta \nabla b_t;$

end

- To see this algorithm in practice let us first derive ∇w and ∇b for our toy neural network

```

X = [0.5, 2.5]
Y = [0.2, 0.9]

def f(w,b,x) : #sigmoid with parameters w,b
    return 1.0 / (1.0 + np.exp(-(w*x + b)))

def error (w, b) :
    err = 0.0
    for x,y in zip(X,Y) :
        fx = f(w,b,x)
        err += 0.5 * (fx - y) ** 2
    return err

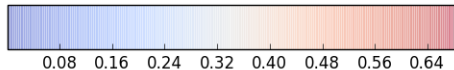
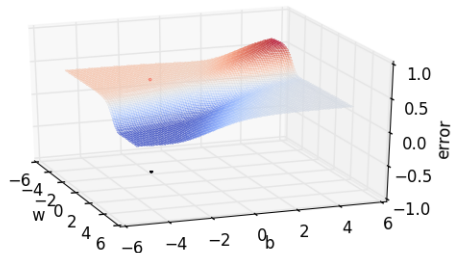
def grad_b(w,b,x,y) :
    fx = f(w,b,x)
    return (fx - y) * fx * (1 - fx)

def grad_w(w,b,x,y) :
    fx = f(w,b,x)
    return (fx - y) * fx * (1 - fx) * x

def do_gradient_descent() :
    w, b, eta, max_epochs = -2, -2, 1.0, 1000
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)
        w = w - eta * dw
        b = b - eta * db

```

Gradient descent on the error surface



- *Let's see what happens when we start from a different point*

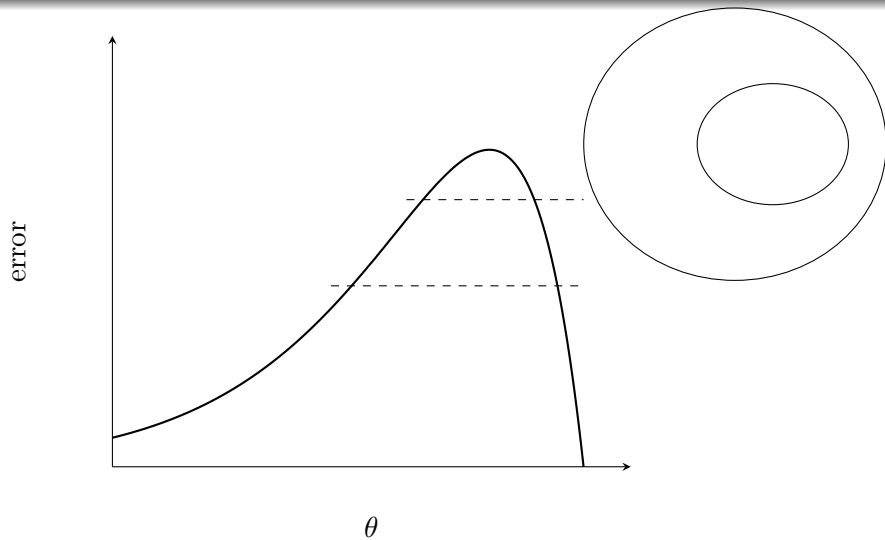


Figure: Front view of a 3d error surface

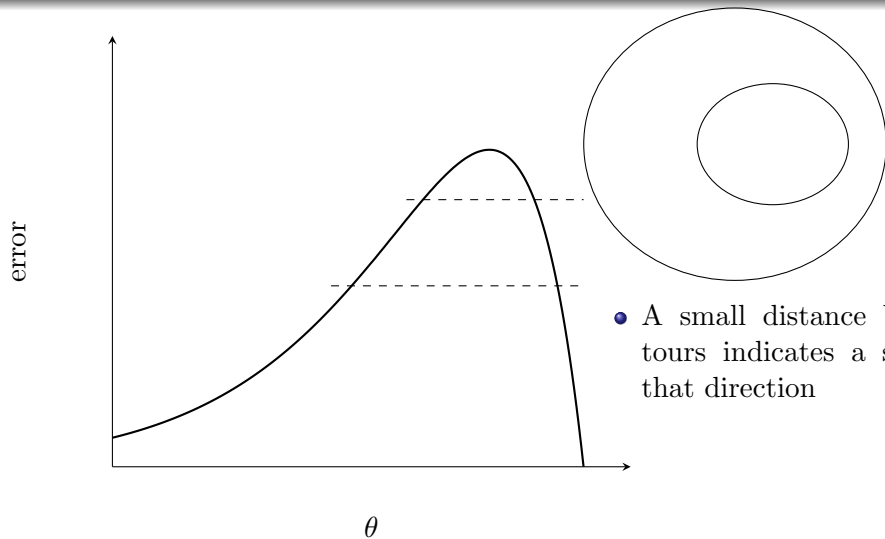


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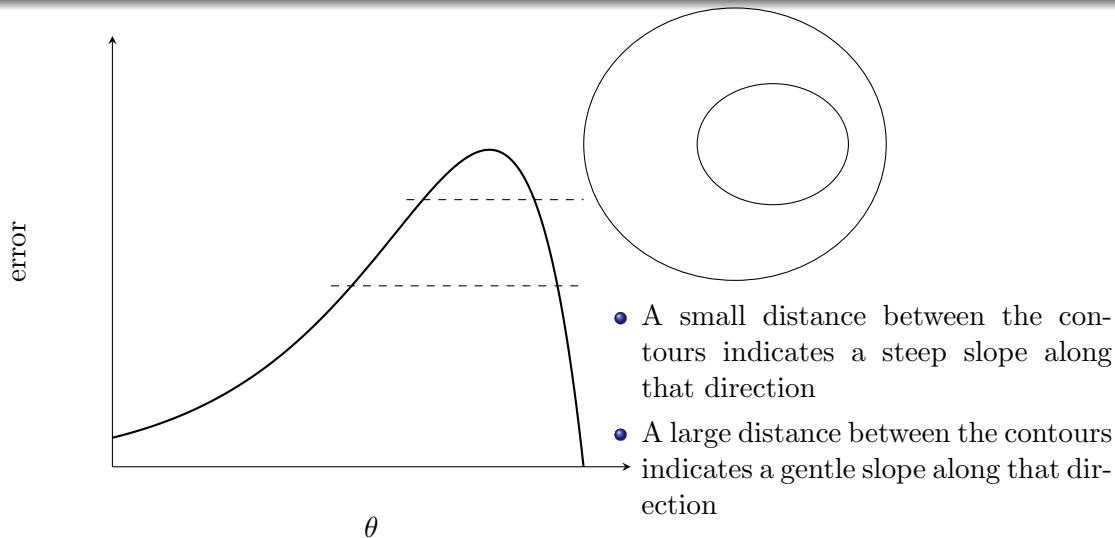
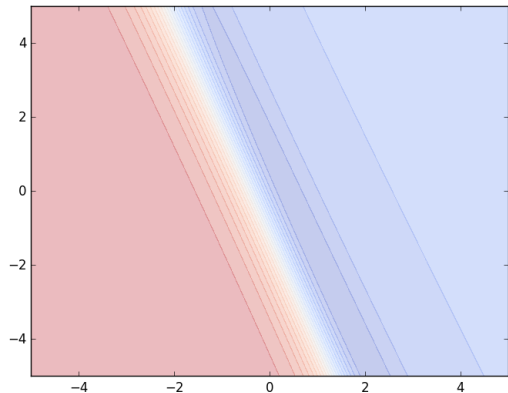
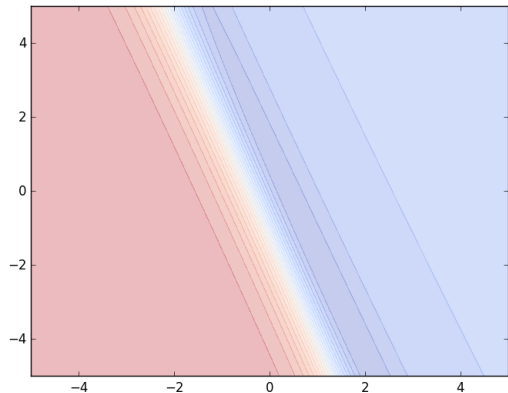


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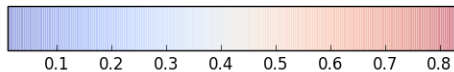
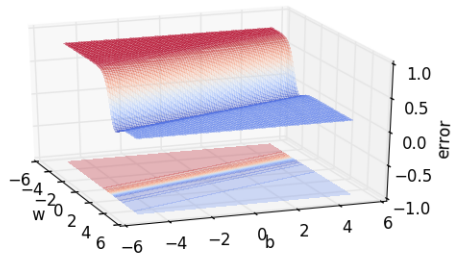
- *Just to ensure that we understand this properly let us do a few exercises ...*

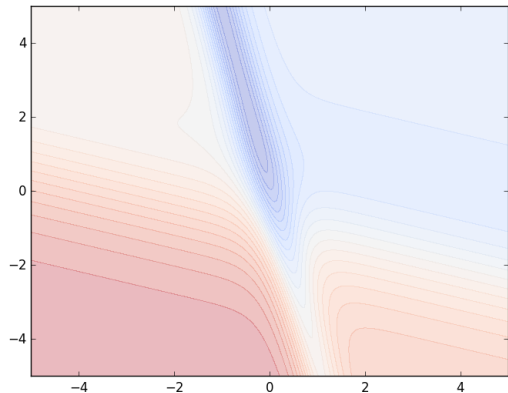


Guess the 3d surface

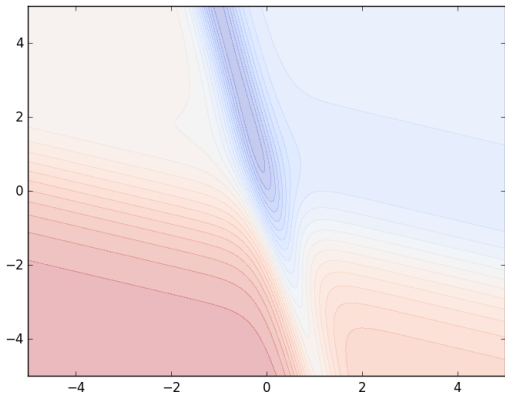


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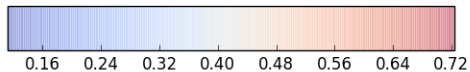
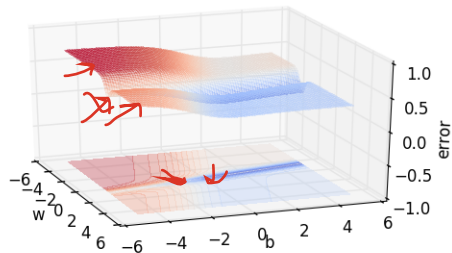


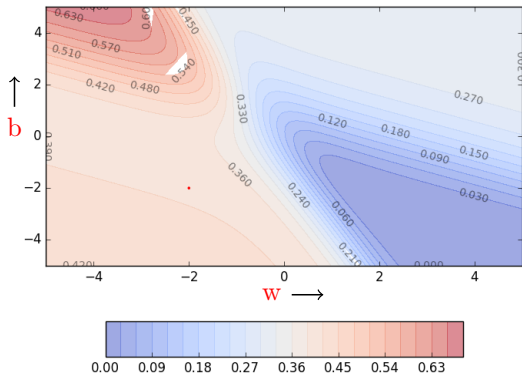


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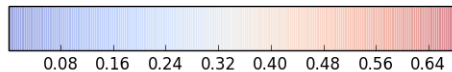
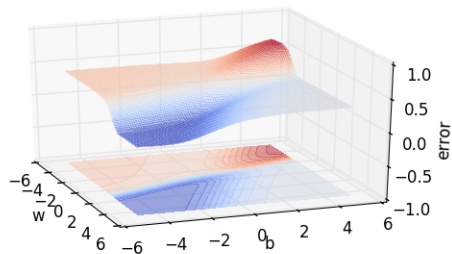


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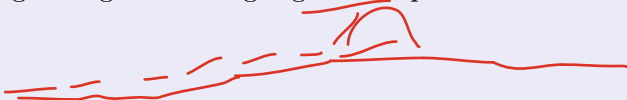


Gradient descent on the error surface



Some observations about gradient descent

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- Can we do something better ?
- Yes, let's take a look at 'Momentum based gradient descent'

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Update rule for momentum based gradient descent

$$\begin{aligned} \text{update}_t &= \gamma \cdot \text{update}_{t-1} + \eta \nabla w_t \\ \underline{w_{t+1}} &= \underline{w_t} - \text{update}_t \end{aligned}$$

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Update rule for momentum based gradient descent

$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t$$

$$w_{t+1} = w_t - update_t$$

- In addition to the current update, also look at the history of updates.

$$update_t = \gamma \cdot update_{t-1} + \eta \nabla w_t$$

$$w_{t+1} = w_t - update_t$$

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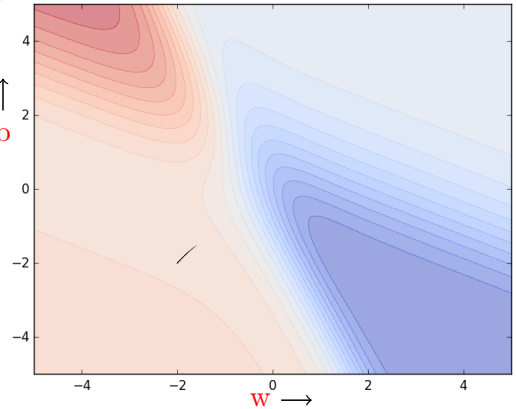
```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
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        v_w = gamma * prev_v_w + eta * dw
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```

↑
b

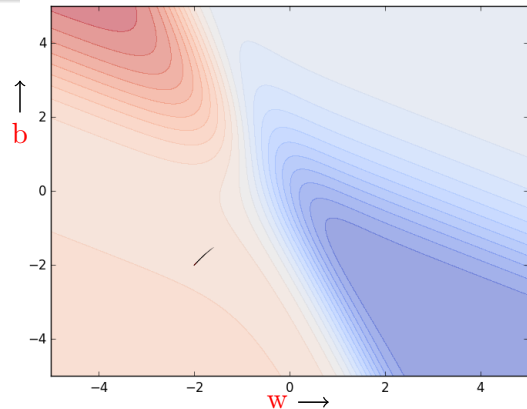


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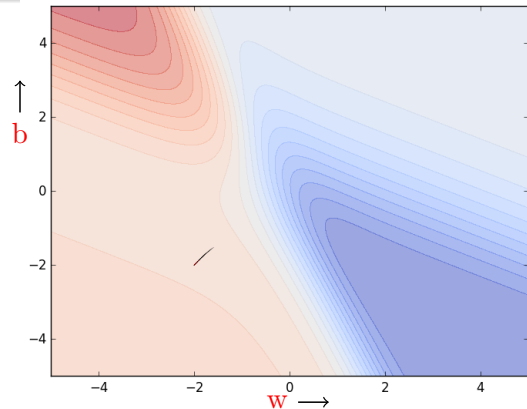


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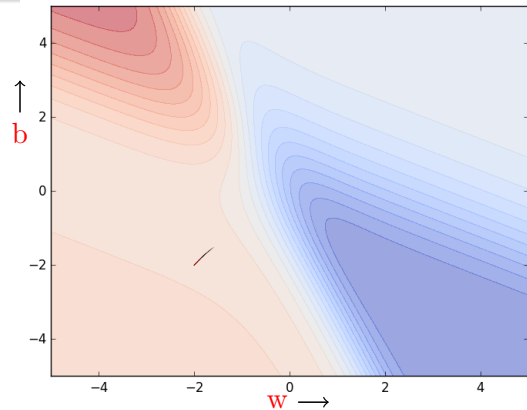


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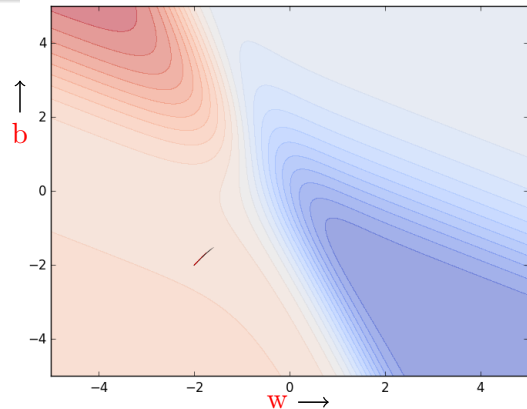


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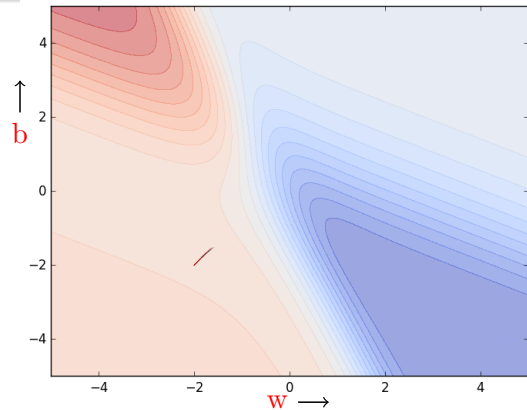


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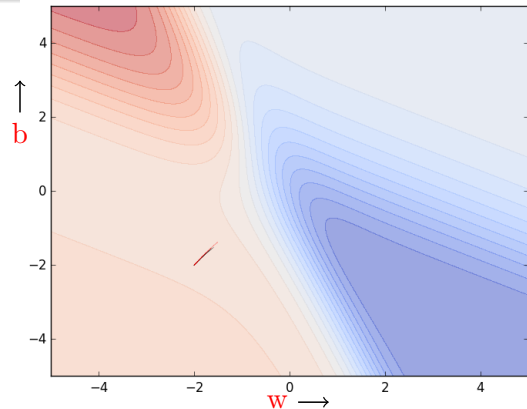


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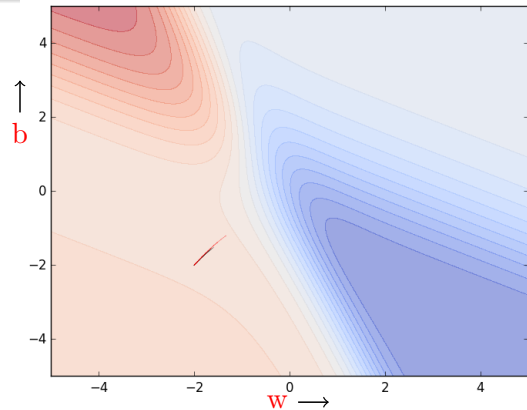



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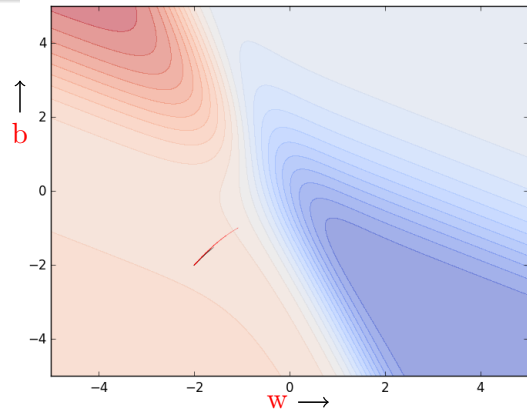


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        prev_v_b = v_b

```

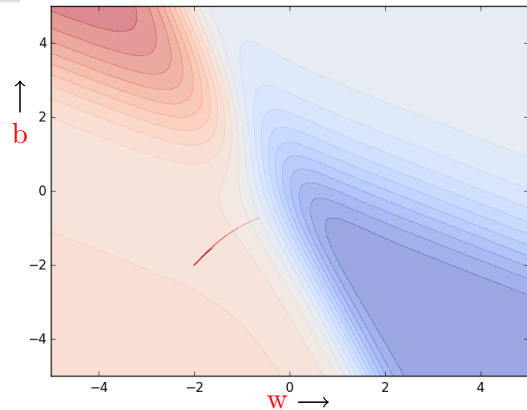


```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = gamma * prev_v_w + eta * dw
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```

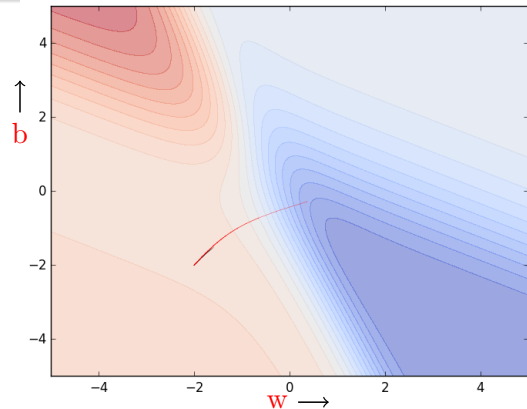


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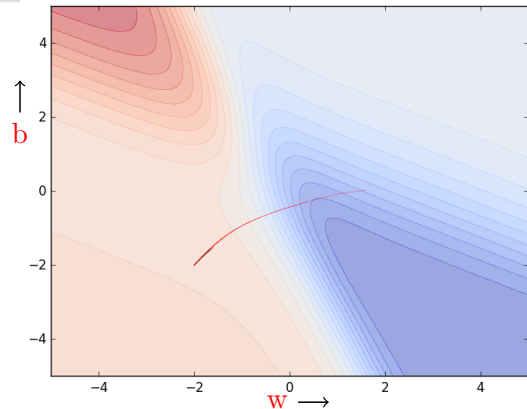


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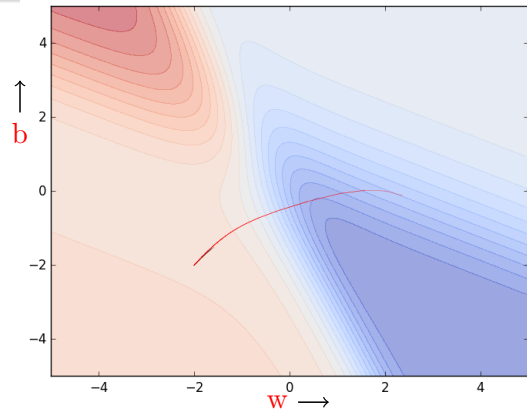


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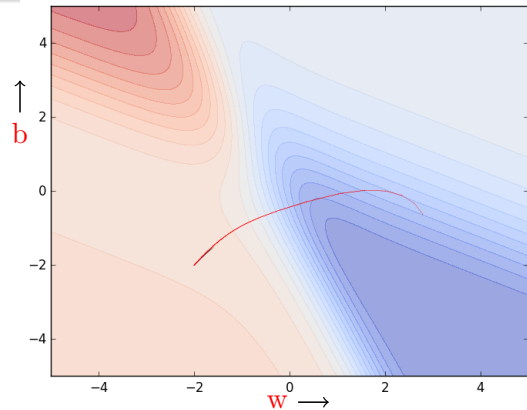


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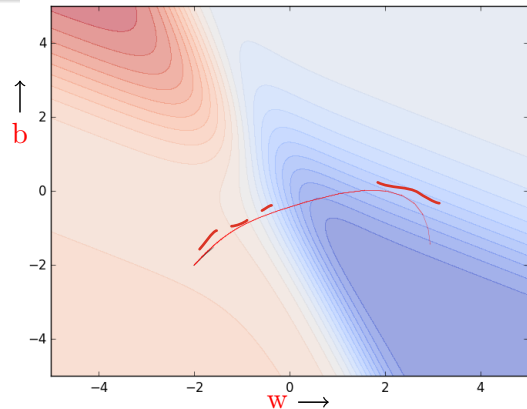


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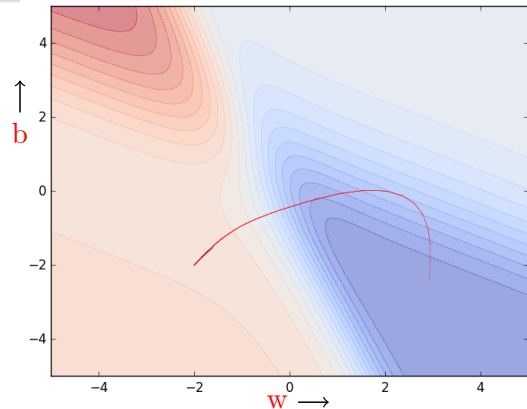



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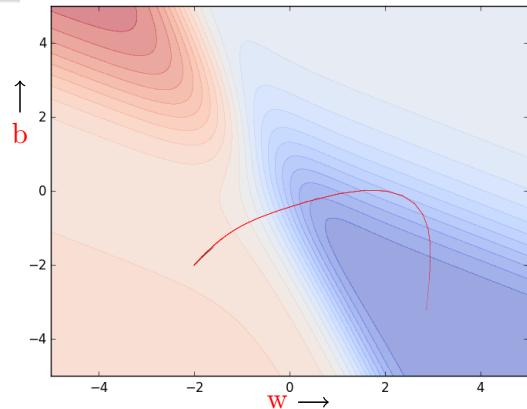


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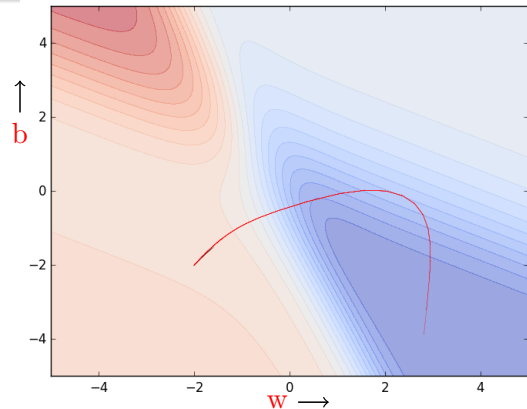


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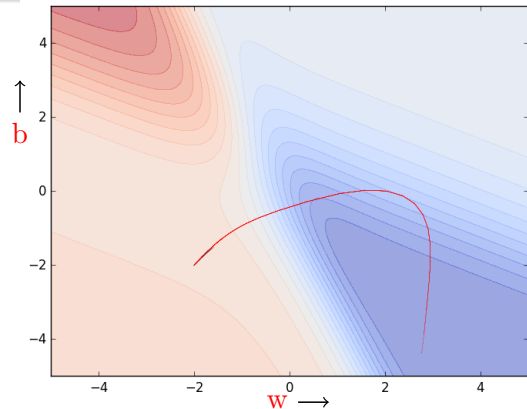


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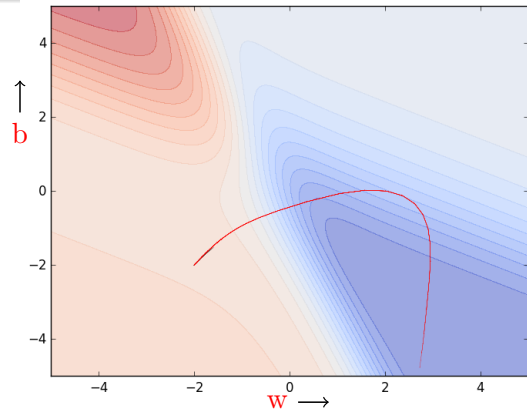


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Some observations and questions

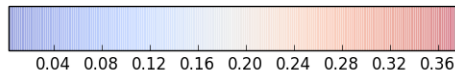
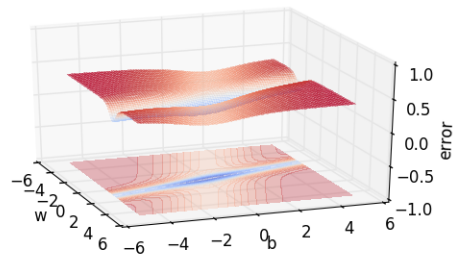
- Even in the regions having gentle slopes, momentum based gradient descent is able to take large steps because the momentum carries it along

Some observations and questions

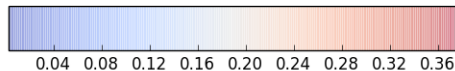
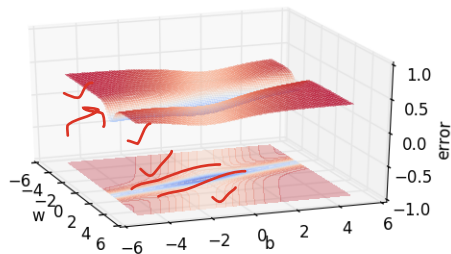
- Even in the regions having gentle slopes, momentum based gradient descent is able to take large steps because the momentum carries it along
- ✓ • Is moving fast always good? Would there be a situation where momentum would cause us to run pass our goal?

Some observations and questions

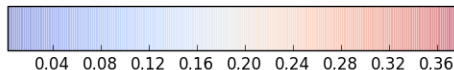
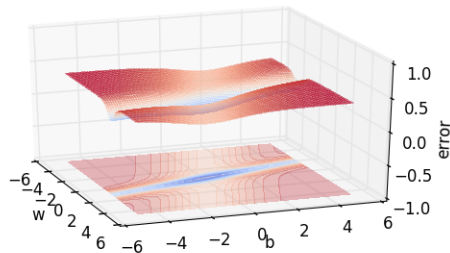
- Even in the regions having gentle slopes, momentum based gradient descent is able to take large steps because the momentum carries it along
- Is moving fast always good? Would there be a situation where momentum would cause us to run pass our goal?
- Let us change our input data so that we end up with a different error surface and then see what happens ...

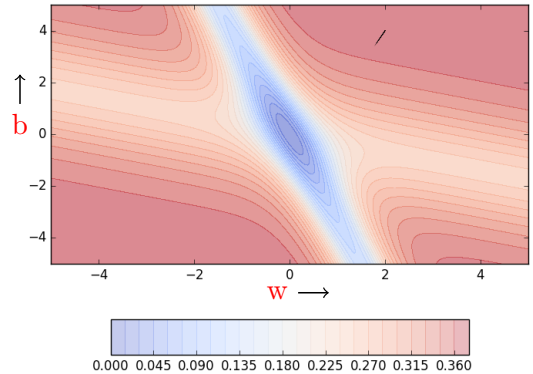


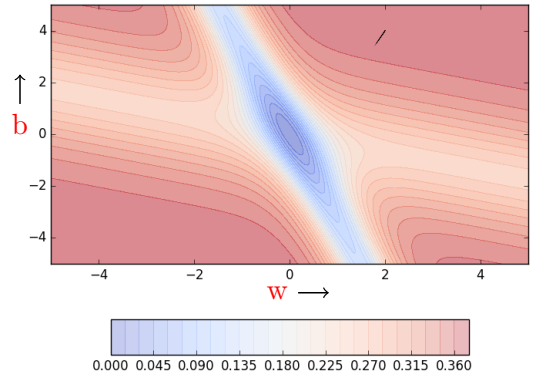
- In this case, the error is high on either side of the minima valley

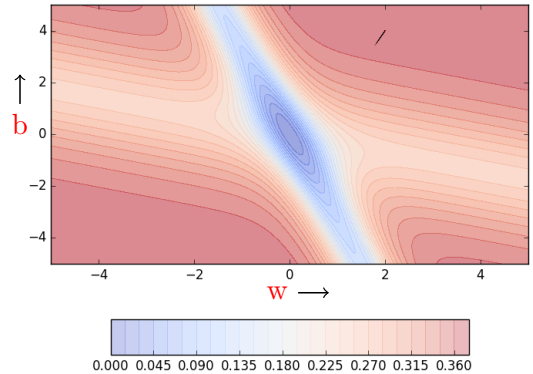


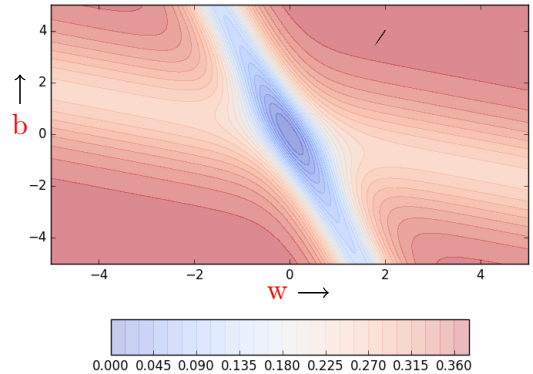
- In this case, the error is high on either side of the minima valley
- Could momentum be detrimental in such cases... let's see....

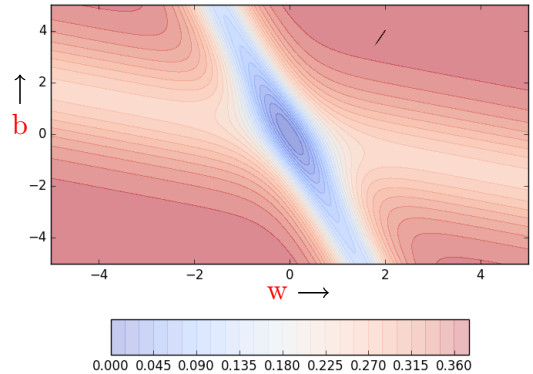


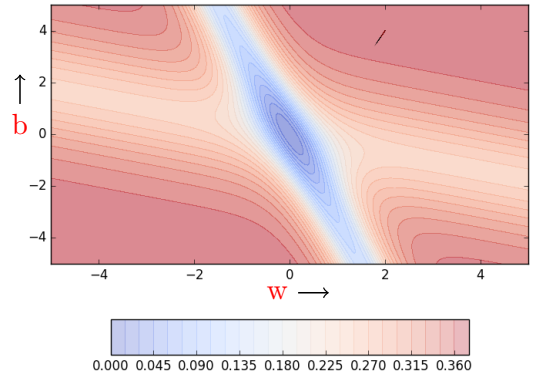


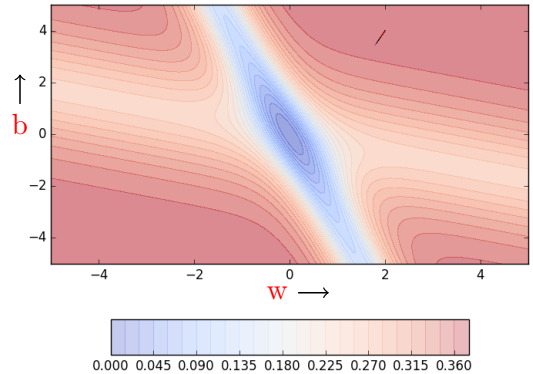


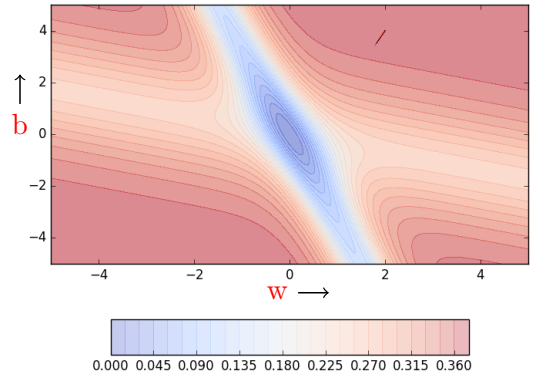


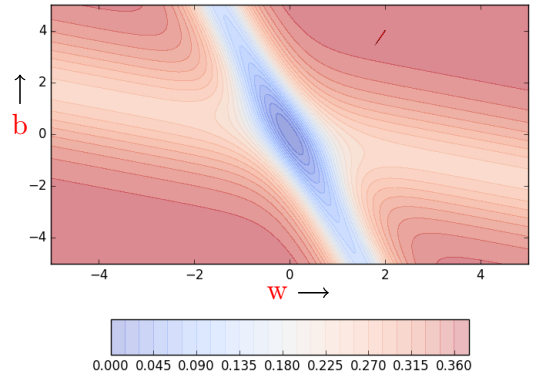


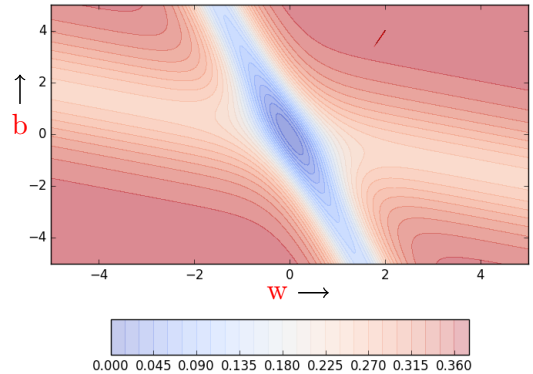


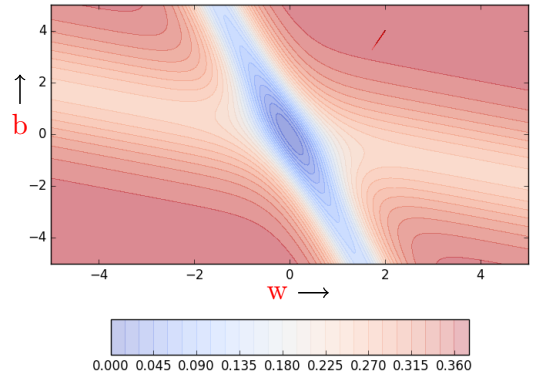


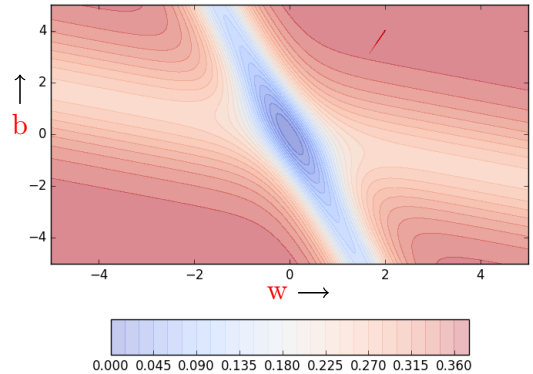


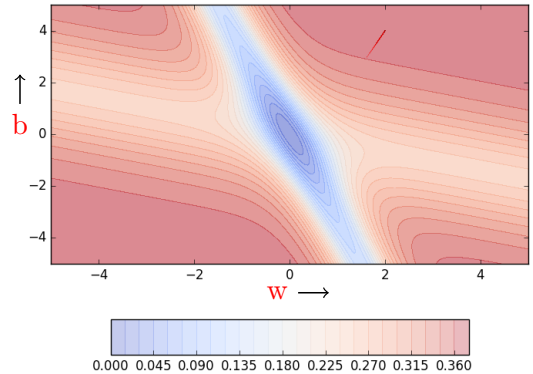


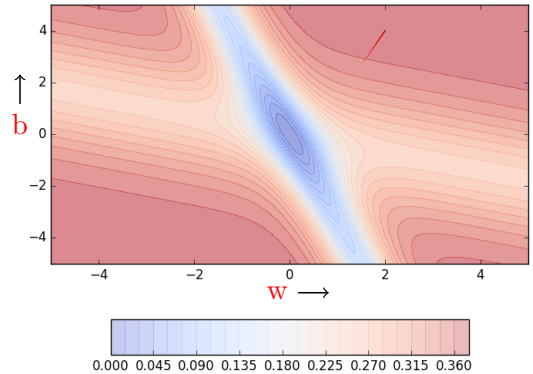


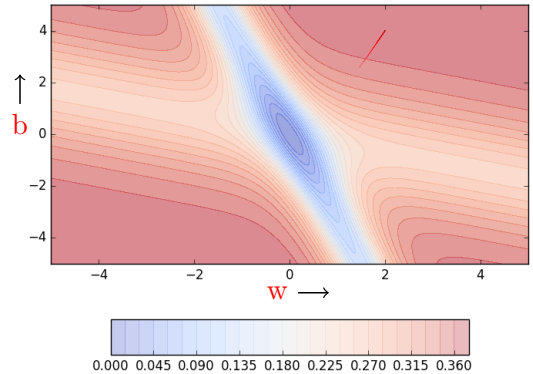


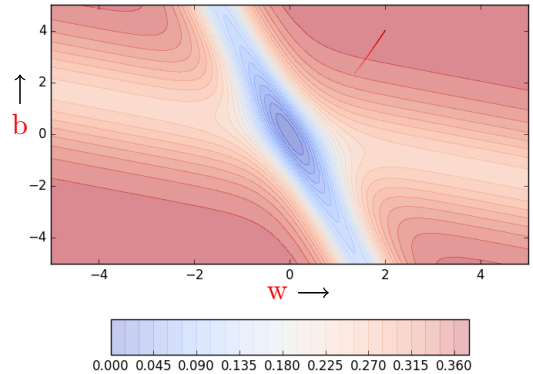


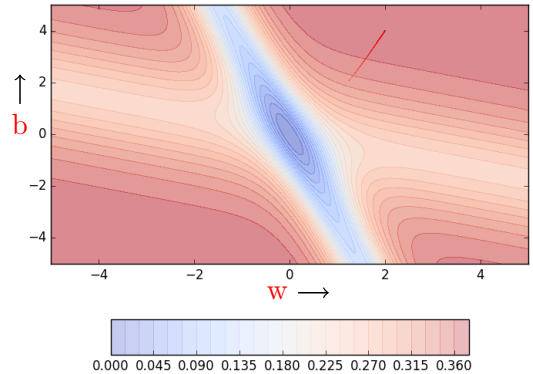


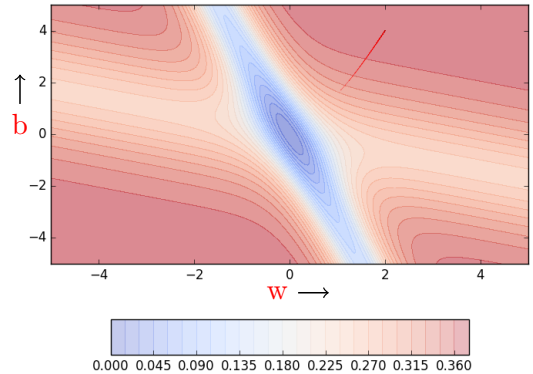


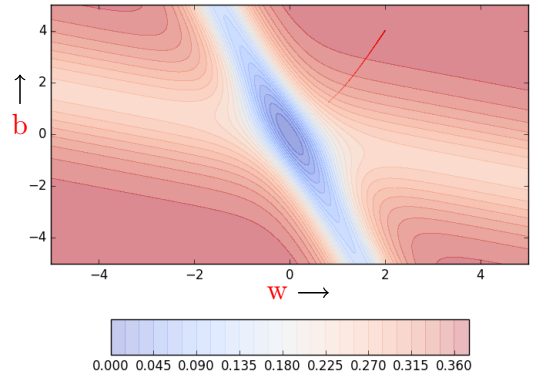


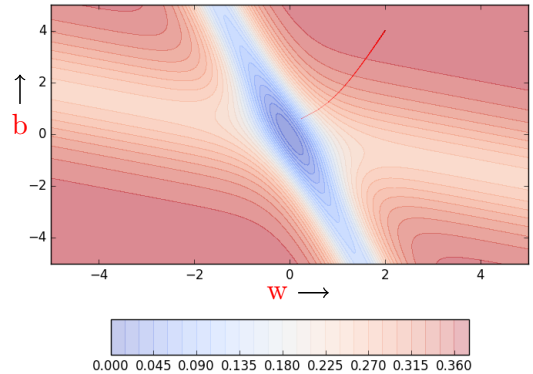


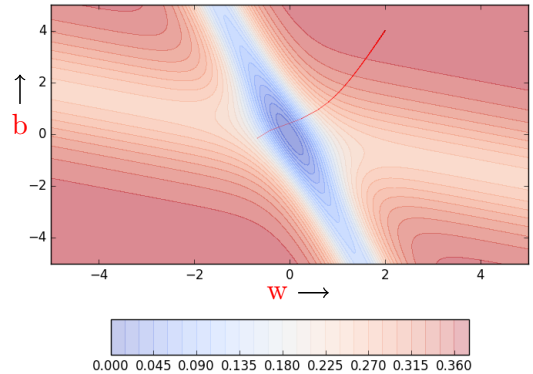


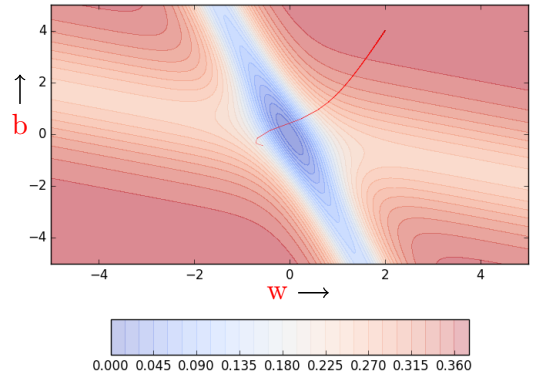


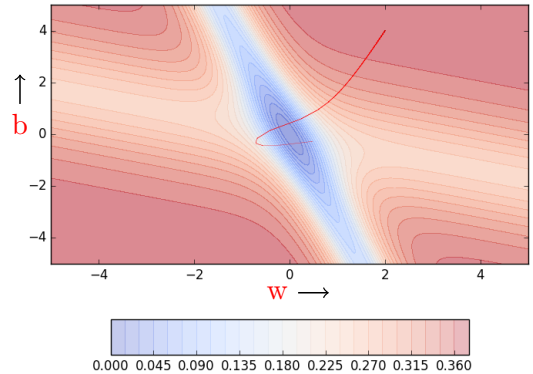


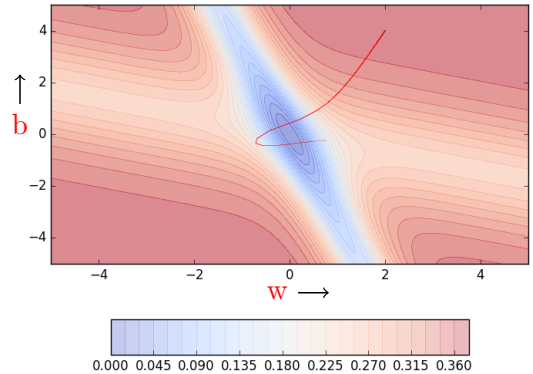


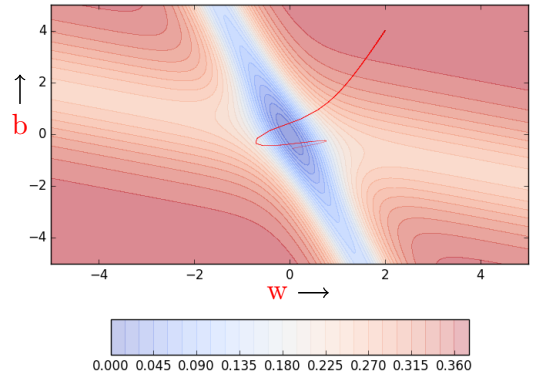


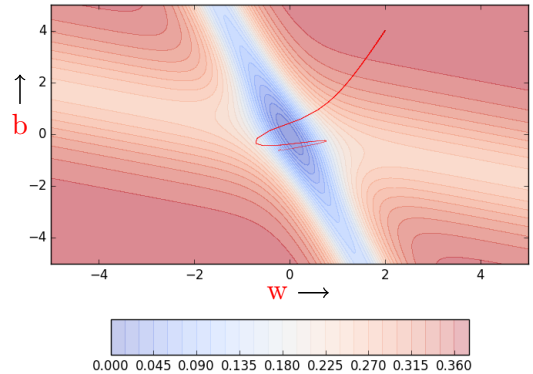


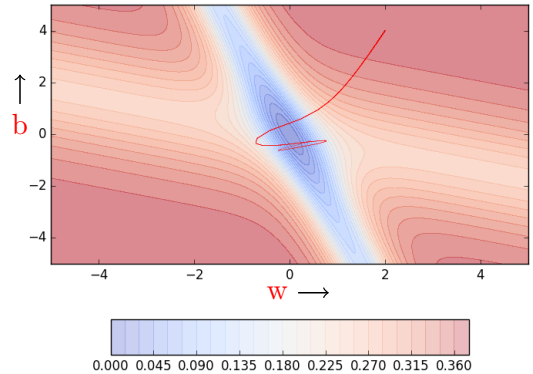


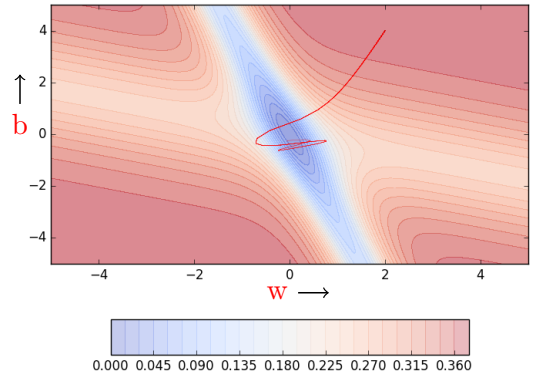


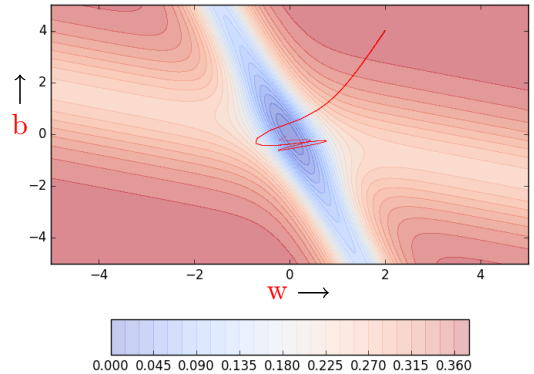


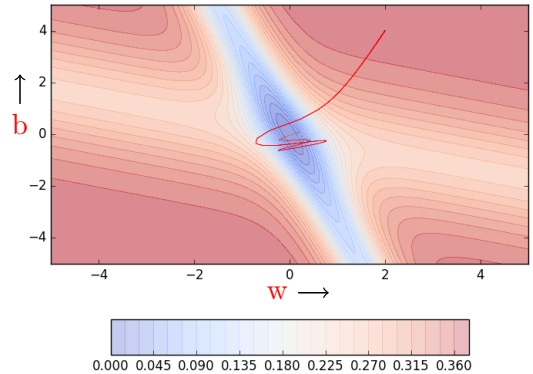


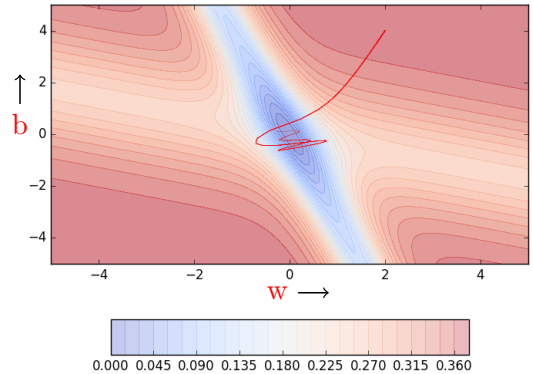


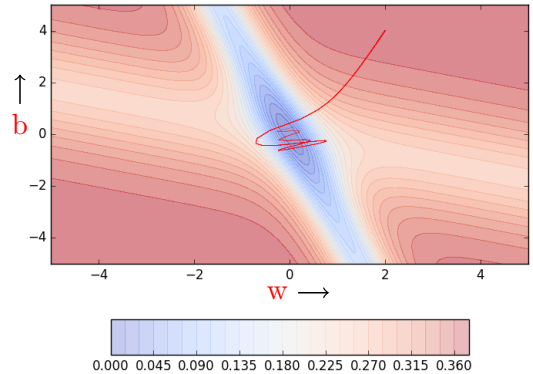


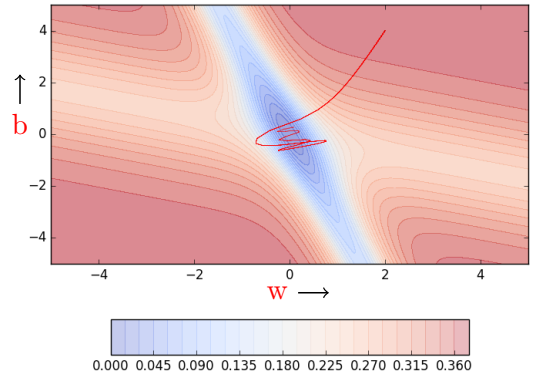


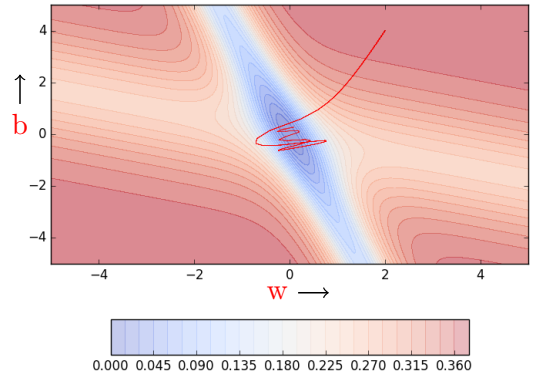


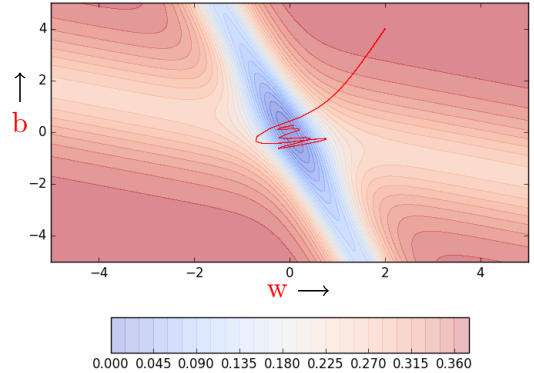


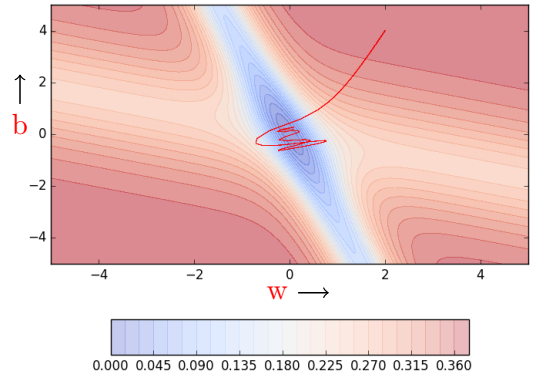




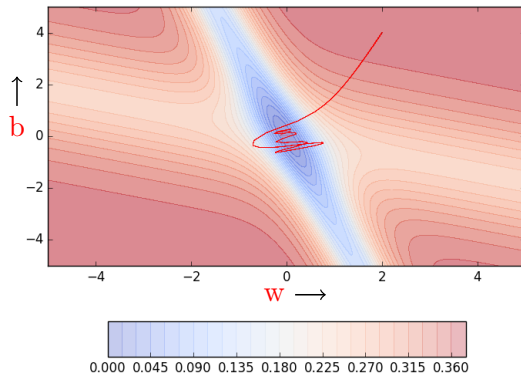




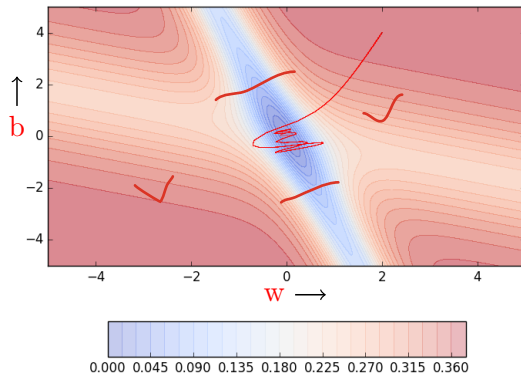




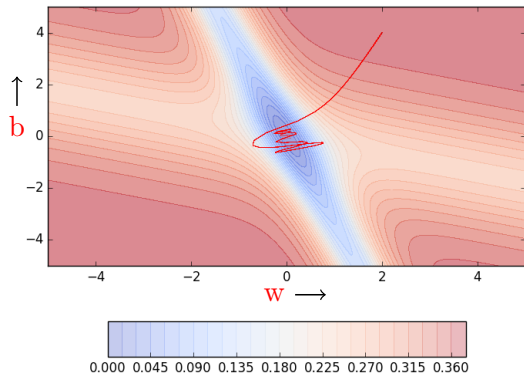
- Momentum based gradient descent oscillates in and out of the minima valley as the momentum carries it out of the valley



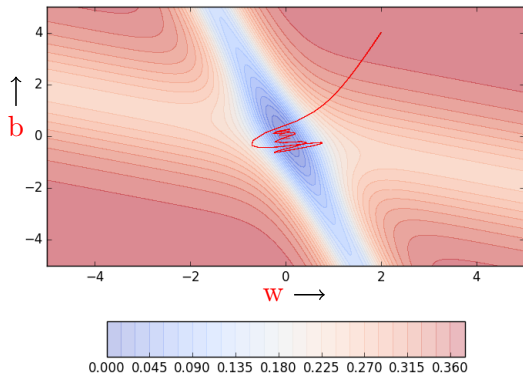
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- Despite these *u*-turns it still converges faster than vanilla gradient descent
- After 100 iterations momentum based method has reached an error of 0.00001 whereas vanilla gradient descent is still stuck at an error of 0.36



Question

- Can we do something to reduce these oscillations ?



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- Can we do something to reduce these oscillations ?
- Yes, let's look at Nesterov accelerated gradient

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- Why not calculate the gradient (∇w_{look_ahead}) at this partially updated value of w ($w_{look_ahead} = w_t - \gamma \cdot update_{t-1}$) instead of calculating it using the current value w_t

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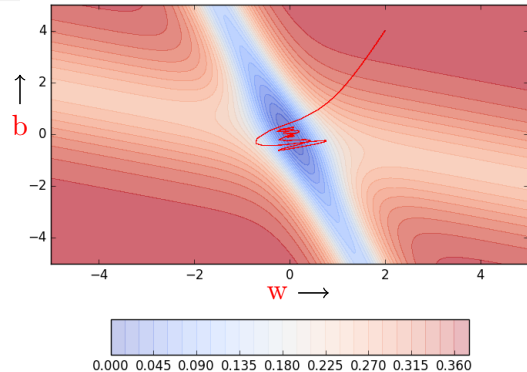
Update rule for NAG

$$\begin{aligned} \checkmark w_{look_ahead} &= \underbrace{w_t}_{\text{P}} - \gamma \cdot update_{t-1} \\ update_t &= \gamma \cdot update_{t-1} + \eta \nabla w_{look_ahead} \\ w_{t+1} &= w_t - update_t \end{aligned}$$

We will have similar update rule for b_t

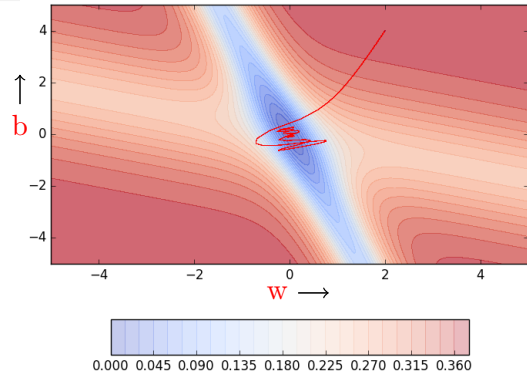
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



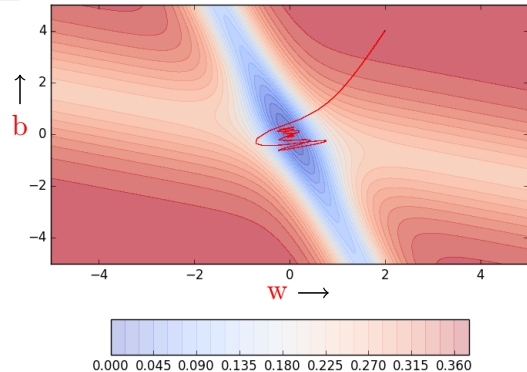
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
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        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
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            db += grad_b(w - v_w, b - v_b, x, y)

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        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
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```



```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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        dw, db = 0, 0
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        v_b = gamma * prev_v_b
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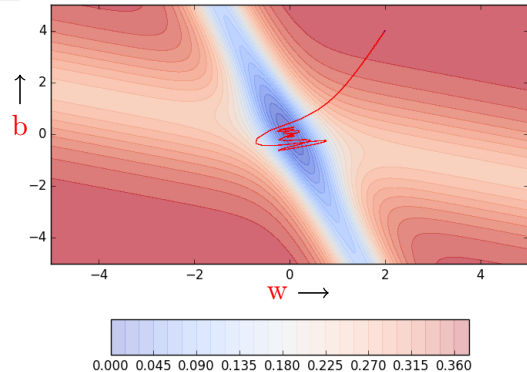
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
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        prev_v_b = v_b
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```
def do_nesterov_accelerated_gradient_descent() :

    w, b, eta = init_w, init_b , 1.0
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            db += grad_b(w - v_w, b - v_b, x, y)

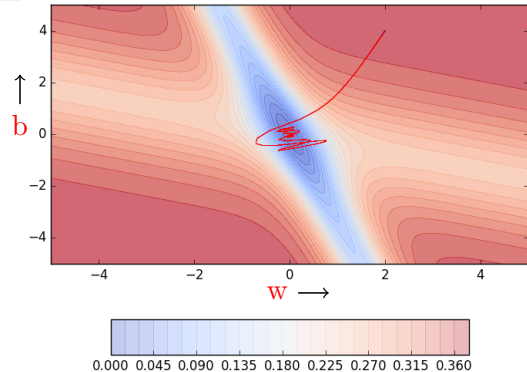
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :

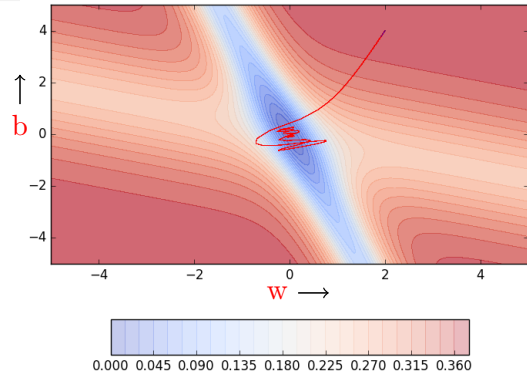
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
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```



```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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        dw, db = 0, 0
        #do partial updates
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        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

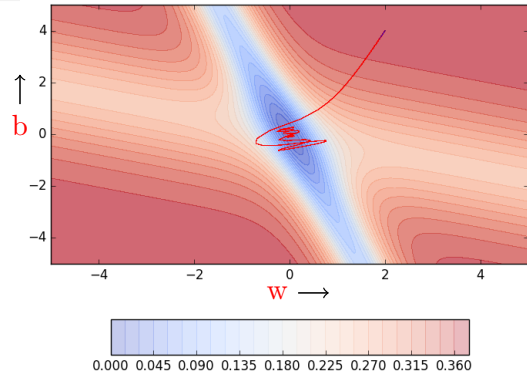
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```




```
def do_nesterov_accelerated_gradient_descent() :

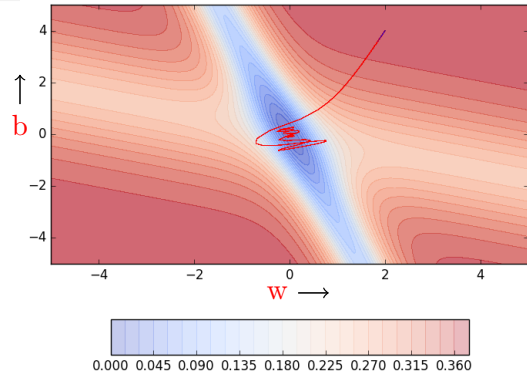
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



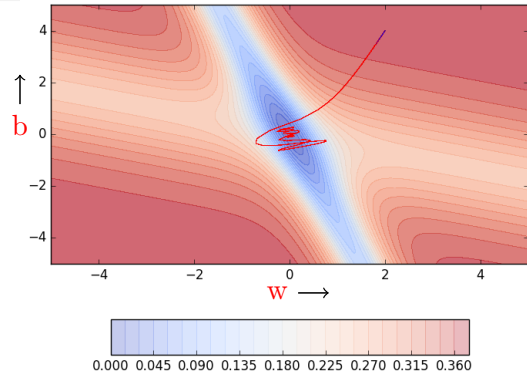
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
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            dw += grad_w(w - v_w, b - v_b, x, y)
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        v_w = gamma * prev_v_w + eta * dw
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```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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        dw, db = 0, 0
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        v_w = gamma * prev_v_w
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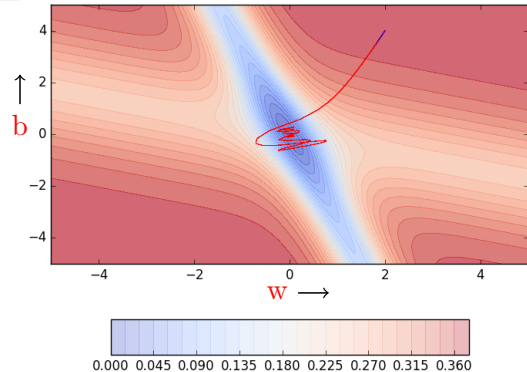
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :

    w, b, eta = init_w, init_b , 1.0
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    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

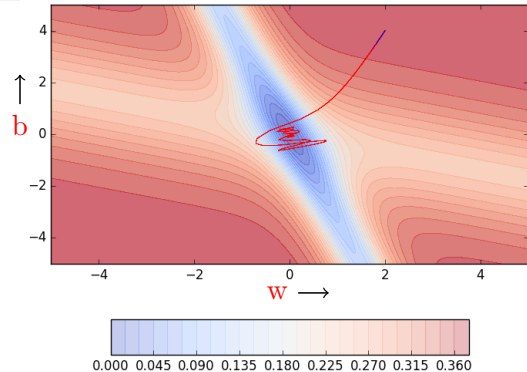
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :

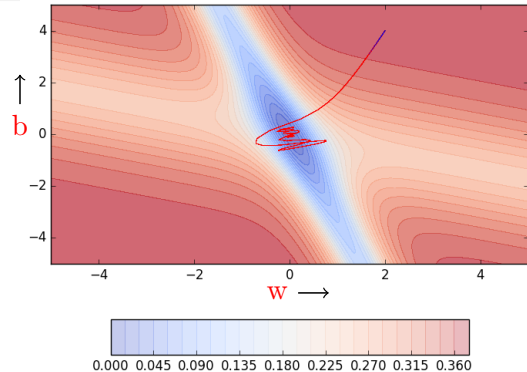
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



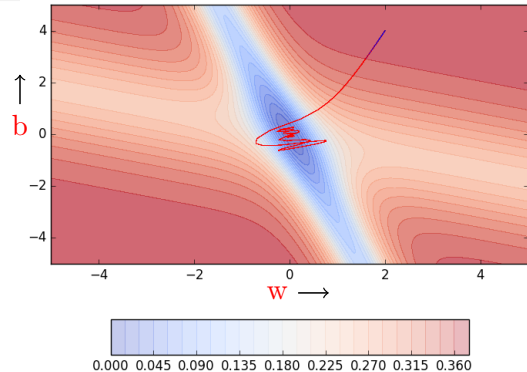
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
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        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



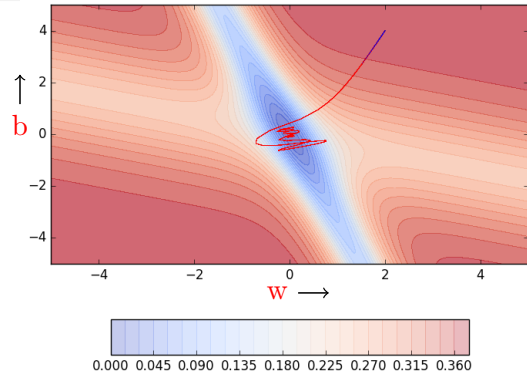
```
def do_nesterov_accelerated_gradient_descent() :
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        b = b - v_b
        prev_v_w = v_w
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```



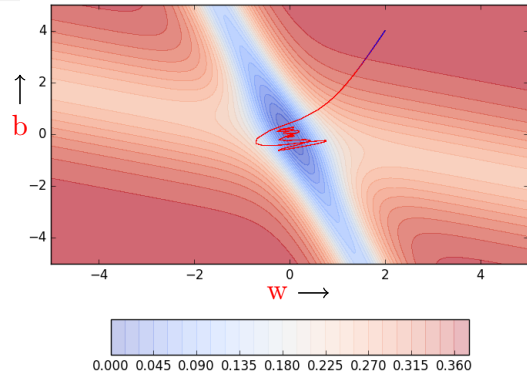
```
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        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



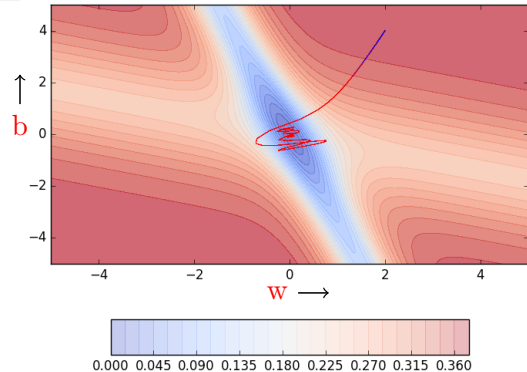

```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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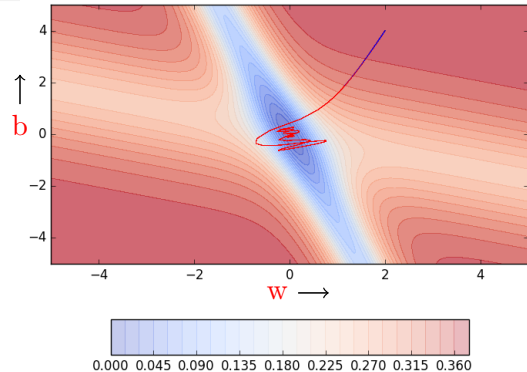
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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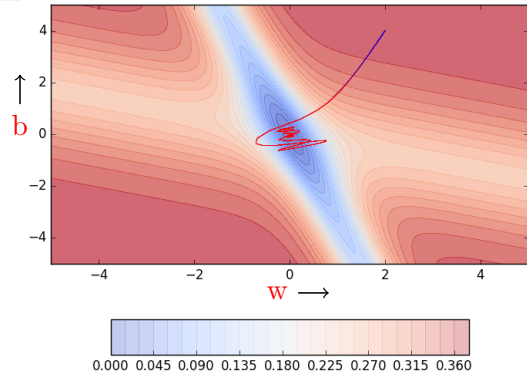
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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```
def do_nesterov_accelerated_gradient_descent() :

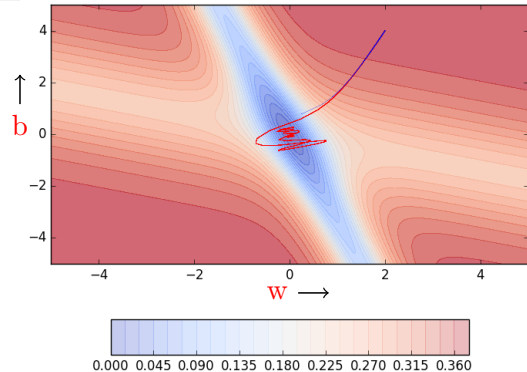
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
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```



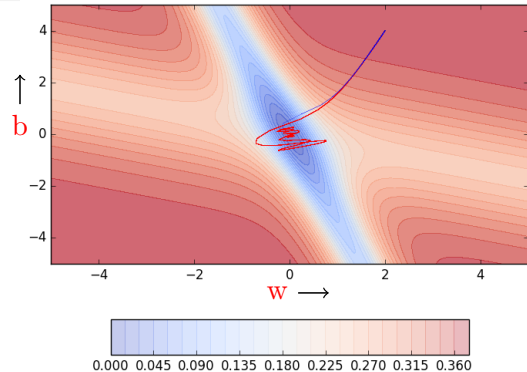
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        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



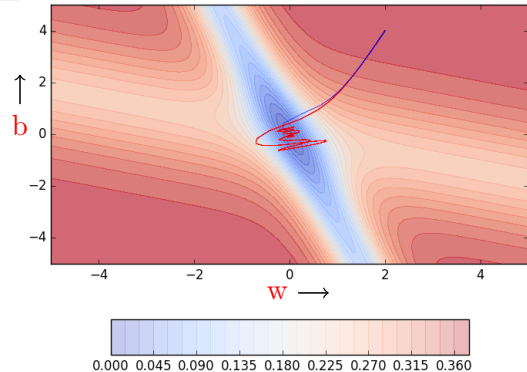
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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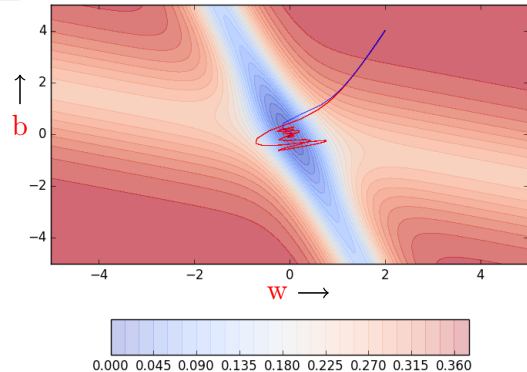
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        b = b - v_b
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```
def do_nesterov_accelerated_gradient_descent() :

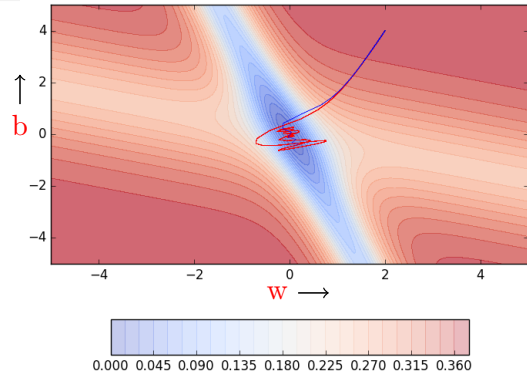
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
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```




```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
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        for x,y in zip(X, Y) :
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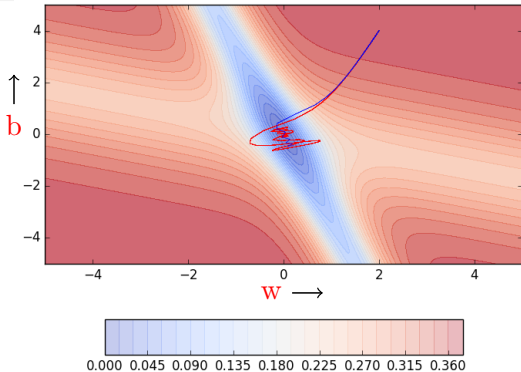
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
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```



```
def do_nesterov_accelerated_gradient_descent() :

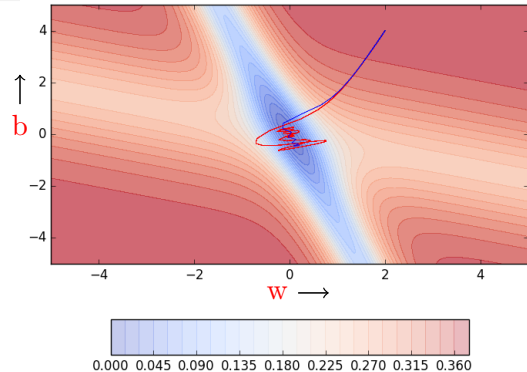
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
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        v_w = gamma * prev_v_w
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        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



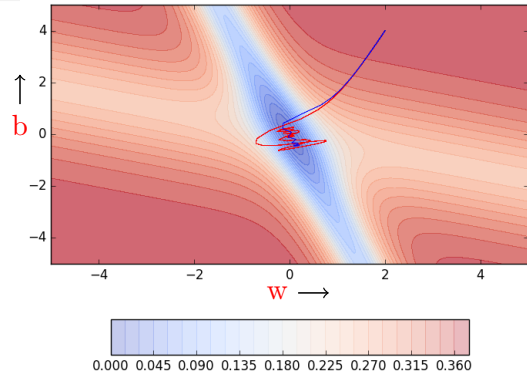
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
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        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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        dw, db = 0, 0
        #do partial updates
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        for x,y in zip(X, Y) :
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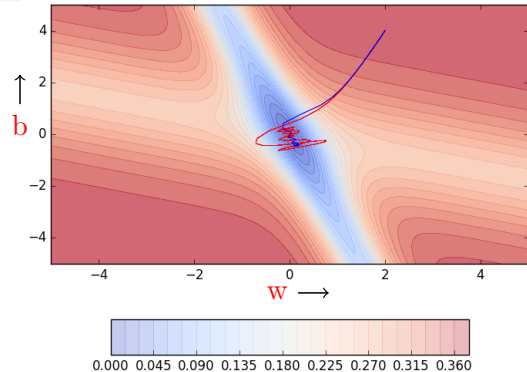
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
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```
def do_nesterov_accelerated_gradient_descent() :

    w, b, eta = init_w, init_b , 1.0
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        for x,y in zip(X, Y) :
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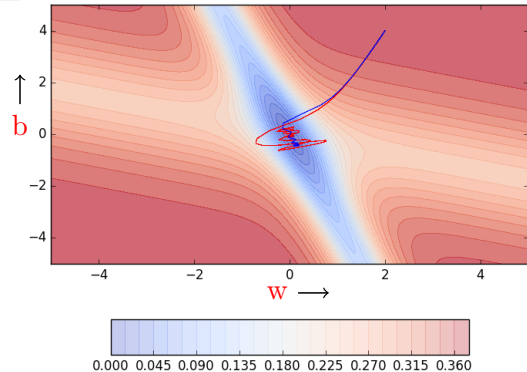
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :

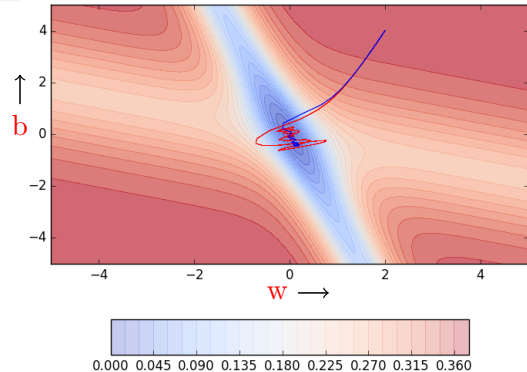
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
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        for x,y in zip(X, Y) :
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        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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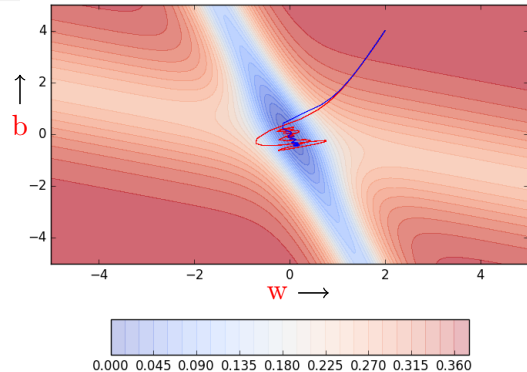
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
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        prev_v_b = v_b
```



```
def do_nesterov_accelerated_gradient_descent() :

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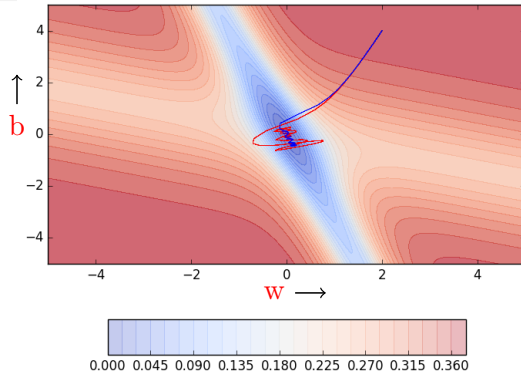
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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```




```
def do_nesterov_accelerated_gradient_descent() :
```

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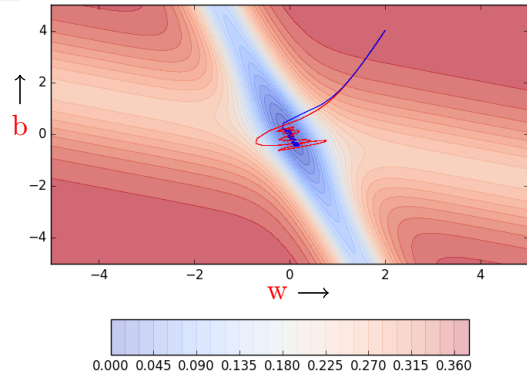
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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        prev_v_w = v_w
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```



```
def do_nesterov_accelerated_gradient_descent() :

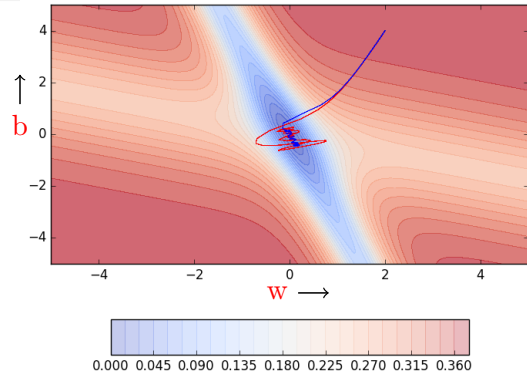
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
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        for x,y in zip(X, Y) :
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            dw += grad_w(w - v_w, b - v_b, x, y)
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        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```



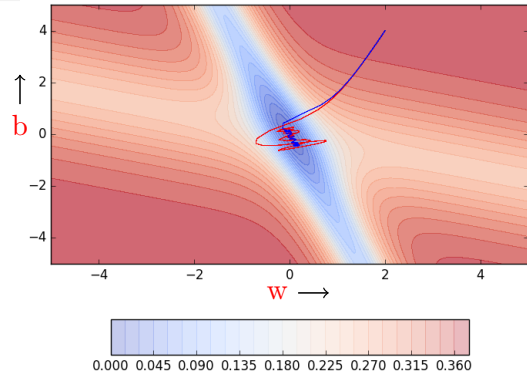
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
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```



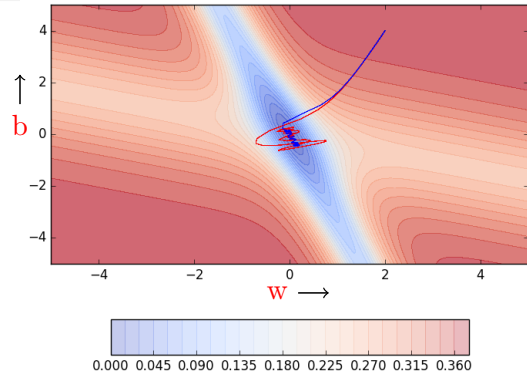
```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
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        b = b - v_b
        prev_v_w = v_w
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```



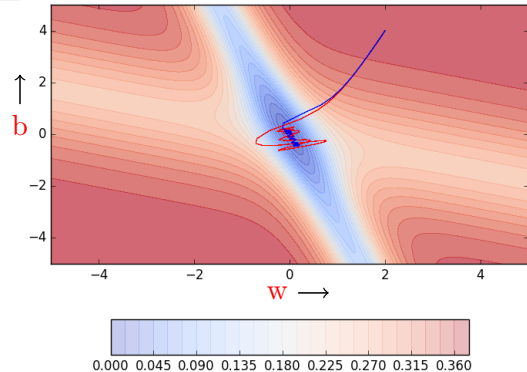
```
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```
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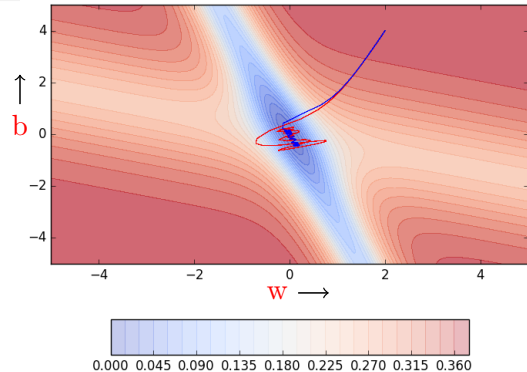
        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
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        prev_v_w = v_w
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```



```
def do_nesterov_accelerated_gradient_descent() :

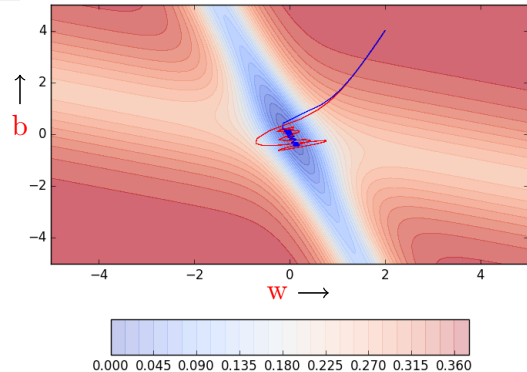
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
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        v_w = gamma * prev_v_w
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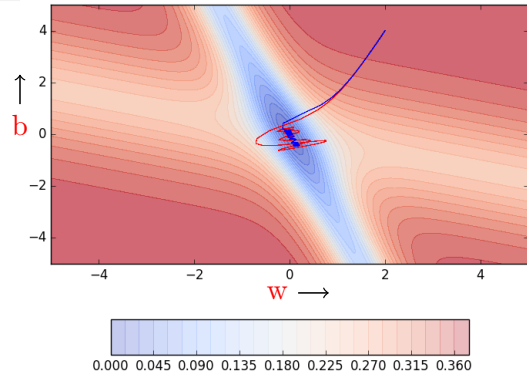
```
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    w, b, eta = init_w, init_b , 1.0
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        dw, db = 0, 0
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        for x,y in zip(X, Y) :
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```



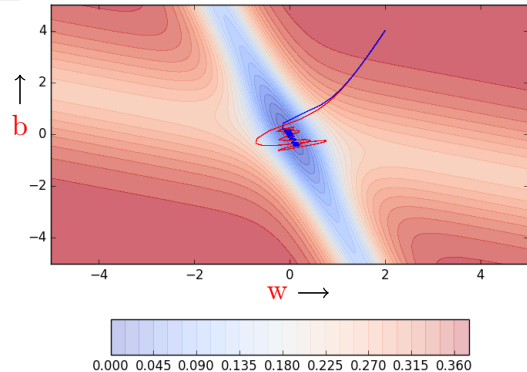

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```
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```



Observations about NAG

- Looking ahead helps NAG in correcting its course quicker than momentum based gradient descent

Observations about NAG

- Looking ahead helps NAG in correcting its course quicker than momentum based gradient descent
- Hence the oscillations are smaller and the chances of escaping the minima valley also smaller

Module 5.6 : Stochastic And Mini-Batch Gradient Descent

Let's digress a bit and talk about the stochastic version of these algorithms...

```
X = [0.5, 2.5]
Y = [0.2, 0.9]

def f(w, b, x): #sigmoid with parameters w,b
    return 1.0 / (1.0 + np.exp(-(w*x +b)))

def error(w, b):
    err = 0.0
    for x,y in zip(X,Y):
        fx = f(w,b,x)
        err += 0.5* (fx - y) ** 2
    return err

def grad_b(w, b, x, y):
    fx = f(w, b, x)
    return (fx - y) * fx * (1 - fx)

def grad_w(w, b, x, y):
    fx = f(w, b, x)
    return (fx - y) * fx * (1 - fx) * x

def do_gradient_descent():
    w, b, eta, max_epochs = -2, -2, 1.0, 1000
    for i in range(max_epochs):
        dw, db = 0, 0
        for x, y in zip(X, Y):
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)
        w = w - eta * dw
        b = b - eta * db
```

```

X = [0.5, 2.5]
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    return err

def grad_b(w, b, x, y):
    fx = f(w, b, x)
    return (fx - y) * fx * (1 - fx)

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    fx = f(w, b, x)
    return (fx - y) * fx * (1 - fx) * x

def do_gradient_descent():
    w, b, eta, max_epochs = -2, -2, 1.0, 1000
    for i in range(max_epochs):
        dw, db = 0, 0
        for x, y in zip(X, Y):
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)
        w = w - eta * dw
        b = b - eta * db

```

- Notice that the algorithm goes over the entire data once before updating the parameters


```

X = [0.5, 2.5]
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def f(w, b, x): #sigmoid with parameters w,b
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    fx = f(w, b, x)
    return (fx - y) * fx * (1 - fx)

def grad_w(w, b, x, y):
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    w, b, eta, max_epochs = -2, -2, 1.0, 1000
    for i in range(max_epochs):
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- Why?

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X = [0.5, 2.5]
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def do_stochastic_gradient_descent():
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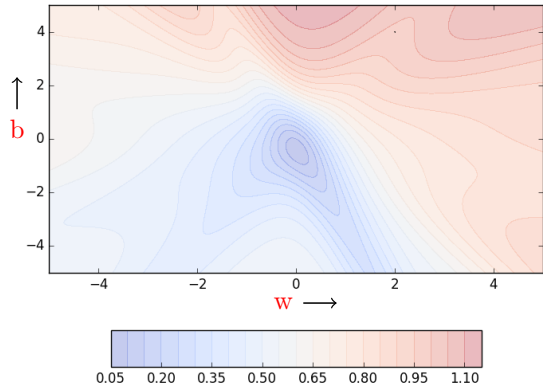
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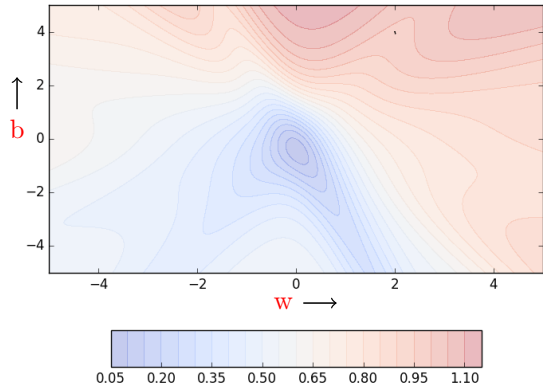
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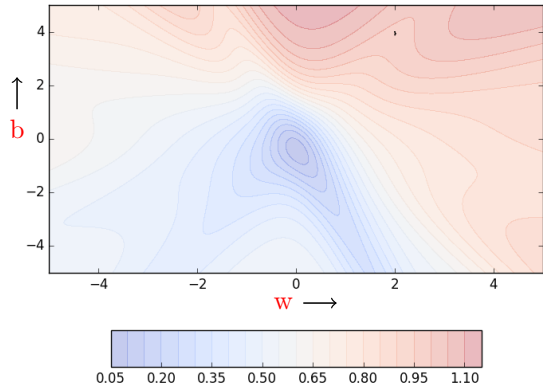
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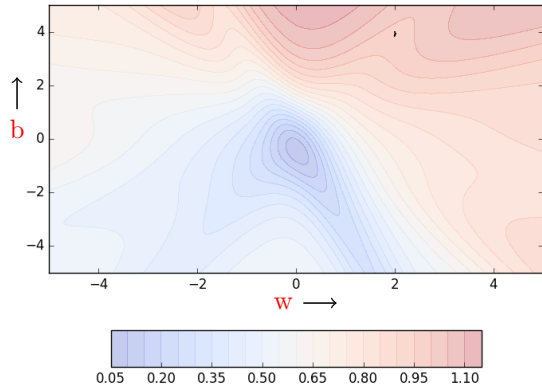
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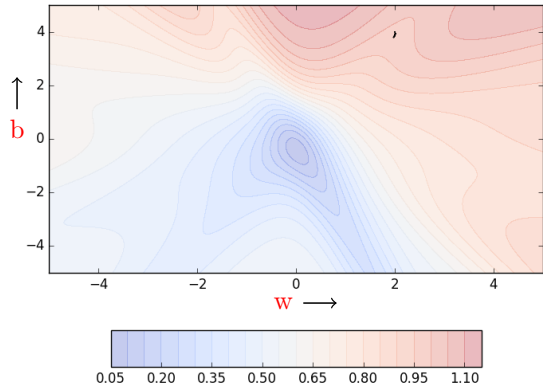
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- Let's see this algorithm in action when we have a few data points

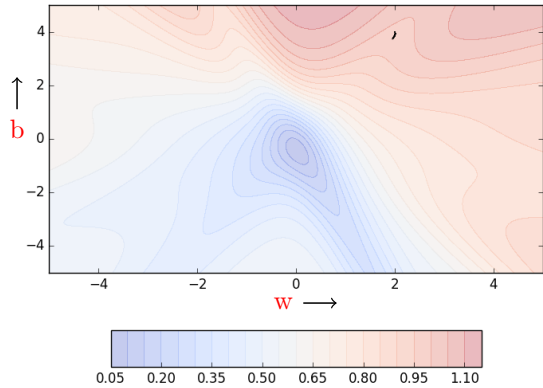


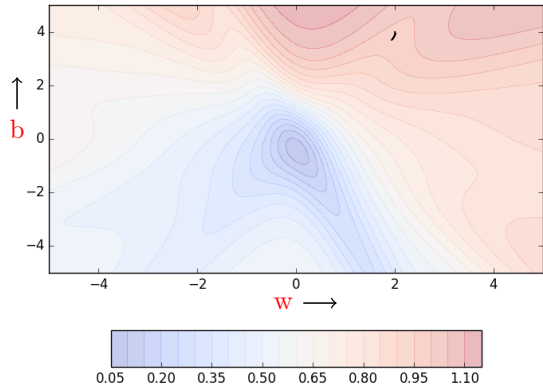


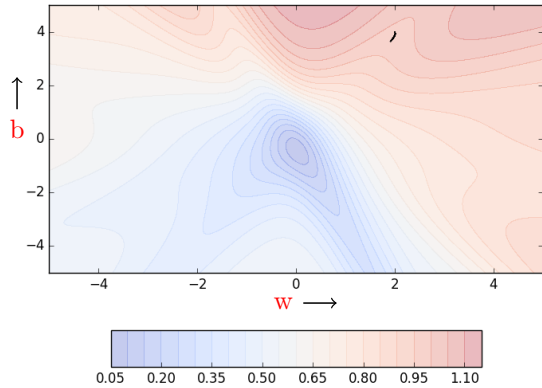


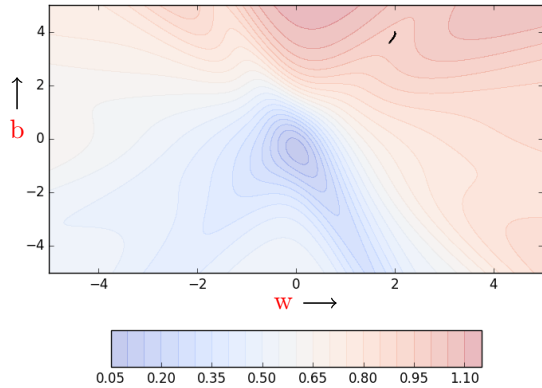


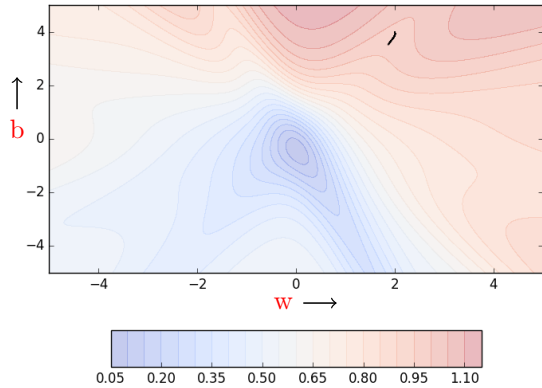


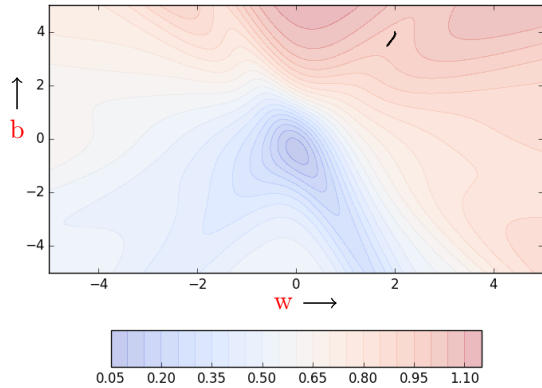


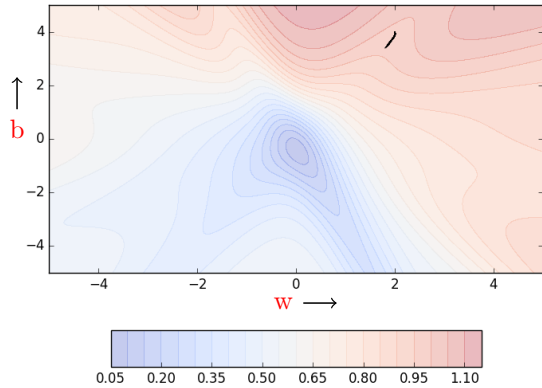


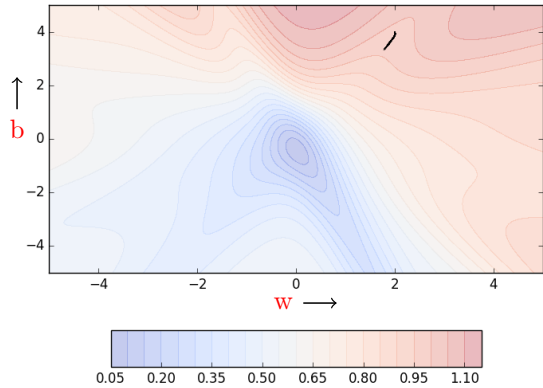


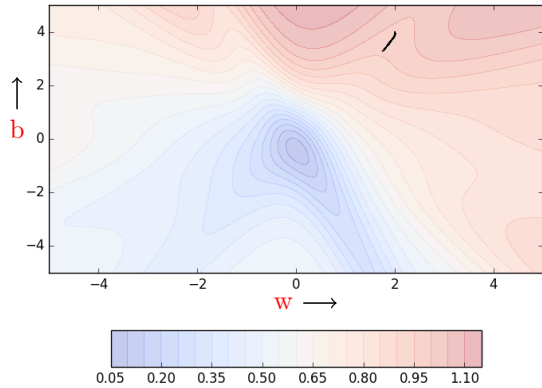


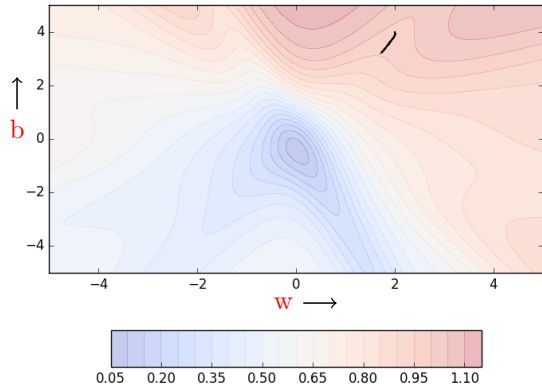


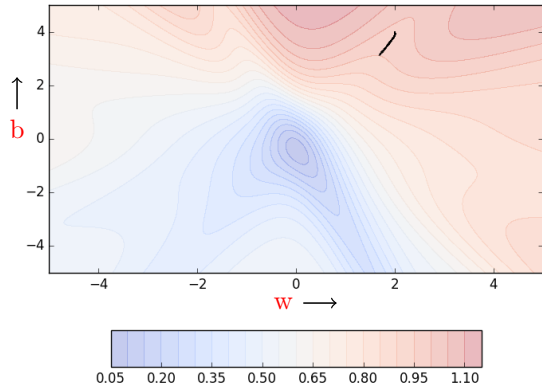


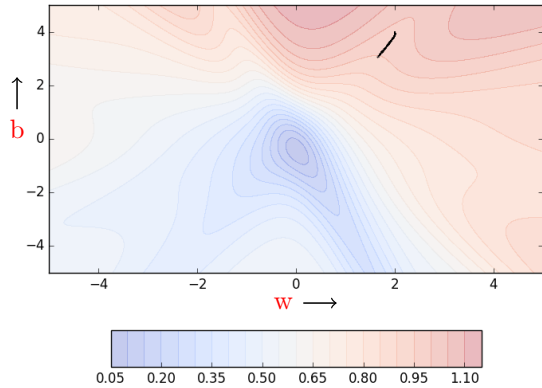


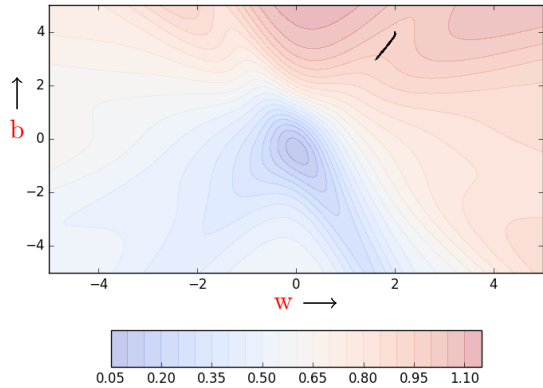


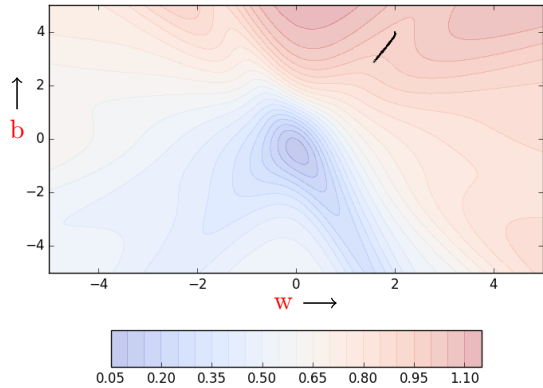


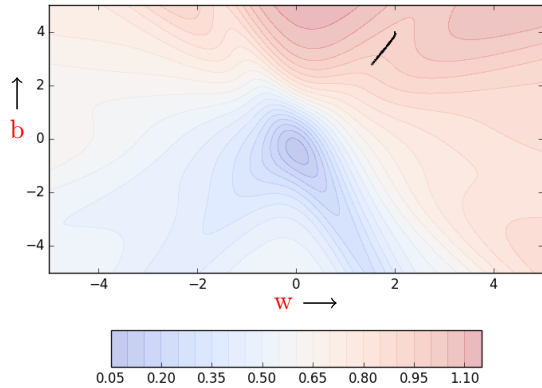


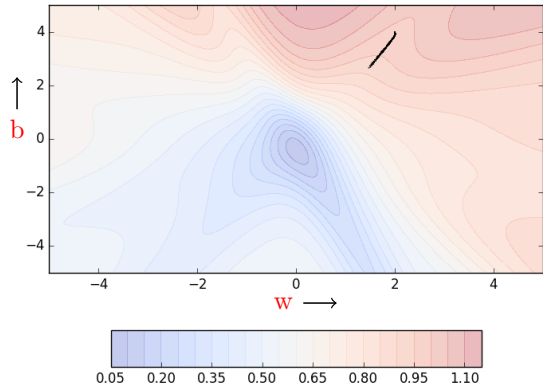


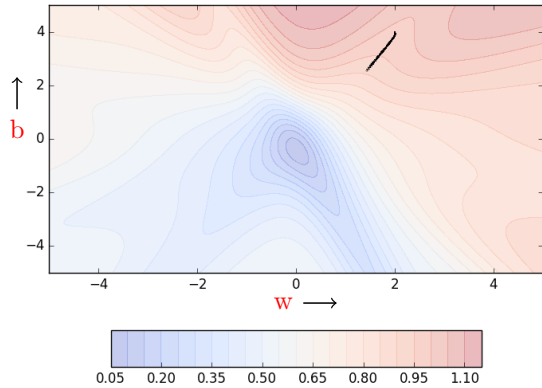


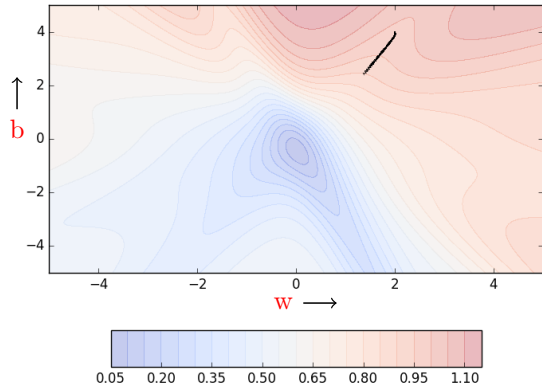


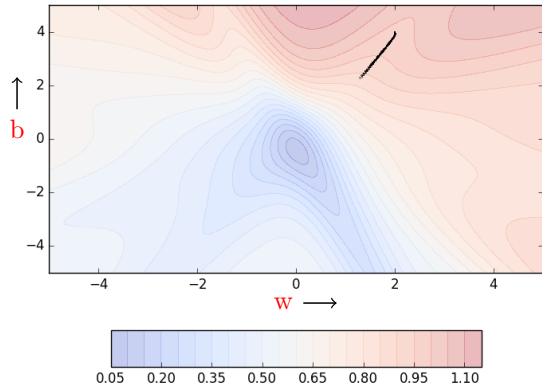


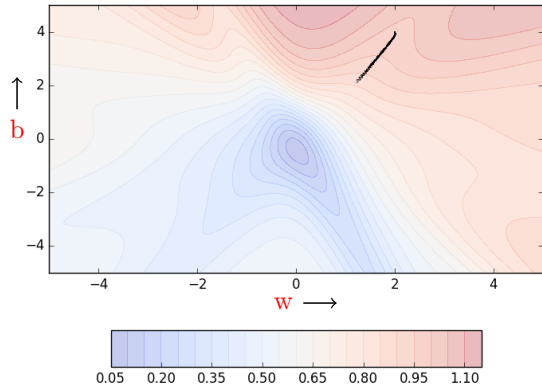


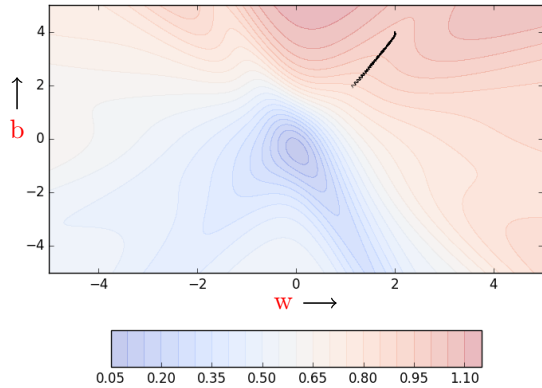


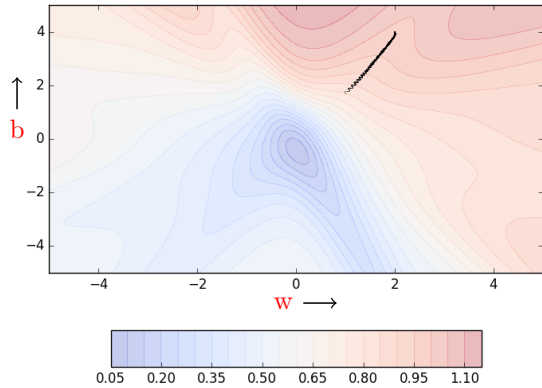


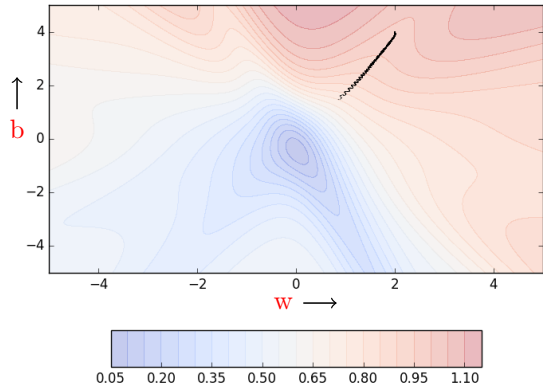


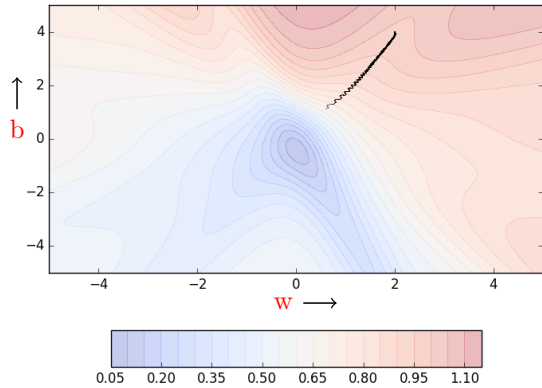


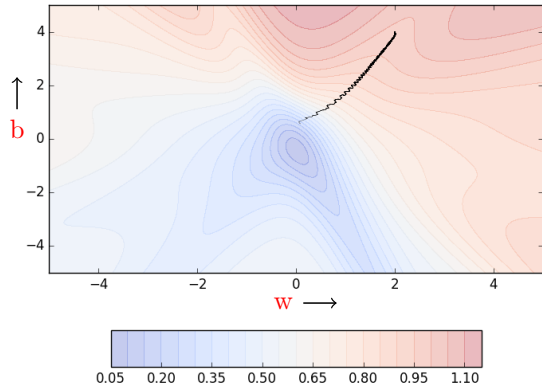


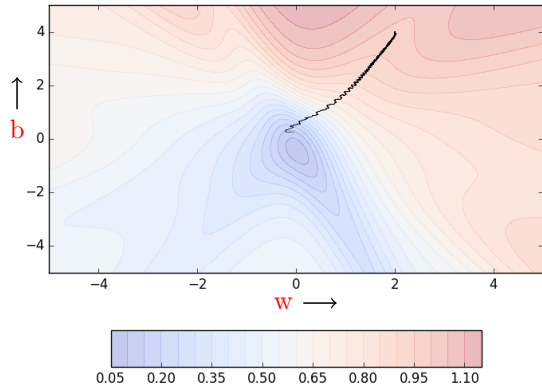


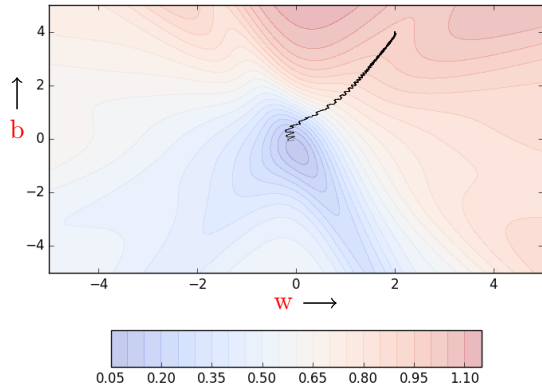


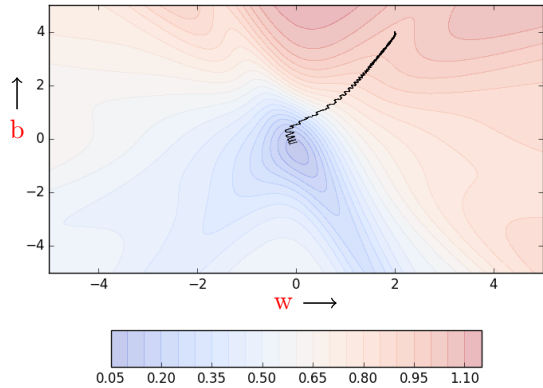


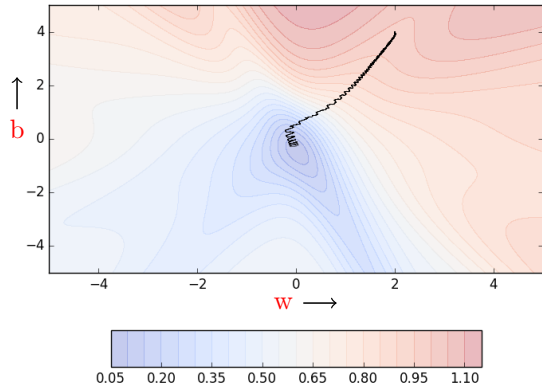


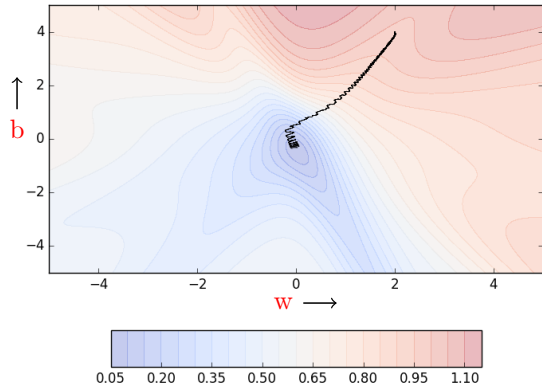


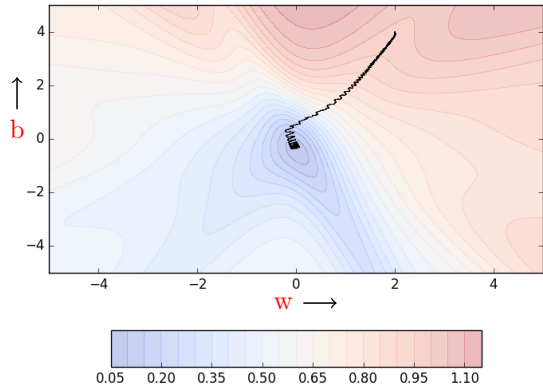




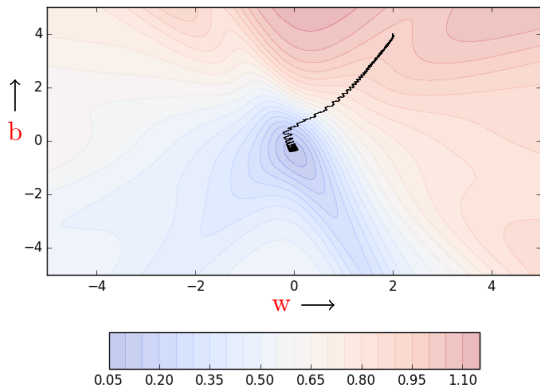




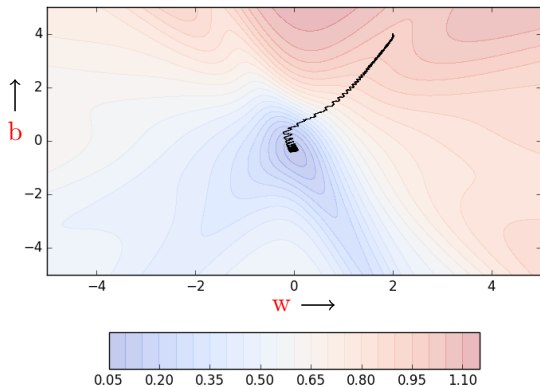




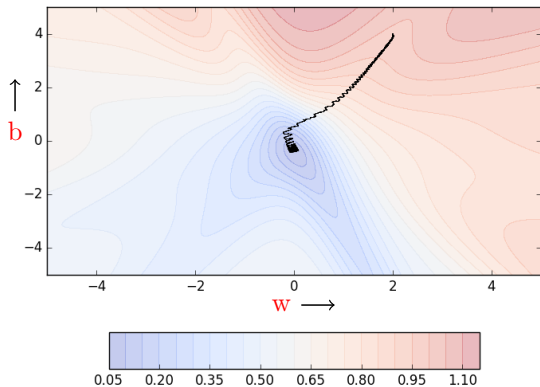
- We see many oscillations. Why ?



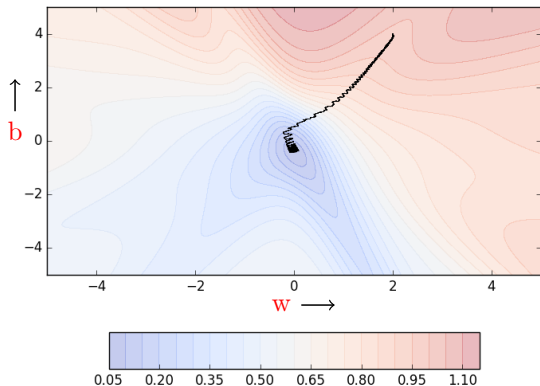
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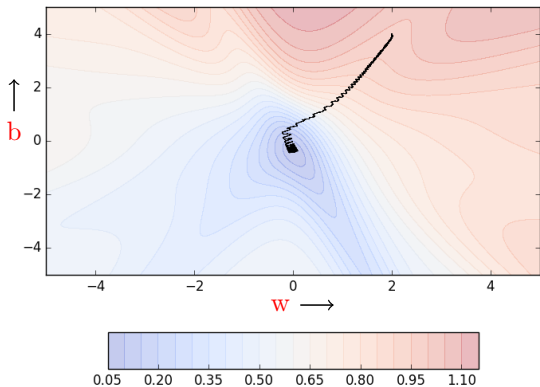
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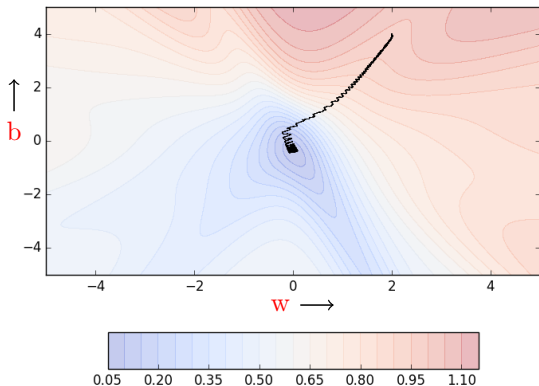
- We see many oscillations. Why? Because we are making greedy decisions.
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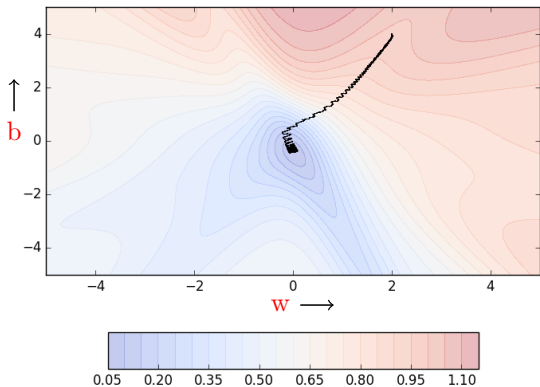
- We see many oscillations. Why? Because we are making greedy decisions.
- Each point is trying to push the parameters in a direction most favorable to it (without being aware of how this affects other points)
- A parameter update which is locally favorable to one point may harm other points



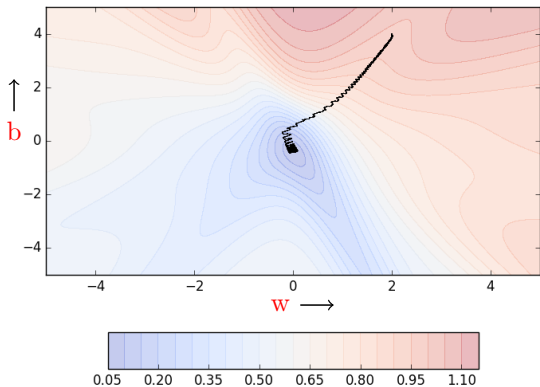
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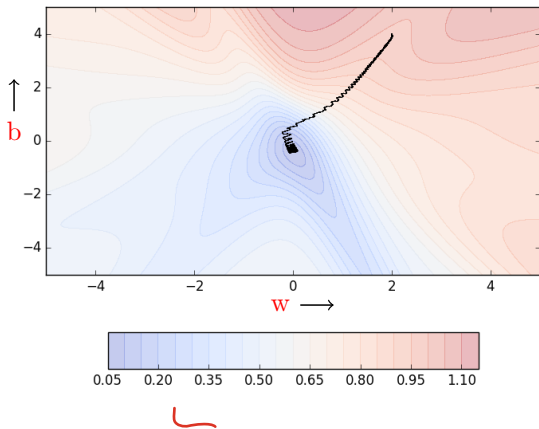
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- Can we reduce the oscillations by improving our stochastic estimates of the gradient (currently estimated from just 1 data point at a time)
- Yes, let's look at mini-batch gradient descent




```
def do_mini_batch_gradient_descent() :
    w, b, eta = -2, -2, 1.0
    mini_batch_size, num_points_seen = 2, 0
    for i in range(max_epochs) :
        dw, db, num_points = 0, 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)
            num_points_seen +=1

        if num_points_seen % mini_batch_size == 0 :
            # seen one mini_batch
            w = w - eta * dw
            b = b - eta * db
            dw, db = 0, 0 #reset gradients
```

- Notice that the algorithm updates the parameters after it sees *mini_batch_size* number of data points

```
def do_stochastic_gradient_descent():
    w, b, eta, max_epochs = -2, -2, 1.0, 1000
    for i in range(max_epochs):
        dw, db = 0, 0
        for x, y in zip(X, Y):
            dw = grad_w(w, b, x, y)
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            w = w - eta * dw
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- Notice that the algorithm updates the parameters after it sees *mini_batch_size* number of data points
- The stochastic estimates are now slightly better

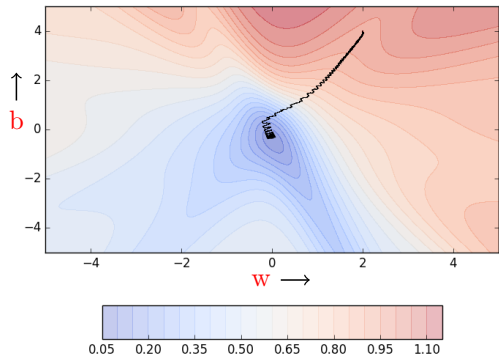
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```

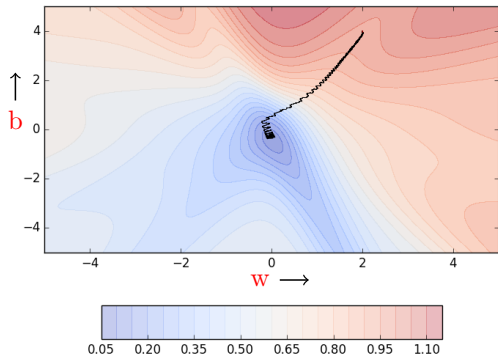
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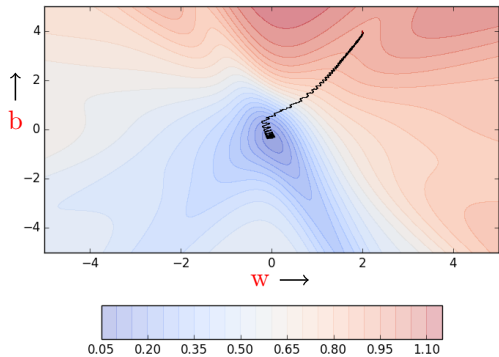
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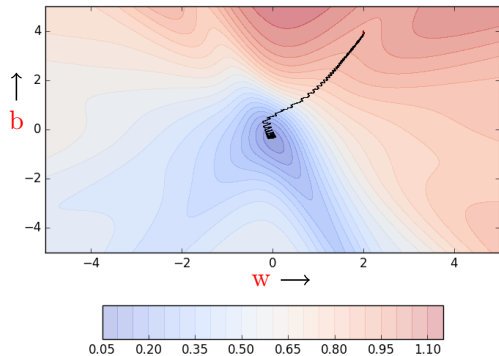
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def do_stochastic_gradient_descent():
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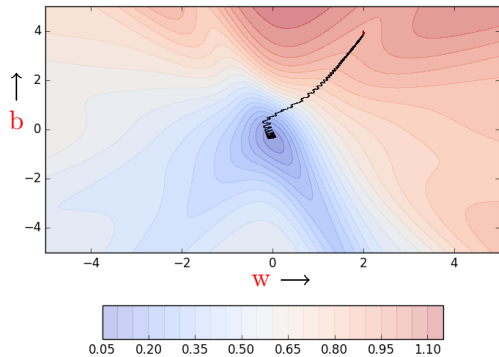
- Notice that the algorithm updates the parameters after it sees *mini_batch_size* number of data points
- The stochastic estimates are now slightly better
- Let's see this algorithm in action when we have $k = 2$

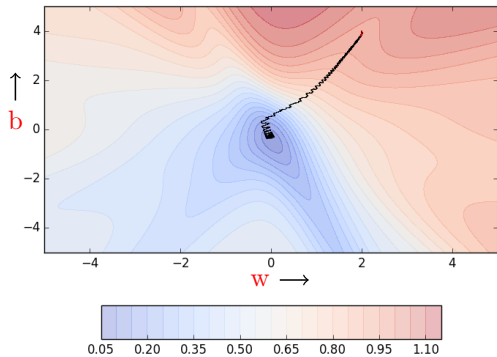


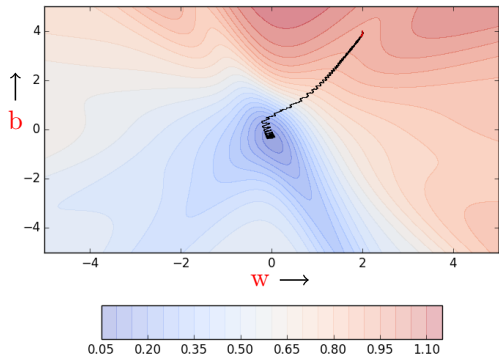


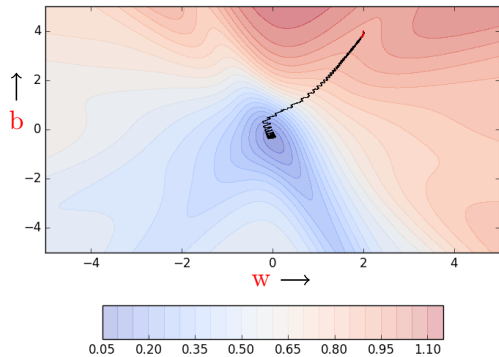


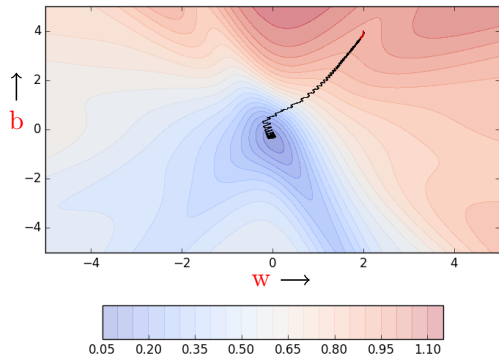


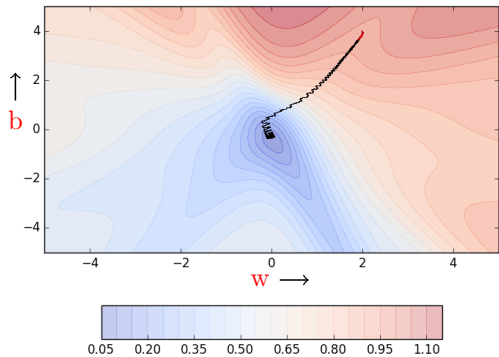


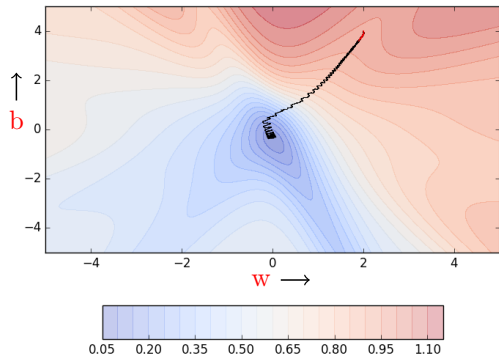


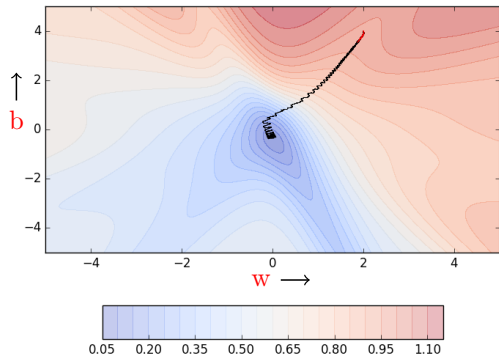


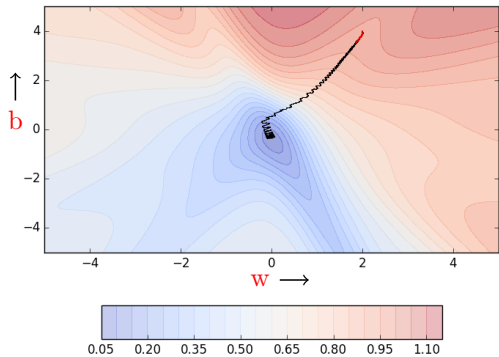


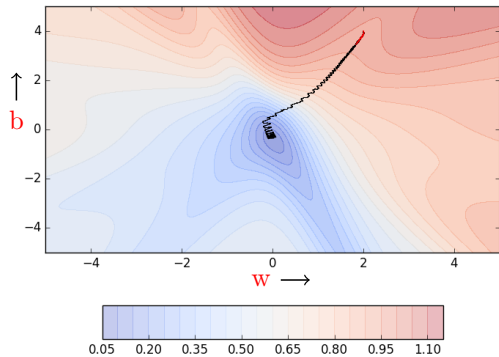


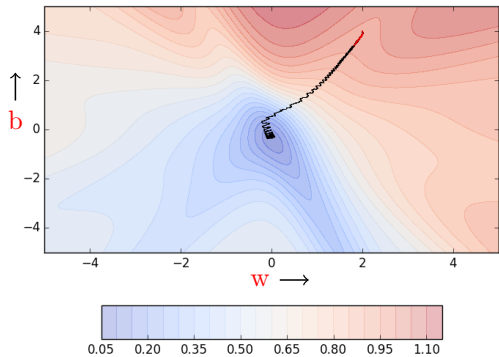


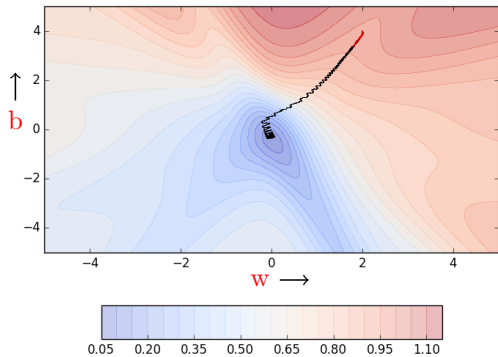


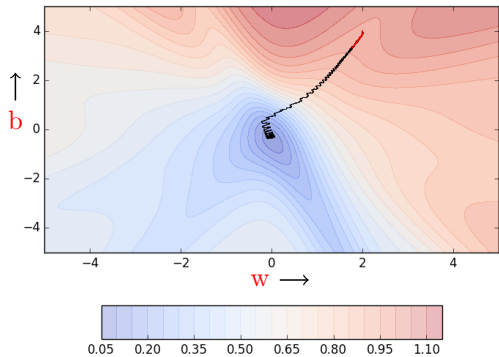


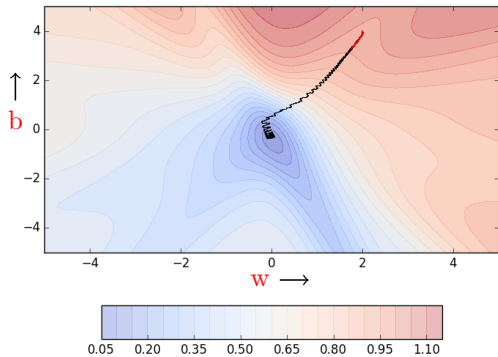


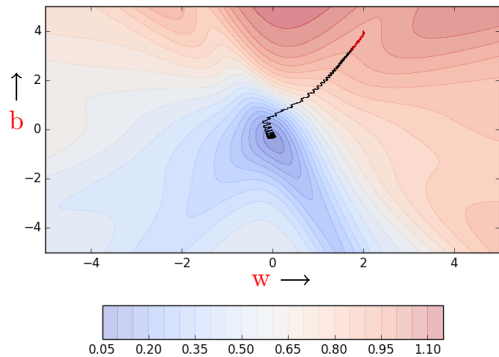


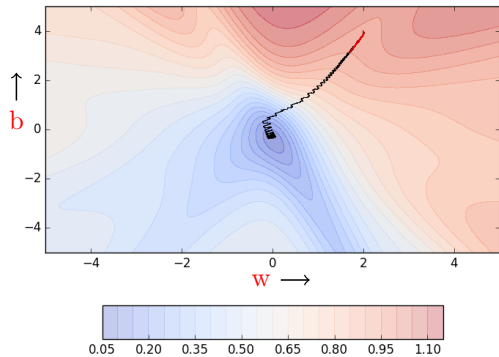


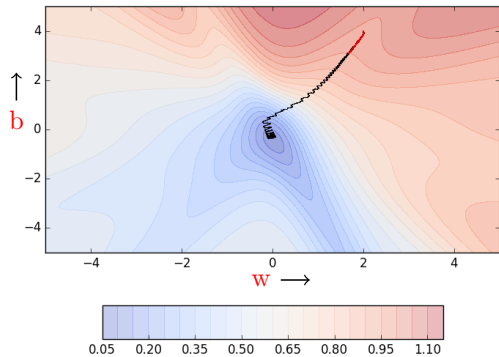


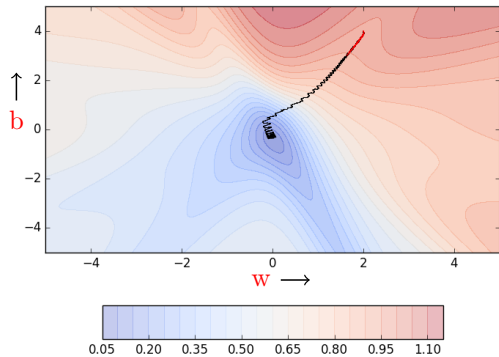


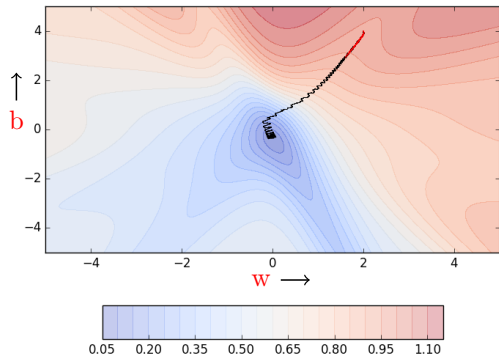


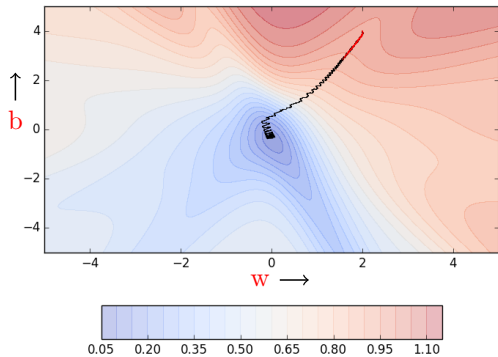


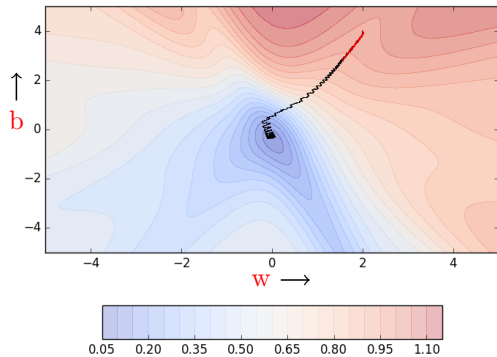


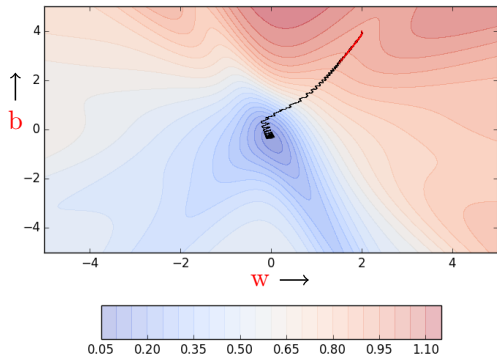


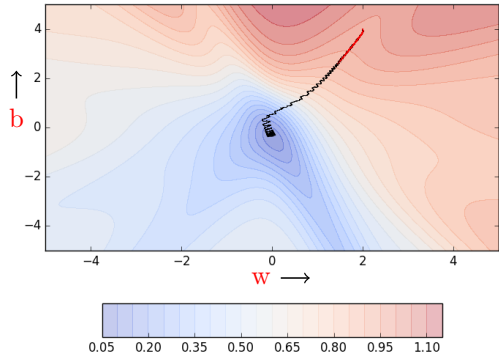


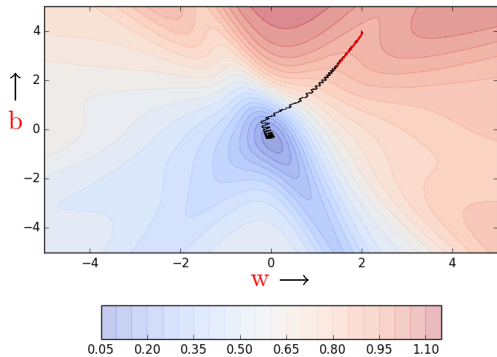


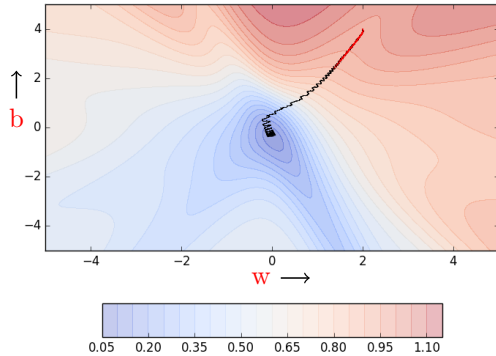


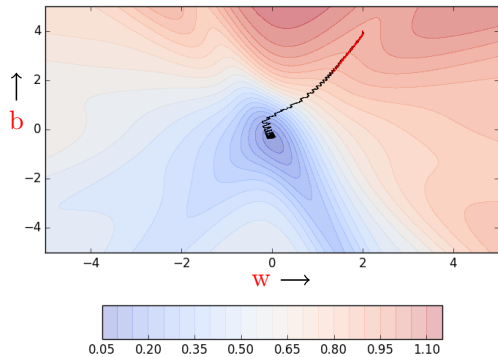


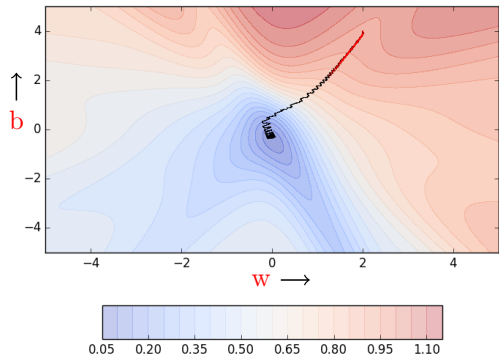


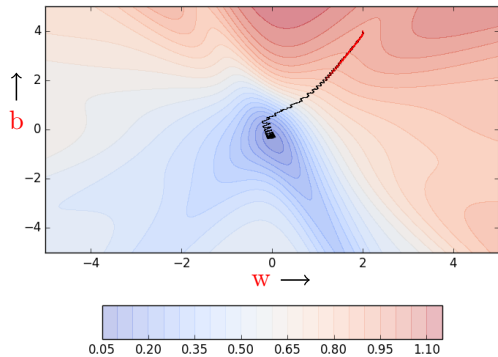


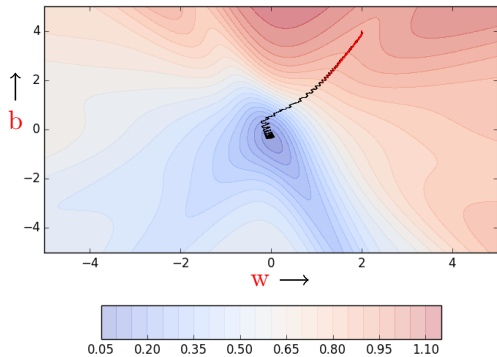


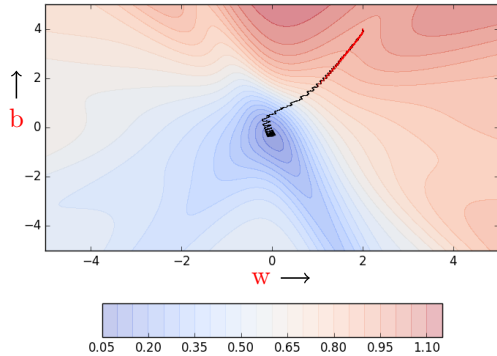


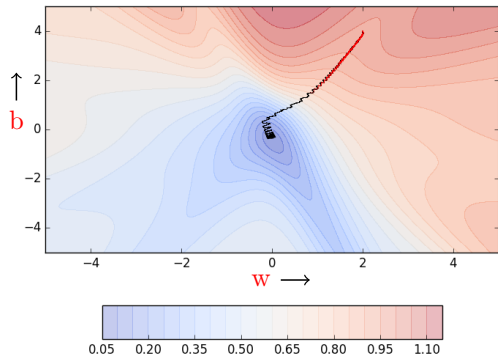


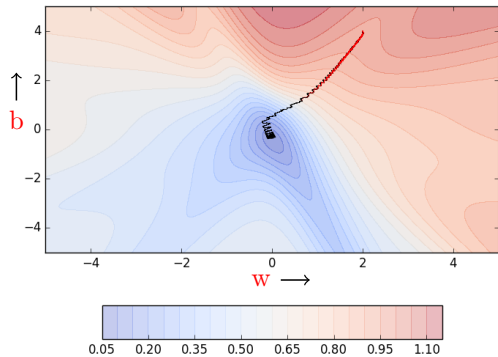


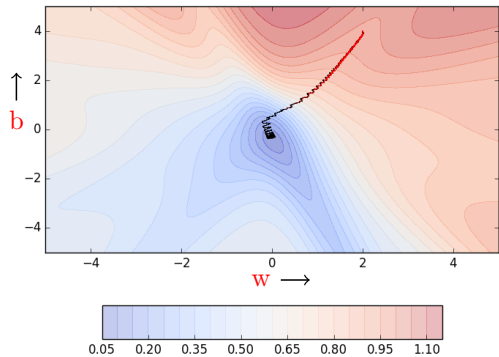


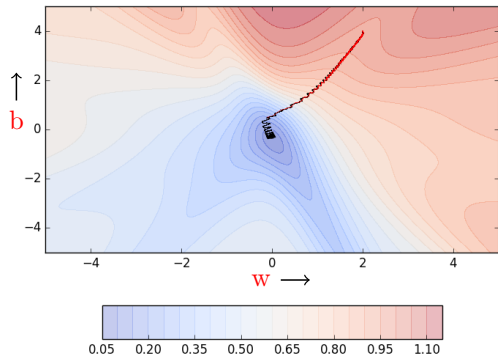


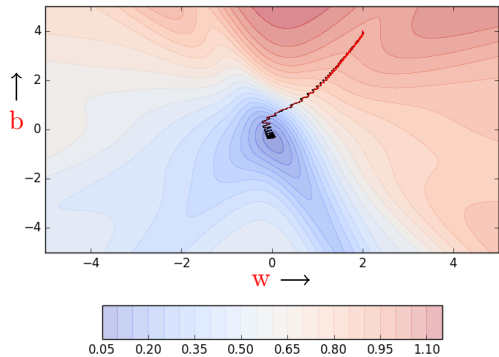


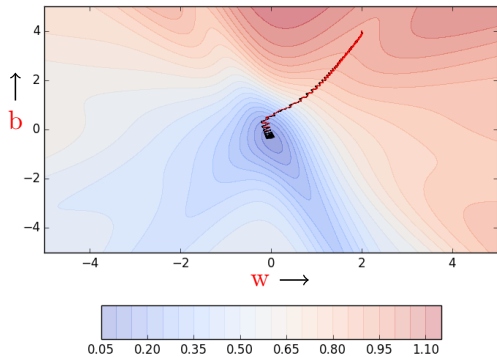


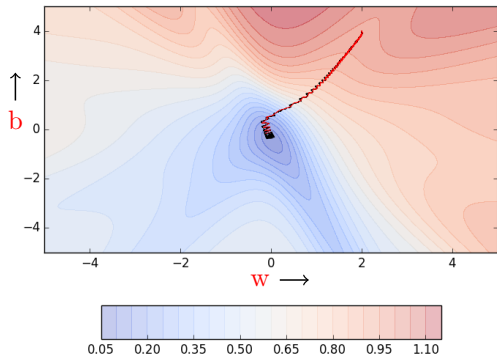


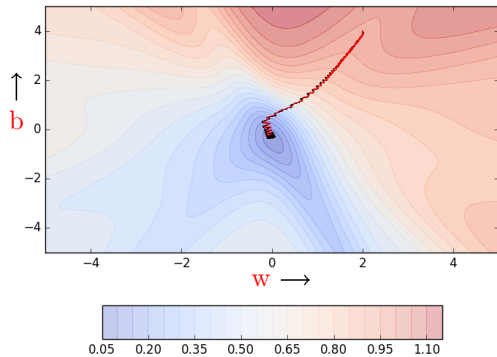


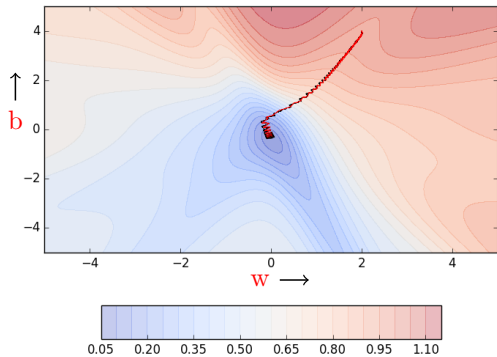


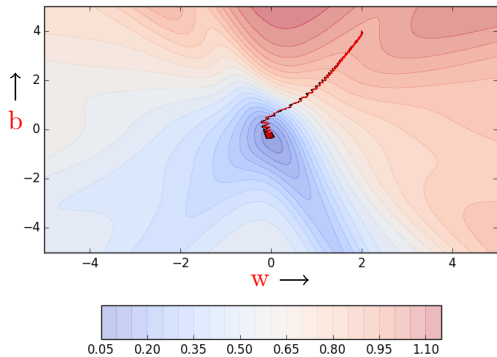




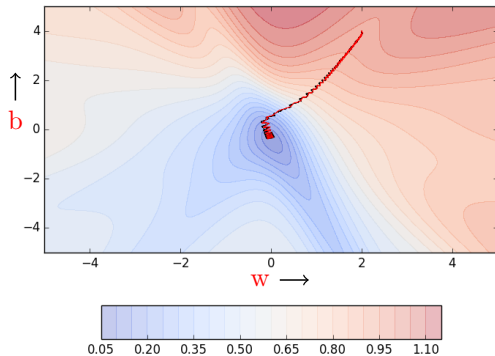




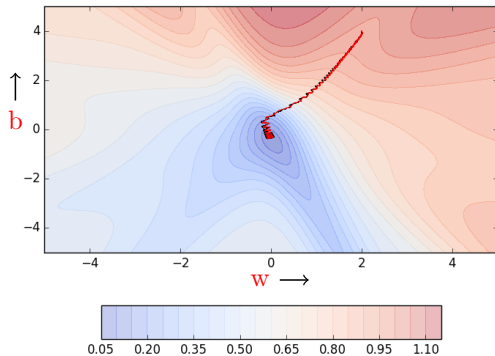




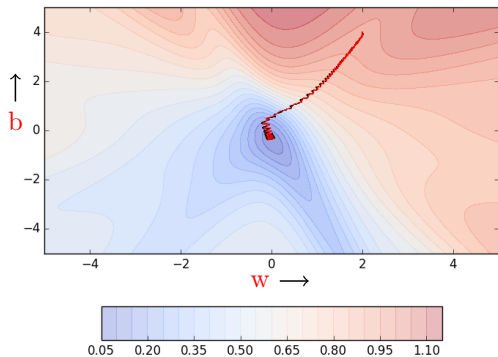
- Even with a batch size of $k=2$ the oscillations have reduced slightly.



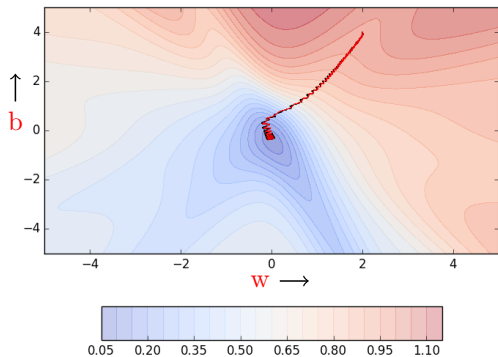
- Even with a batch size of $k=2$ the oscillations have reduced slightly. Why ?



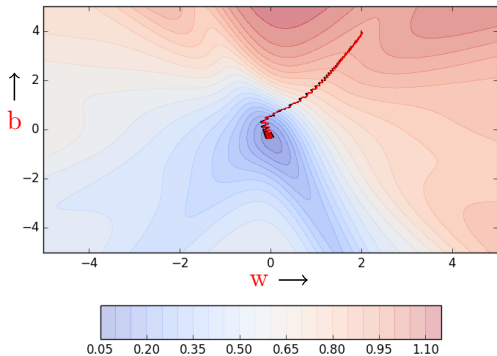
- Even with a batch size of $k=2$ the oscillations have reduced slightly. Why ?
- Because we now have slightly better estimates of the gradient



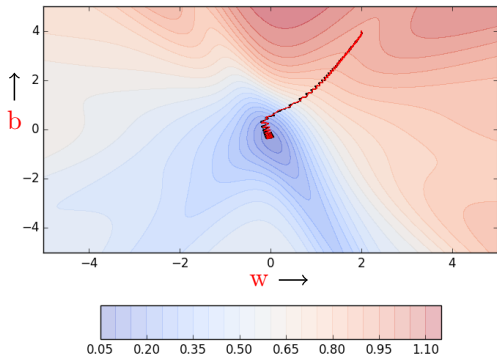
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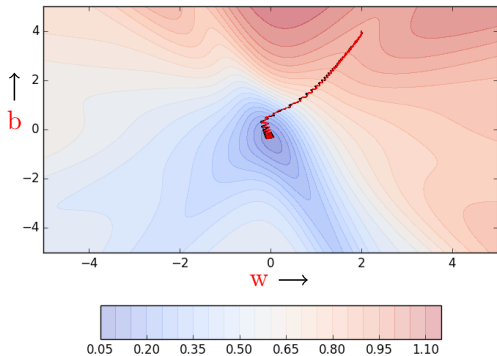
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- The higher the value of k the more accurate are the estimates
- In practice, typical values of k are 16, 32, 64
- Of course, there are still oscillations and they will always be there as long as we are using an approximate gradient as opposed to the true gradient



Some things to remember

- 1 epoch = one pass over the entire data
- 1 step = one update of the parameters
- N = number of data points
- B = Mini batch size

Algorithm	# of steps in 1 epoch
Vanilla (Batch) Gradient Descent	
Stochastic Gradient Descent	
Mini-Batch Gradient Descent	

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Algorithm	# of steps in 1 epoch
Vanilla (Batch) Gradient Descent	1
Stochastic Gradient Descent	N
Mini-Batch Gradient Descent	$\frac{N}{B}$

Similarly, we can have stochastic versions of Momentum based gradient descent and Nesterov accelerated based gradient descent

```

def do_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = gamma * prev_v_w + eta* dw
        v_b = gamma * prev_v_b + eta* db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b

```

```

def do_stochastic_momentum_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw = grad_w(w, b, x, y)
            db = grad_b(w, b, x, y)

        v_w = gamma * prev_v_w + eta* dw
        v_b = gamma * prev_v_b + eta* db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b

```

```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b , 1.0
    prev_v_w, prev_v_b, gamma = 0, 0, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        #do partial updates
        v_w = gamma * prev_v_w
        v_b = gamma * prev_v_b
        for x,y in zip(X, Y) :
            #calculate gradients after partial update
            dw += grad_w(w - v_w, b - v_b, x, y)
            db += grad_b(w - v_w, b - v_b, x, y)

        #now do the full update
        v_w = gamma * prev_v_w + eta * dw
        v_b = gamma * prev_v_b + eta * db
        w = w - v_w
        b = b - v_b
        prev_v_w = v_w
        prev_v_b = v_b
```

```
def do_nesterov_accelerated_gradient_descent() :
    w, b, eta = init_w, init_b, 1.0
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            dw = grad_w(w - v_w, b - v_b, x, y)
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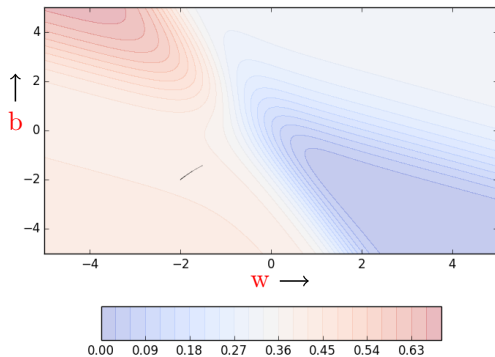
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And, of course, you can also have the mini batch version of Momentum and NAG...

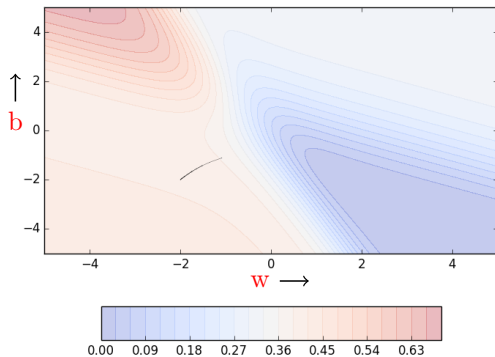
Module 5.7 : Tips for Adjusting learning Rate and Momentum

Before moving on to advanced optimization algorithms let us revisit the problem of learning rate in gradient descent

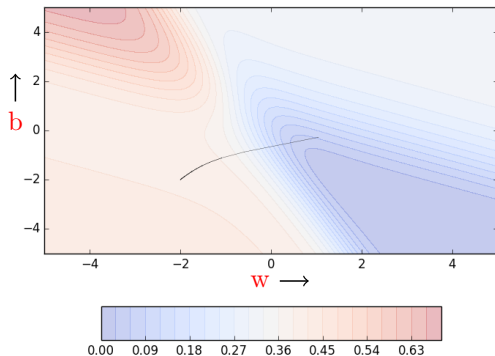
- One could argue that we could have solved the problem of navigating gentle slopes by setting the learning rate high (i.e., blow up the small gradient by multiplying it with a large η)



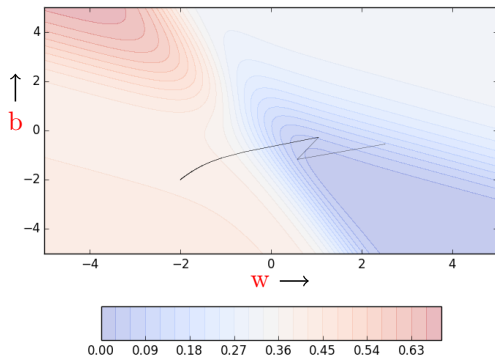
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- Let us see what happens if we set the learning rate to 10



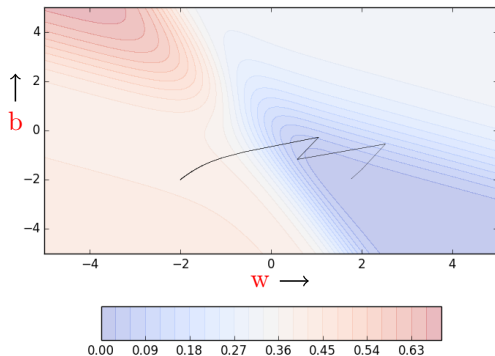
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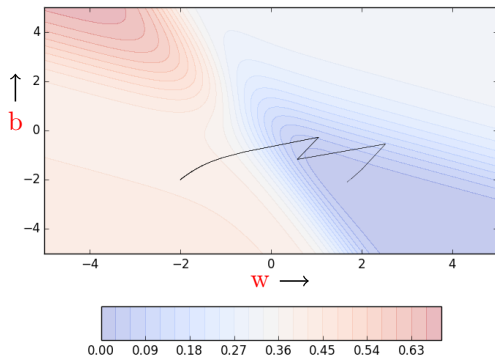
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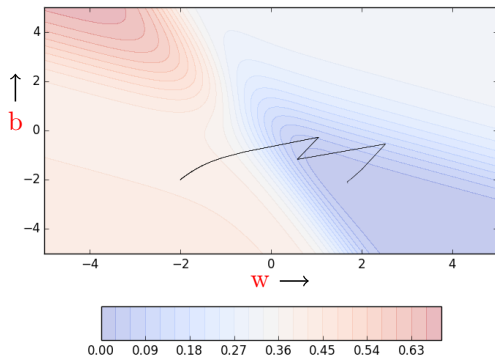
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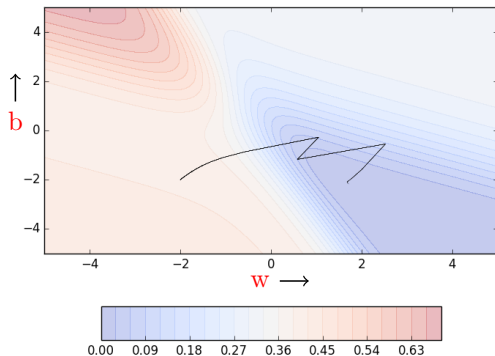
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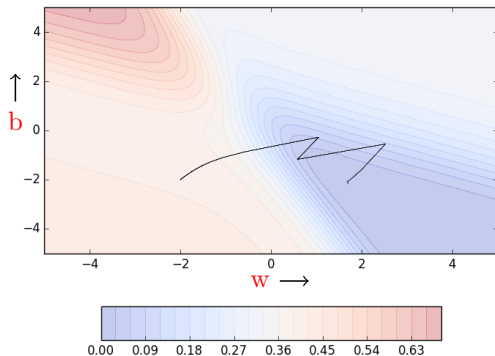
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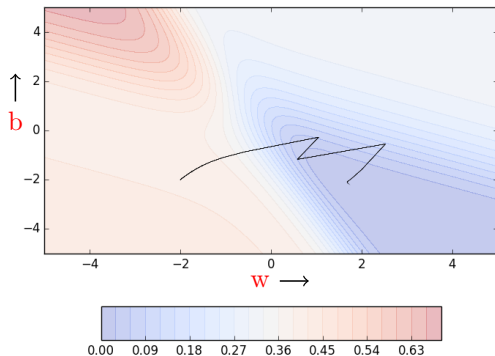
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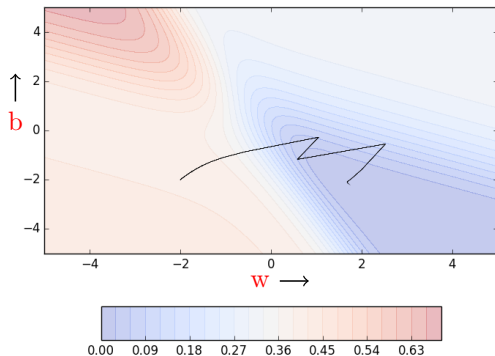
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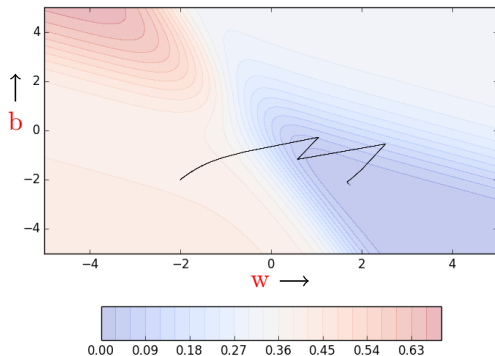
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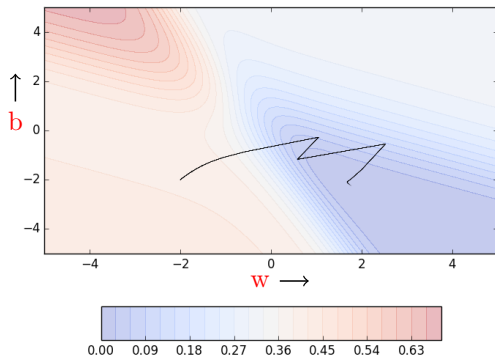
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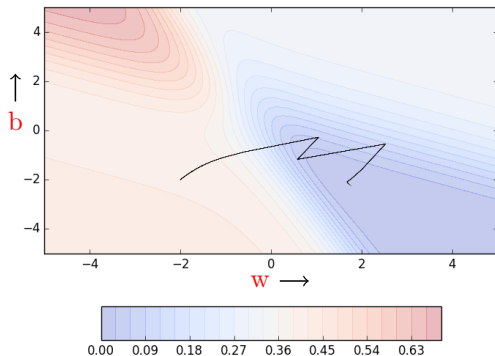
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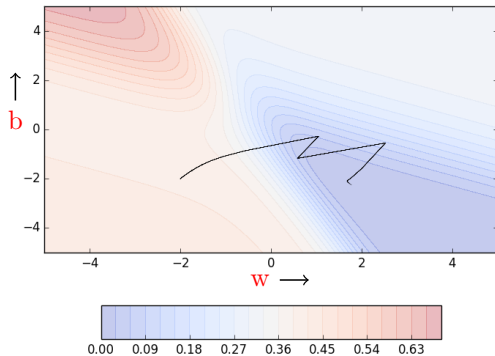
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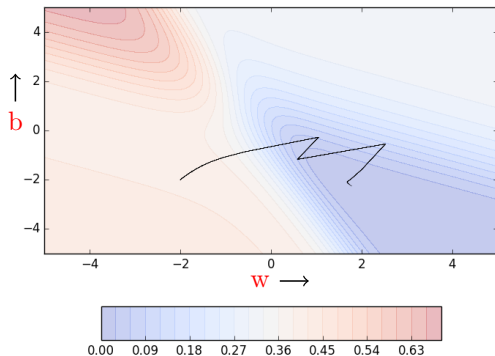
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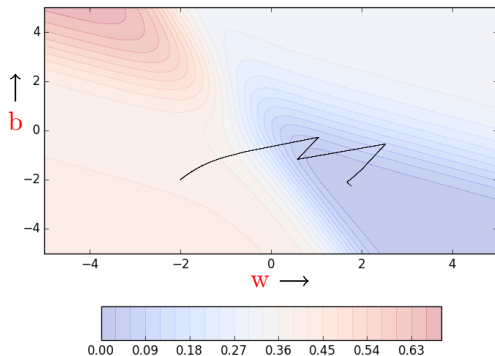
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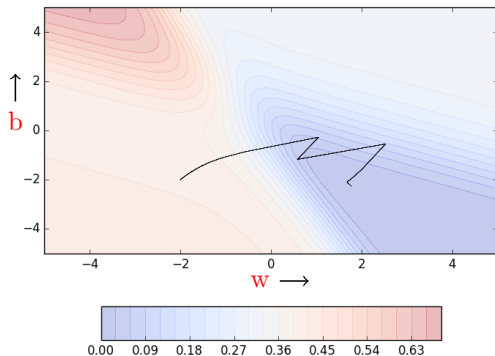
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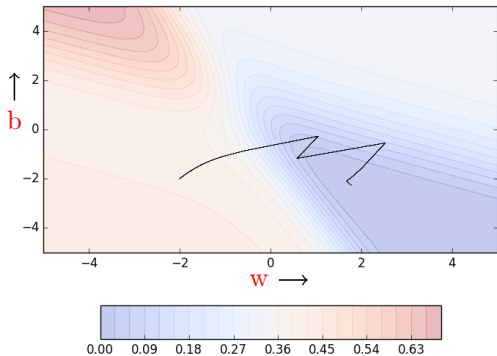
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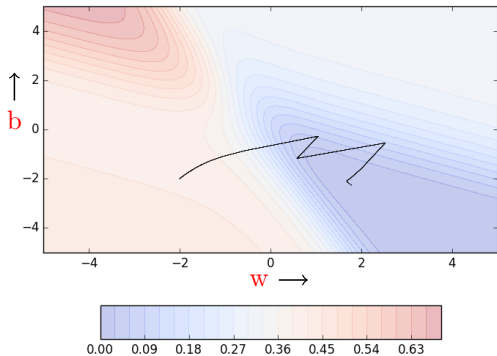
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- It would be good to have a learning rate which could adjust to the gradient ...



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- Let us see what happens if we set the learning rate to 10
- On the regions which have a steep slope, the already large gradient blows up further
- It would be good to have a learning rate which could adjust to the gradient ... we will see a few such algorithms soon



Tips for annealing learning rate

- **Step Decay:**
 - Halve the learning rate after every 5 epochs or

Tips for annealing learning rate

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Tips for annealing learning rate

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- **1/t Decay:** $\eta = \frac{\eta_0}{1+kt}$ where η_0 and k are hyperparameters and t is the step number

Tips for momentum

- The following schedule was suggested by Sutskever *et. al.*, 2013

$$\gamma_t = \min(1 - 2^{-1-\log_2(\lfloor t/250 \rfloor + 1)}, \gamma_{max})$$

where, γ_{max} was chosen from $\{0.999, 0.995, 0.99, 0.9, 0\}$

Module 5.8 : Line Search

Just one last thing before we move on to some other algorithms ...

- In practice, often a line search is done to find a relatively better value of η

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```
def do_line_search_gradient_descent():  
    w, b, etas = init_w, init_b, [0.1, 0.5, 1.0, 5.0, 10.0]  
    for i in range(max_epochs):  
        dw, db = 0, 0  
        for x,y in zip(X, Y):  
            dw += grad_w(w, b, x, y)  
            db += grad_b(w, b, x, y)  
        min_error = 10000 #some large value  
        best_w, best_b = w, b  
        for eta in etas:  
            tmp_w = w - eta * dw  
            tmp_b = b - eta * db  
            if error(tmp_w, tmp_b) < min_error:  
                best_w = tmp_w  
                best_b = tmp_b  
                min_error = error(tmp_w, tmp_b)  
        w, b = best_w, best_b
```

- In practice, often a line search is done to find a relatively better value of η
- Update w using different values of η

```
def do_line_search_gradient_descent():  
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                min_error = error(tmp_w, tmp_b)  
        w, b = best_w, best_b
```

- In practice, often a line search is done to find a relatively better value of η
- Update w using different values of η
- Now retain that updated value of w which gives the lowest loss

```
def do_line_search_gradient_descent():  
    w, b, etas = init_w, init_b, [0.1, 0.5, 1.0, 5.0, 10.0]  
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```

- In practice, often a line search is done to find a relatively better value of η
- Update w using different values of η
- Now retain that updated value of w which gives the lowest loss
- Essentially at each step we are trying to use the best η value from the available choices

```
def do_line_search_gradient_descent():
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        min_error = 10000 #some large value
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            tmp_w = w - eta * dw
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            if error(tmp_w, tmp_b) < min_error:
                best_w = tmp_w
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        w, b = best_w, best_b
```

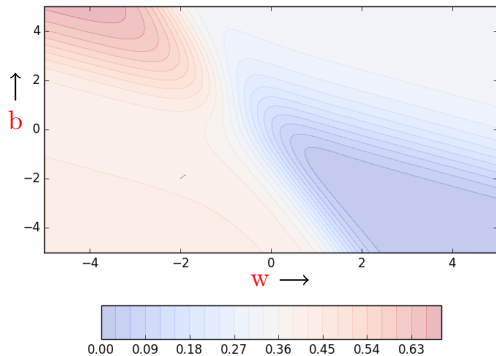

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- What's the flipside?

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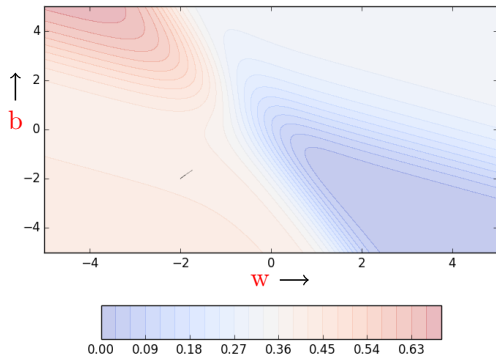
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- Update w using different values of η
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- Essentially at each step we are trying to use the best η value from the available choices
- What's the flipside? We are doing many more computations in each step

```
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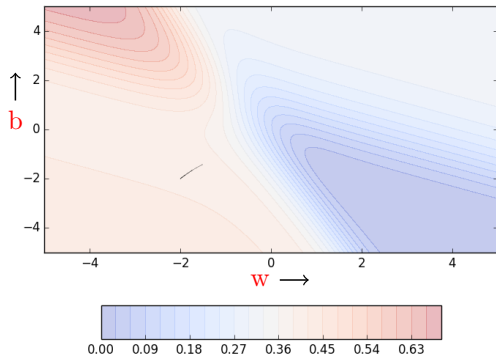
- Let us see line search in action



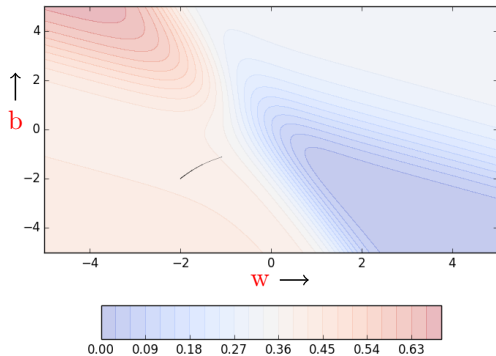
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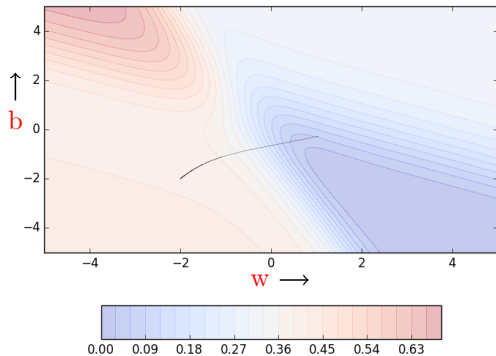
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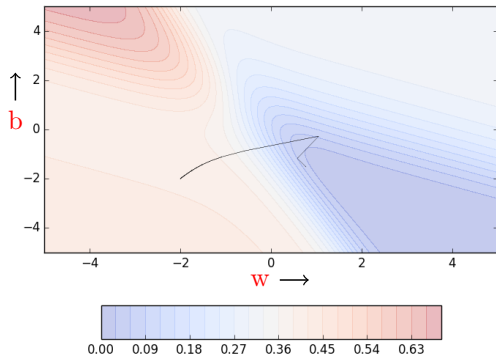
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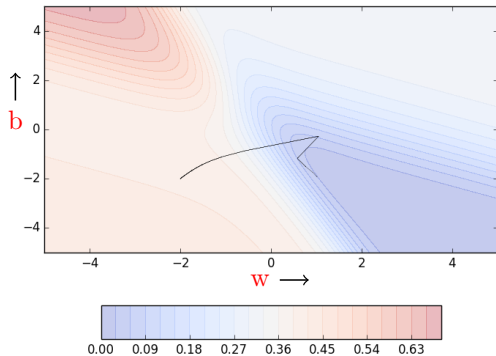
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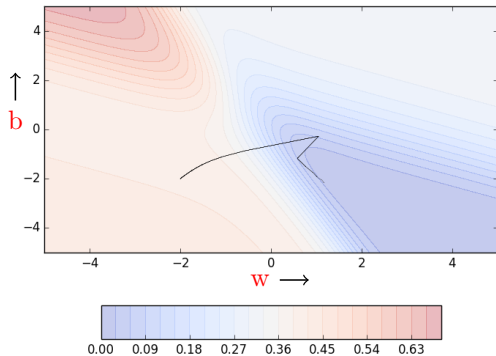
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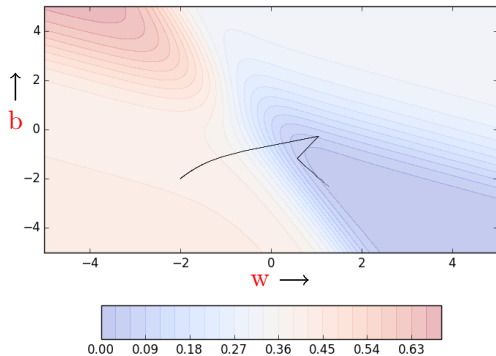
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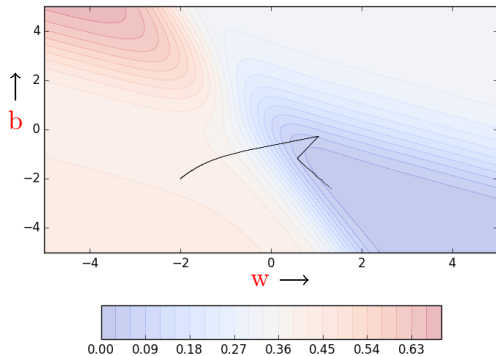
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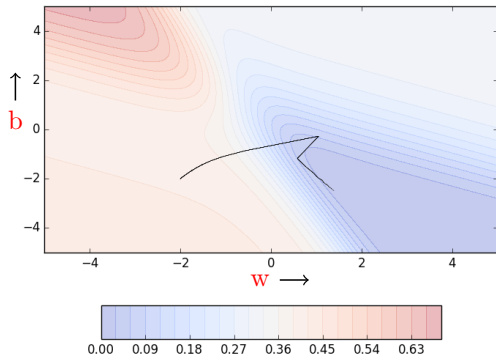
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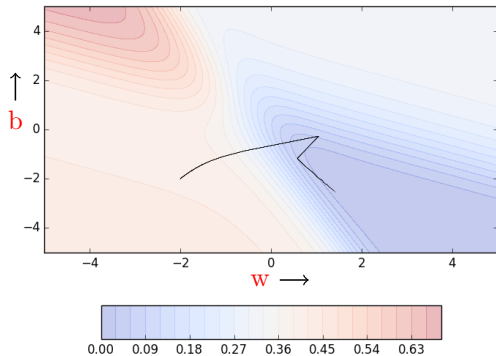
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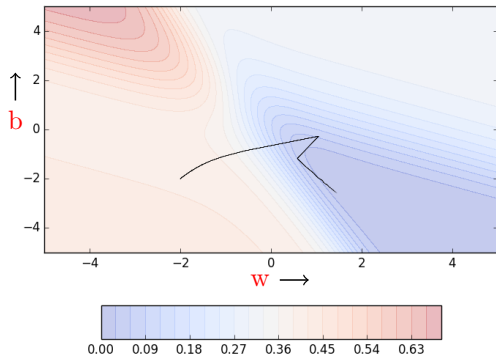
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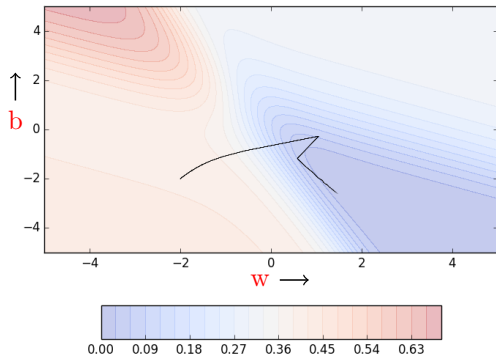
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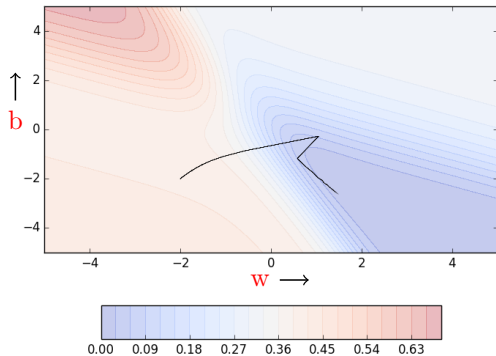
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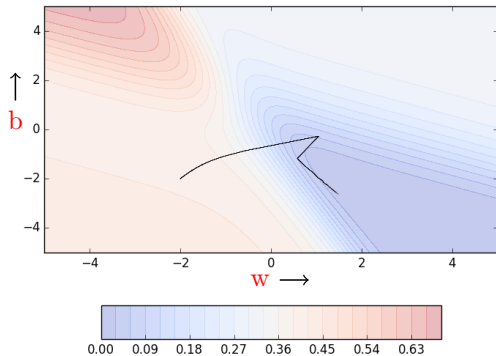
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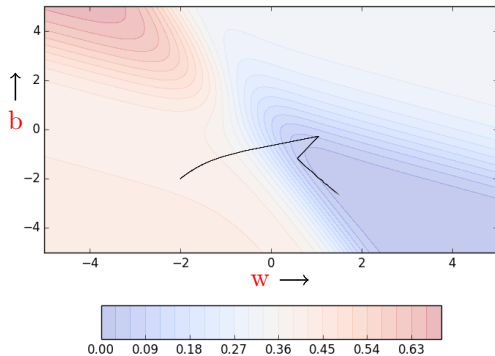
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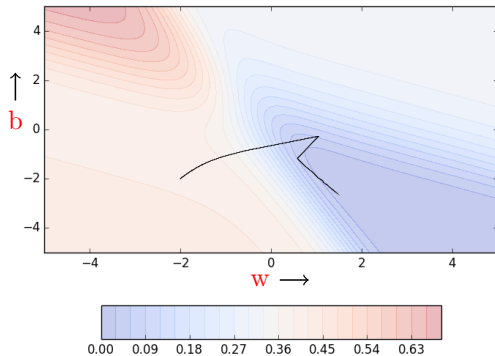
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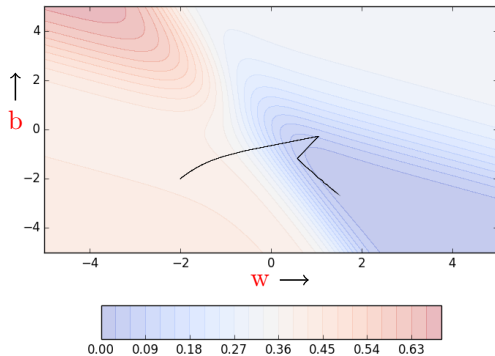
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- Convergence is faster than vanilla gradient descent



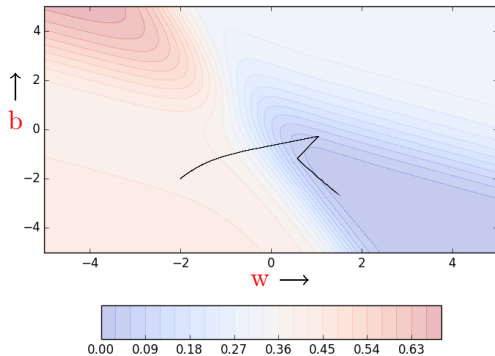
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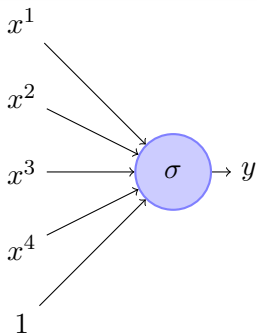
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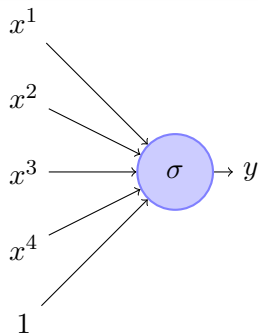
Module 5.9 : Gradient Descent with Adaptive Learning Rate



$$y = f(x) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}}$$

$$\mathbf{x} = \{x^1, x^2, x^3, x^4\}$$

$$\mathbf{w} = \{w^1, w^2, w^3, w^4\}$$

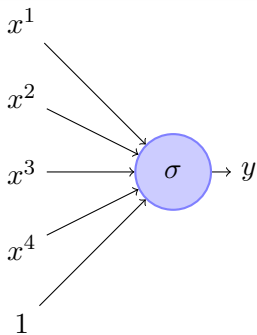


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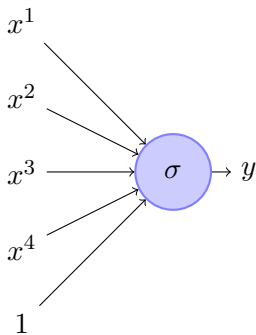


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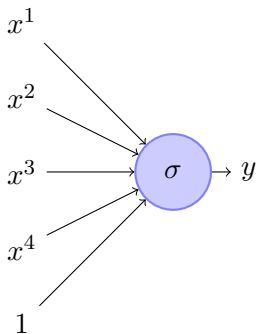


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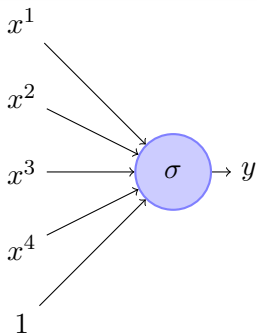


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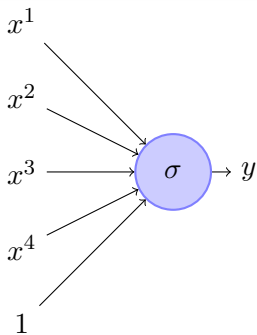


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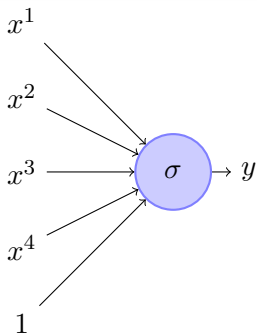


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- Can we have a different learning rate for each parameter which takes care of the frequency of features ?

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- Decay the learning rate for parameters in proportion to their update history

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Update rule for Adagrad

$$v_t = v_{t-1} + (\nabla w_t)^2$$
$$w_{t+1} = w_t - \frac{\eta}{\sqrt{v_t + \epsilon}} * \nabla w_t$$

... and a similar set of equations for b_t

- To see this in action we need to first create some data where one of the features is sparse

```
def do_adagrad():  
    w, b, eta = init_w, init_b, 0.1  
    v_w, v_b, eps = 0, 0, 1e-8  
    for i in range(max_epochs):  
        dw, db = 0, 0  
        for x,y in zip(X, Y):  
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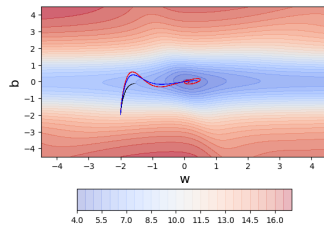
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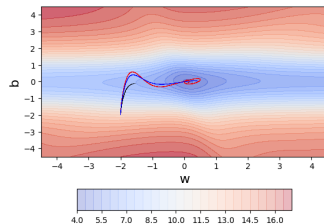
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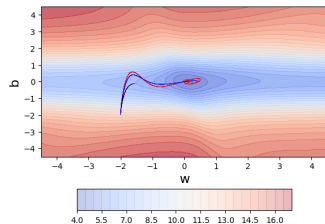
- GD (black), momentum (red) and NAG (blue)



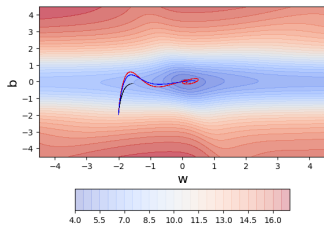
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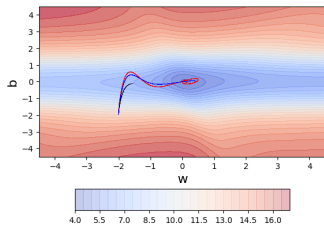
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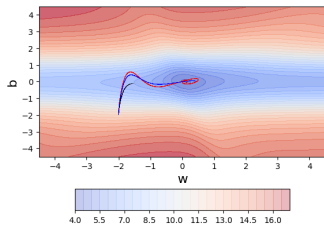
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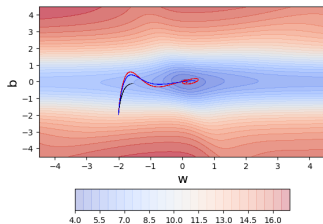
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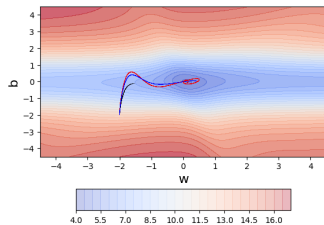
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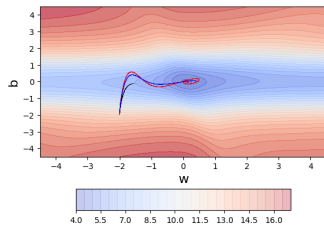
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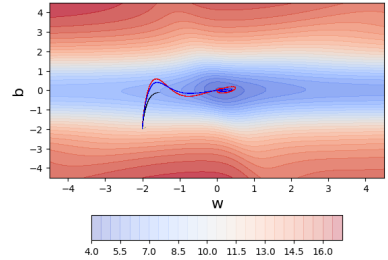
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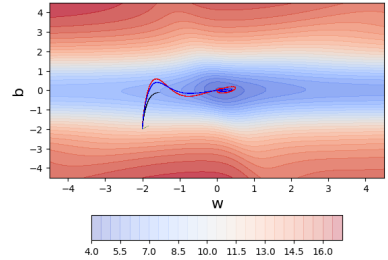


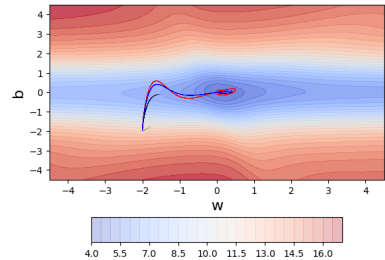
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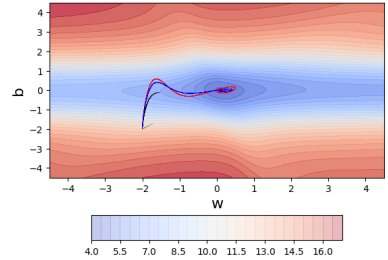


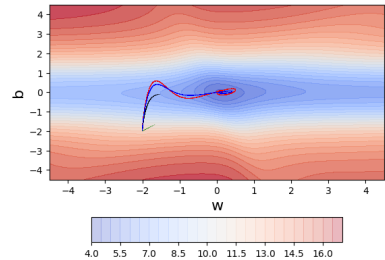
- Let's see what Adagrad does....

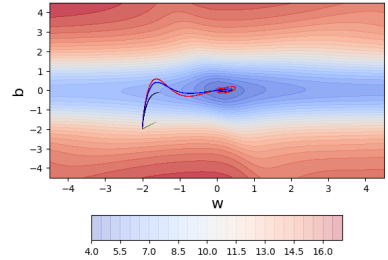


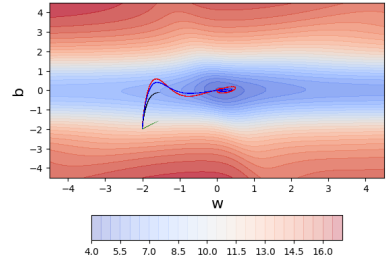


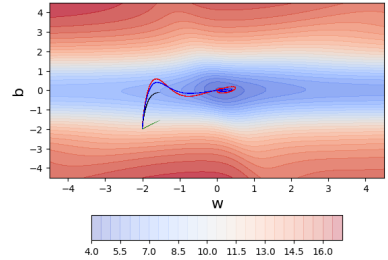


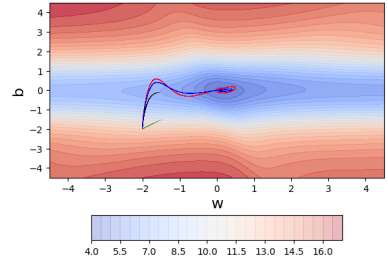


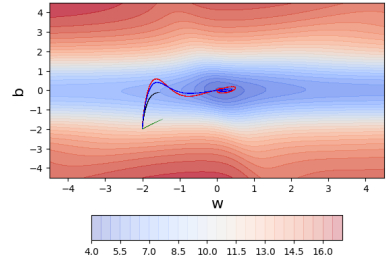


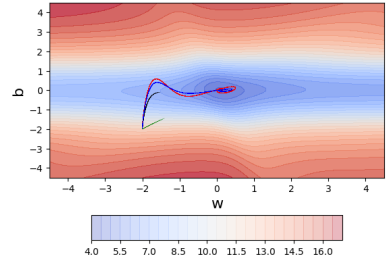


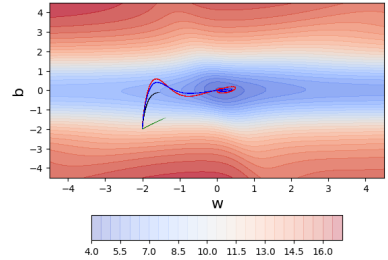


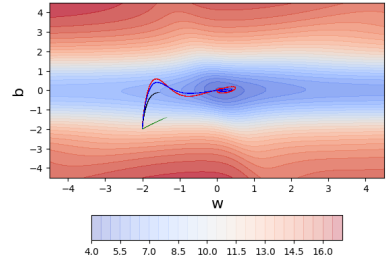


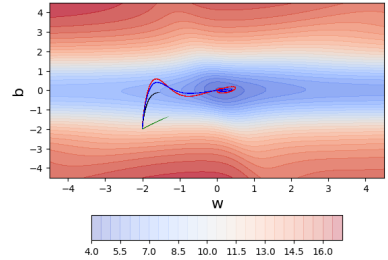


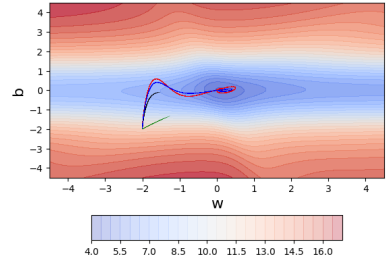


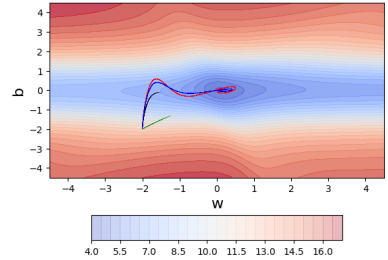


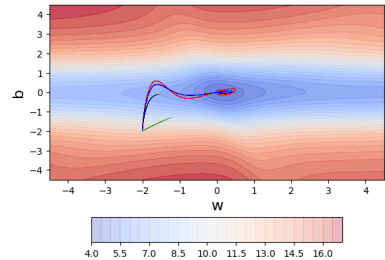


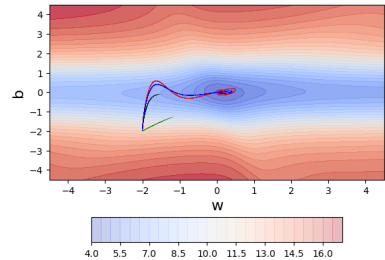


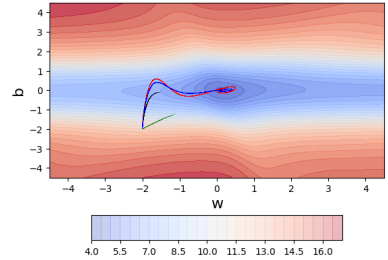


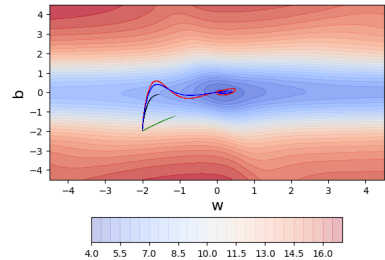


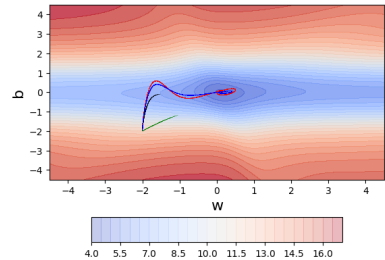


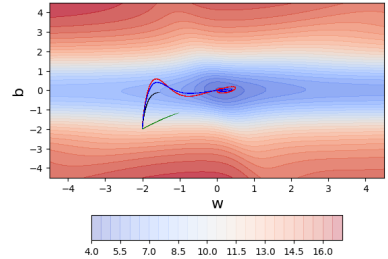


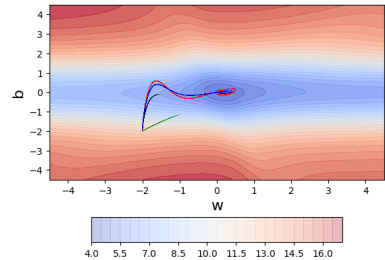


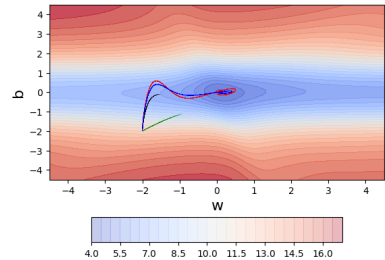


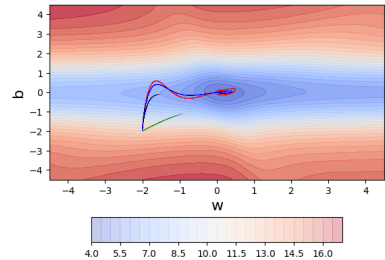


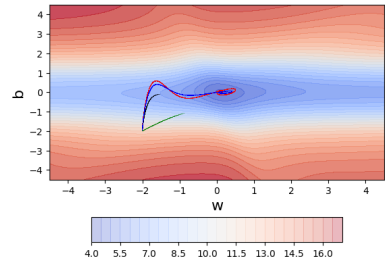


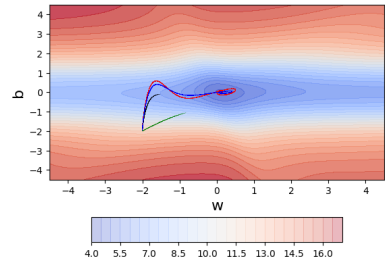


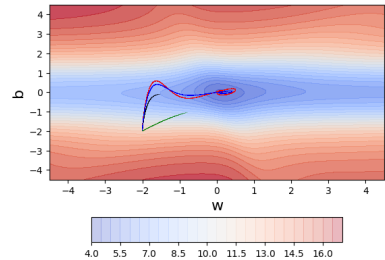


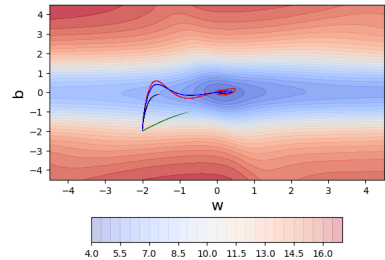


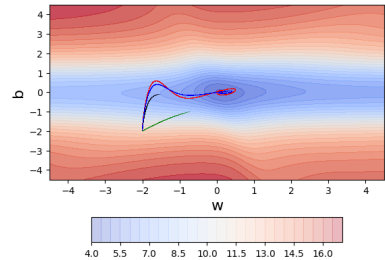


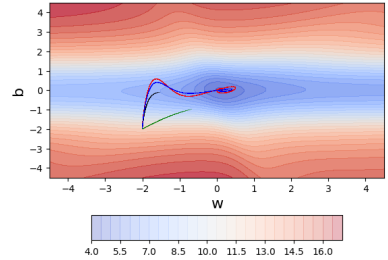


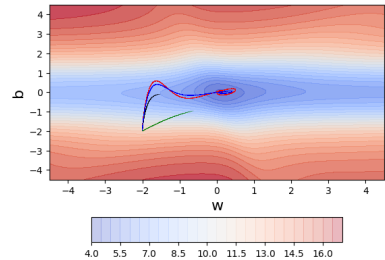


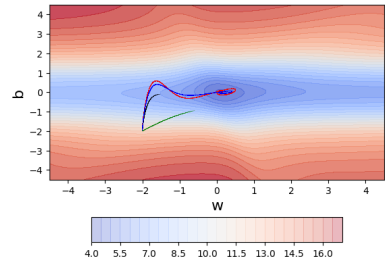


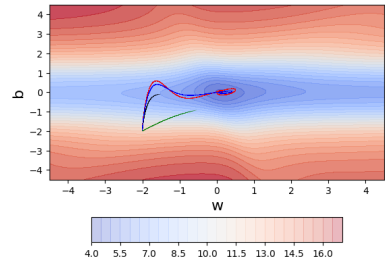


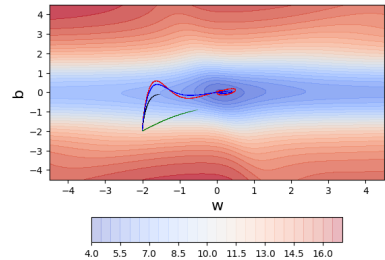


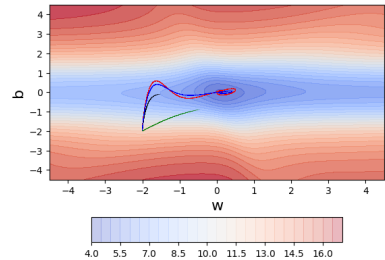


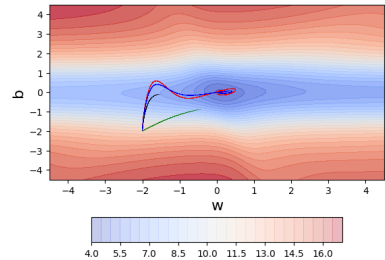


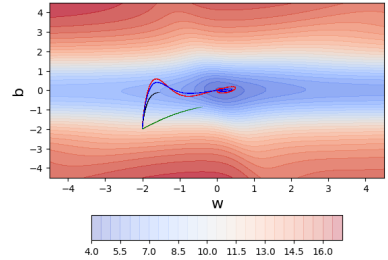


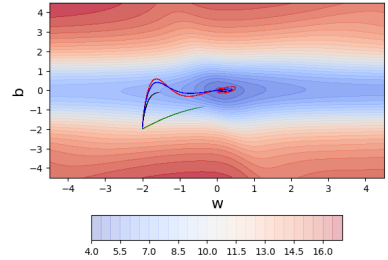


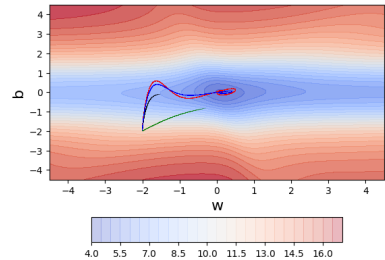


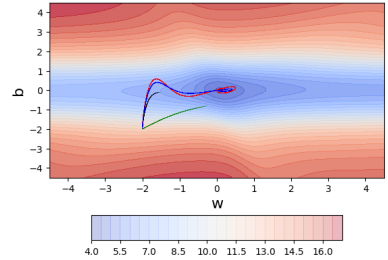


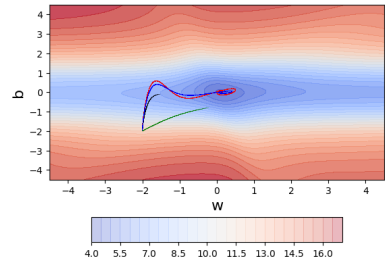


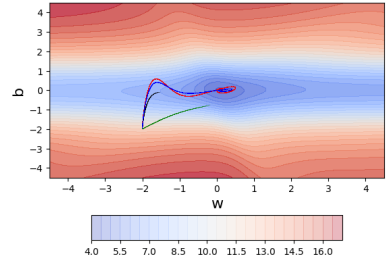


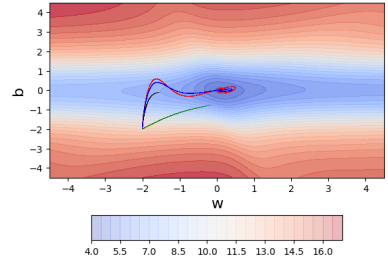


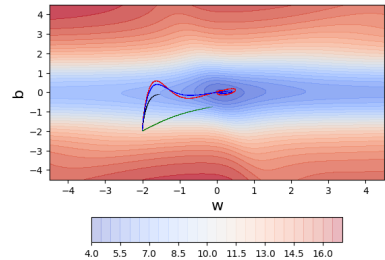


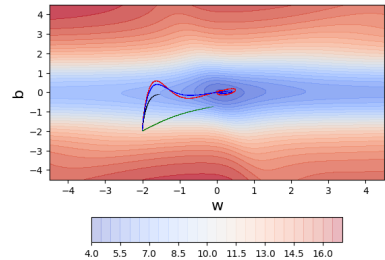


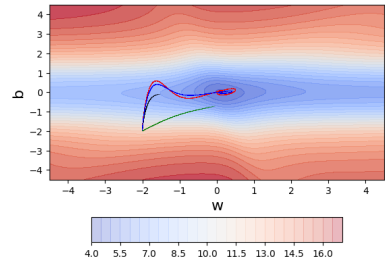


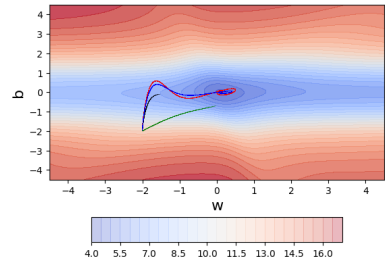


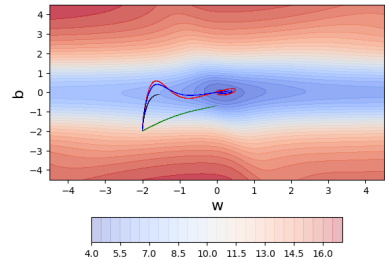


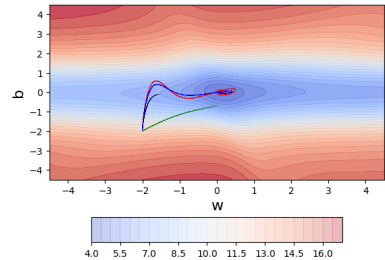


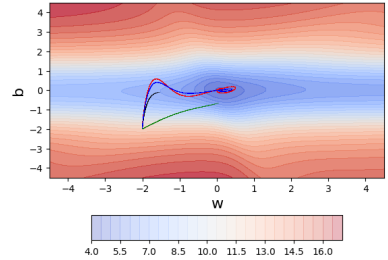


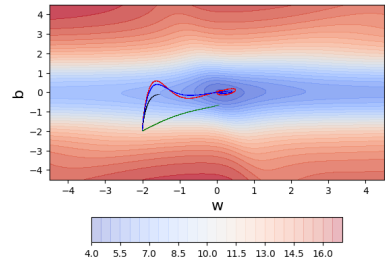


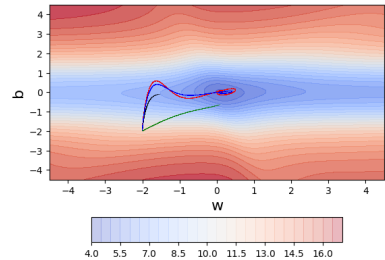


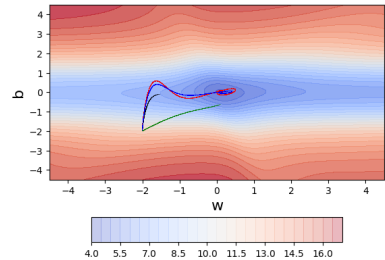


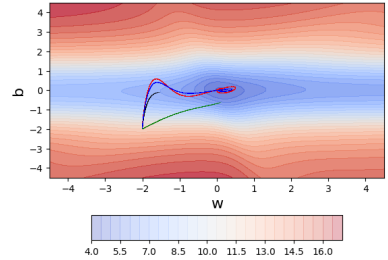


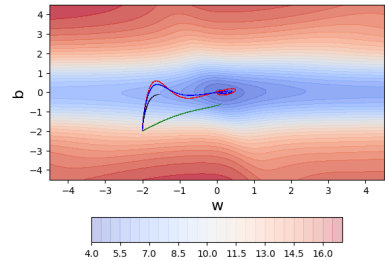


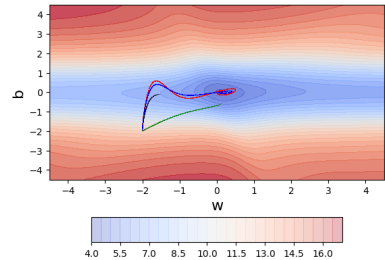


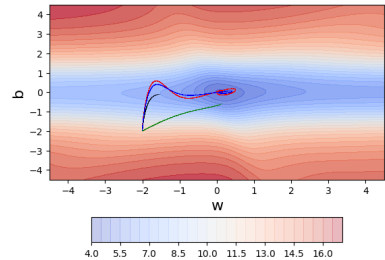


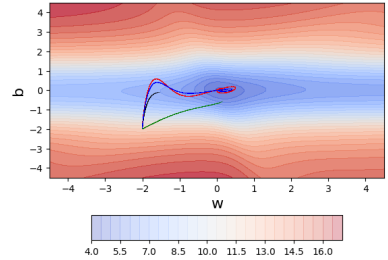


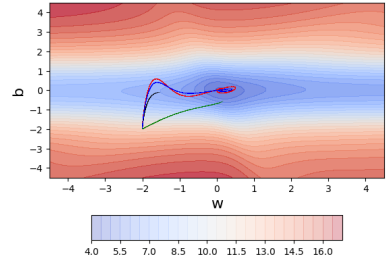


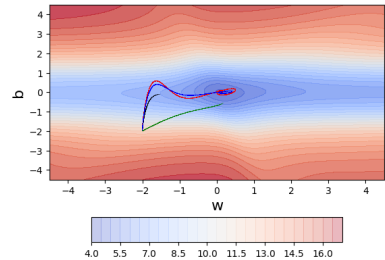


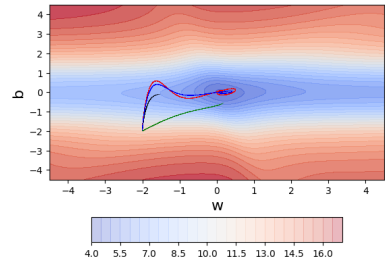


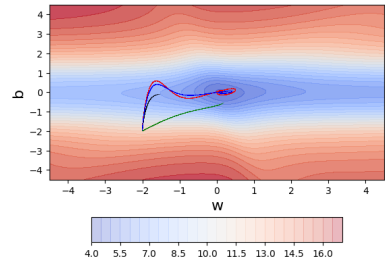


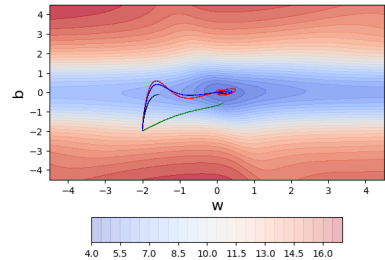


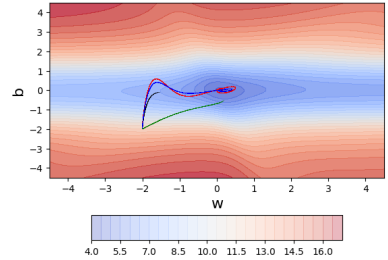


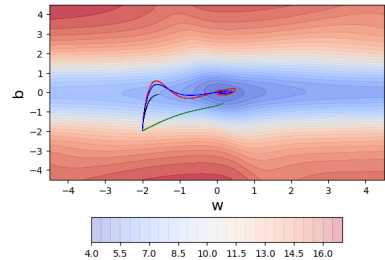


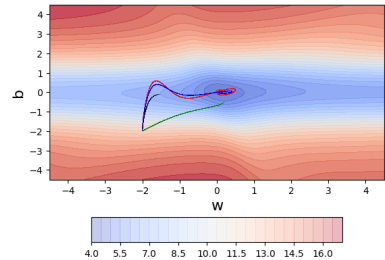


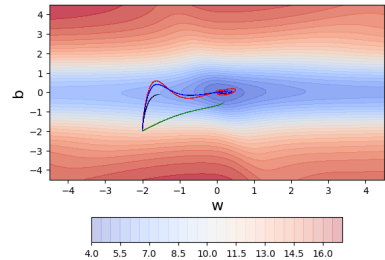


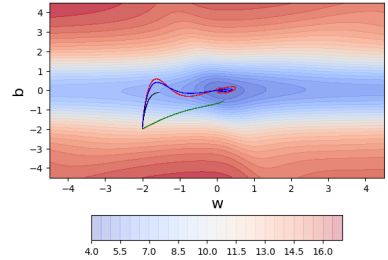


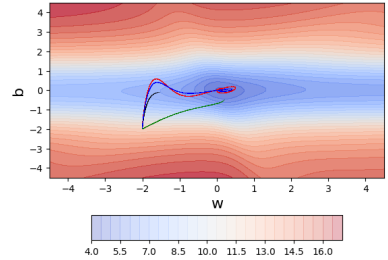


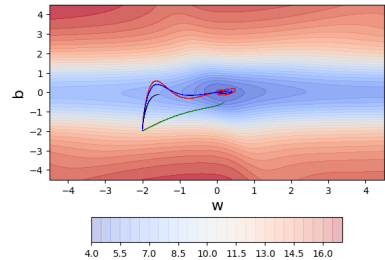


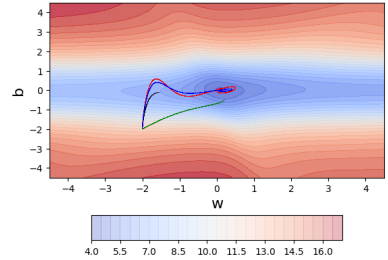


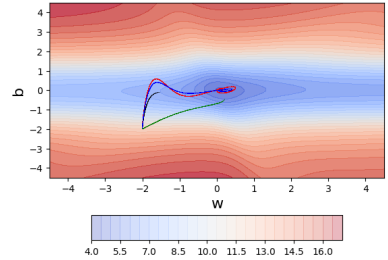


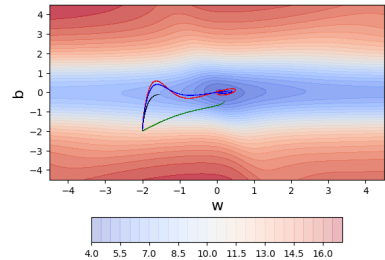


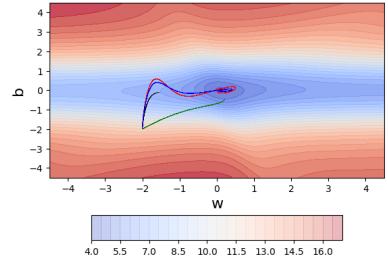


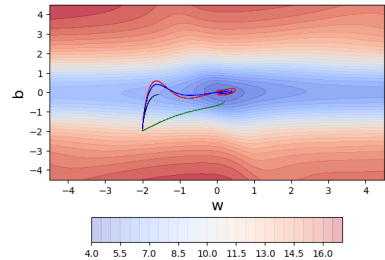


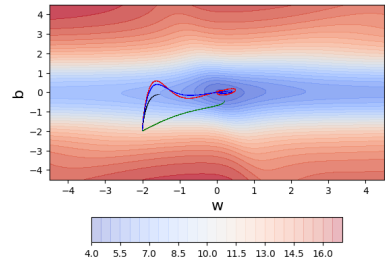


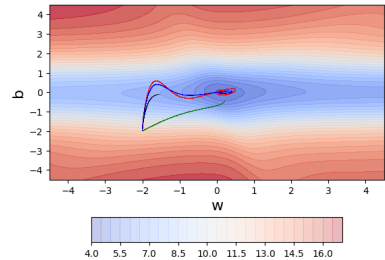


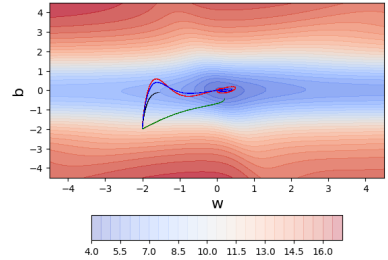


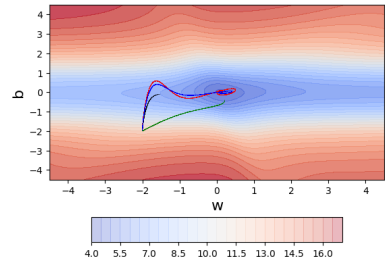


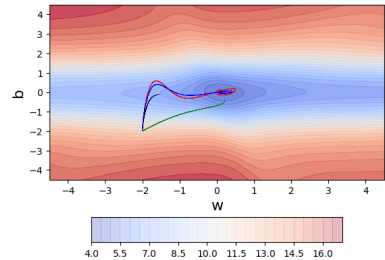


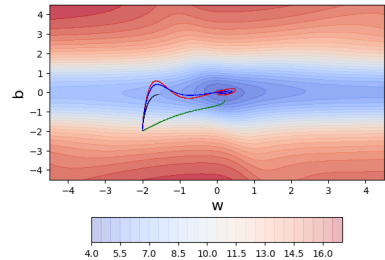


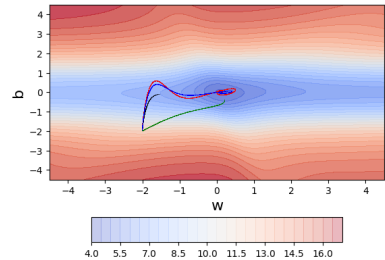


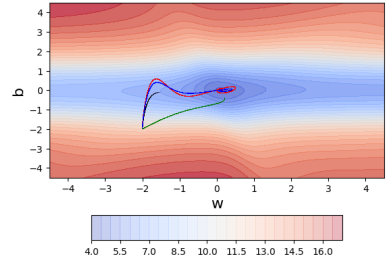


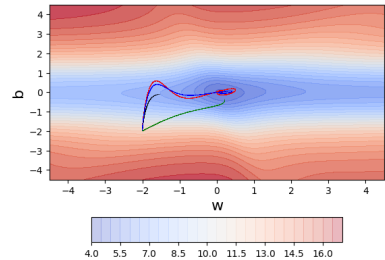


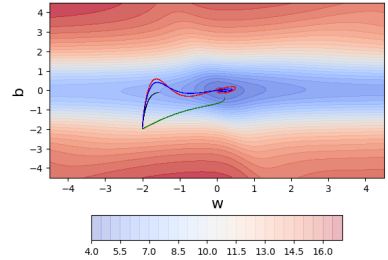


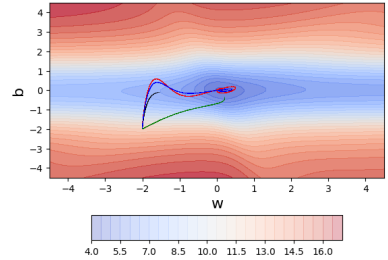


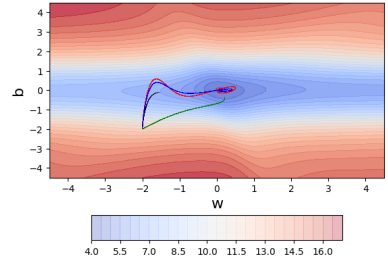


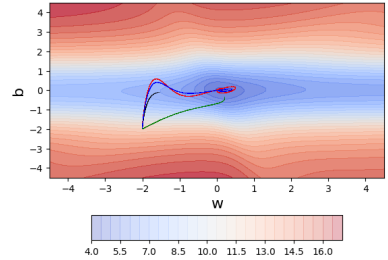


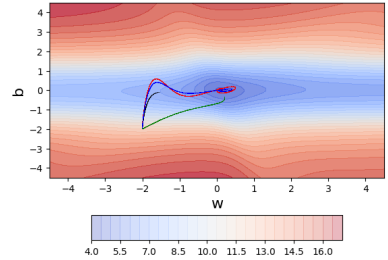




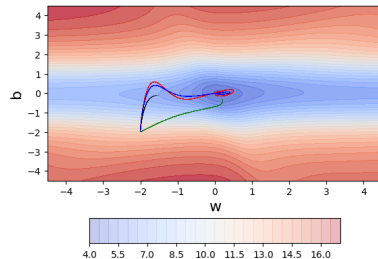




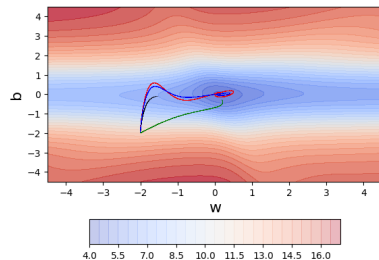




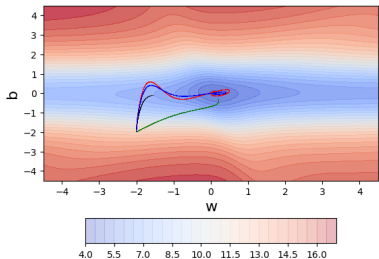
- By using a parameter specific learning rate it ensures that despite sparsity w gets a higher learning rate and hence larger updates



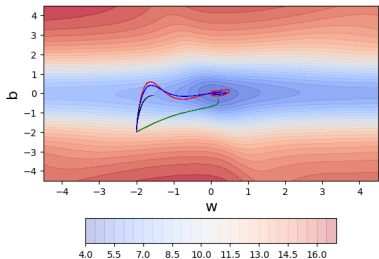
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- Further, it also ensures that if b undergoes a lot of updates its effective learning rate decreases because of the growing denominator



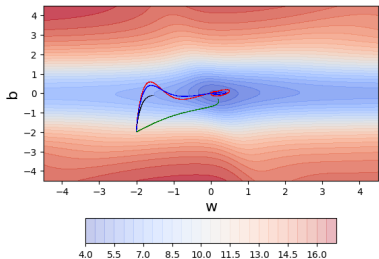
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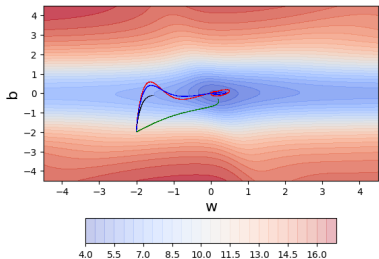
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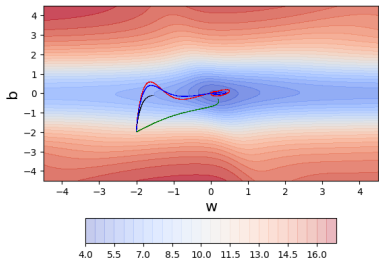
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- What's the flipside?



- By using a parameter specific learning rate it ensures that despite sparsity w gets a higher learning rate and hence larger updates
- Further, it also ensures that if b undergoes a lot of updates its effective learning rate decreases because of the growing denominator
- In practice, this does not work so well if we remove the square root from the denominator (something to ponder about)
- What's the flipside? over time the effective learning rate for b will decay to an extent that there will be no further updates to b



- By using a parameter specific learning rate it ensures that despite sparsity w gets a higher learning rate and hence larger updates
- Further, it also ensures that if b undergoes a lot of updates its effective learning rate decreases because of the growing denominator
- In practice, this does not work so well if we remove the square root from the denominator (something to ponder about)
- What's the flipside? over time the effective learning rate for b will decay to an extent that there will be no further updates to b
- Can we avoid this?



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Update rule for RMSProp

$$v_t = \beta * v_{t-1} + (1 - \beta)(\nabla w_t)^2$$
$$w_{t+1} = w_t - \frac{\eta}{\sqrt{v_t + \epsilon}} * \nabla w_t$$

... and a similar set of equations for b_t

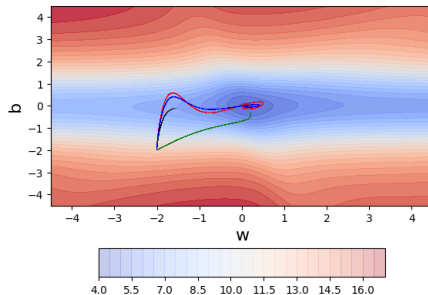
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
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        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
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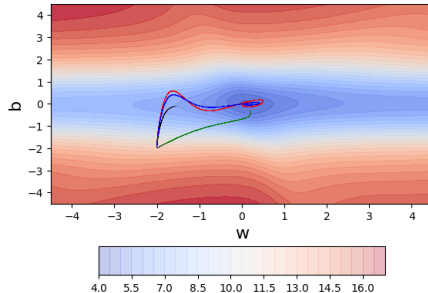
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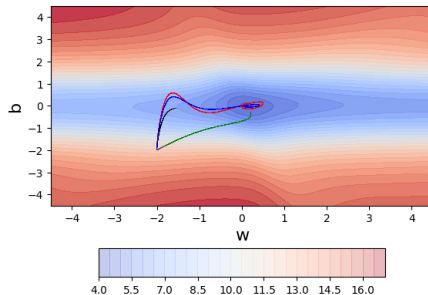
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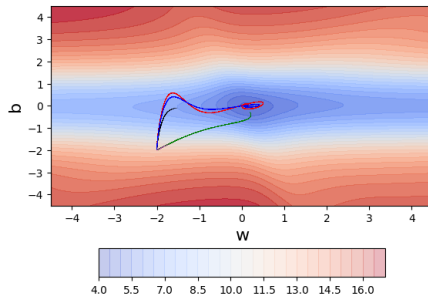
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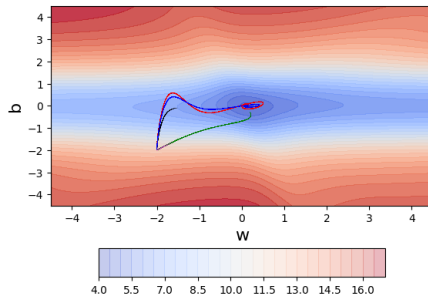
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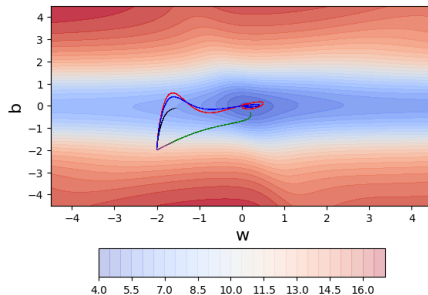
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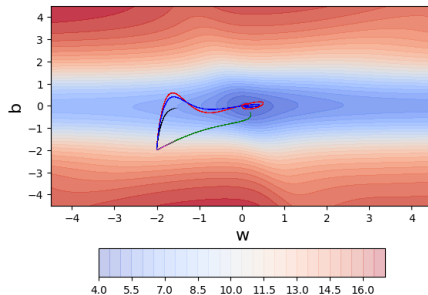
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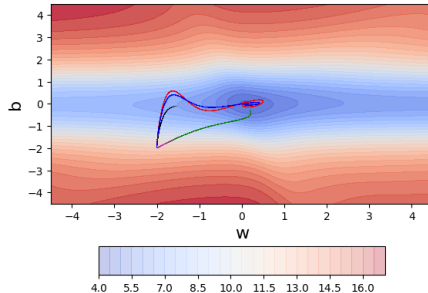

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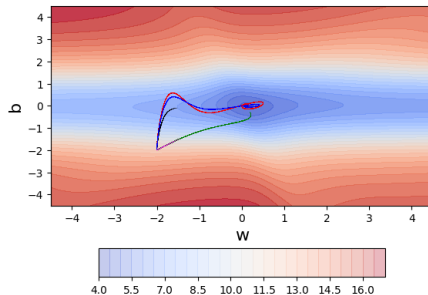
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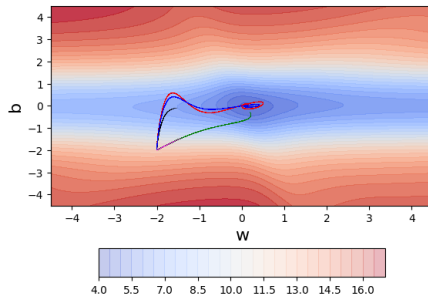
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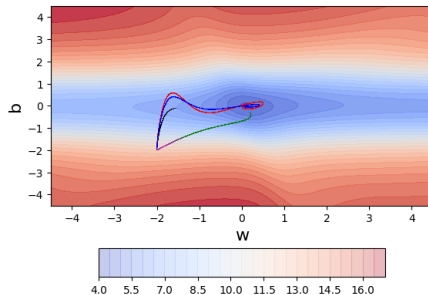
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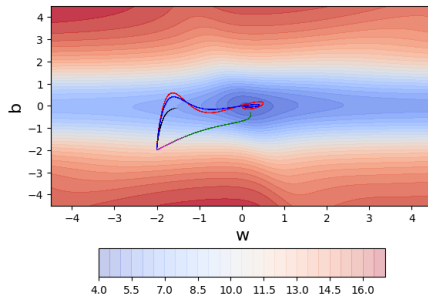
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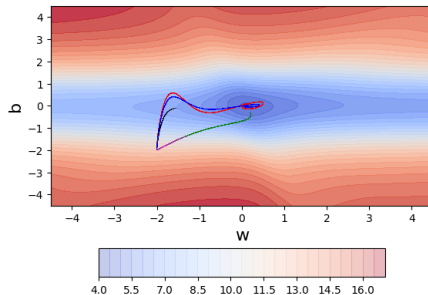
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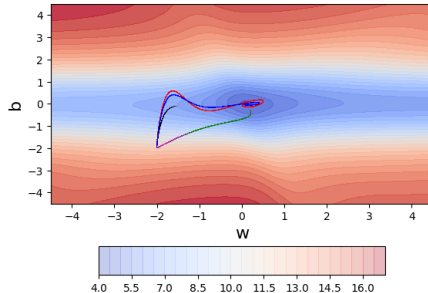
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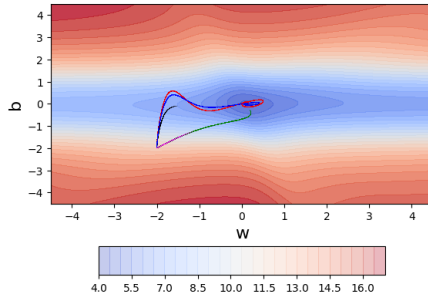
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def do_rmsprop() :
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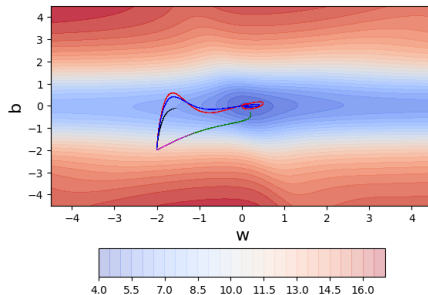

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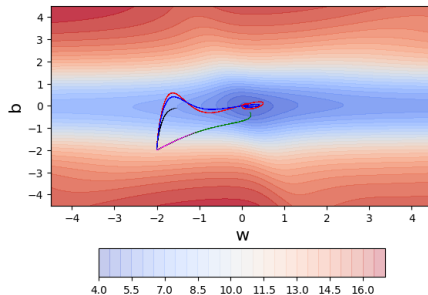
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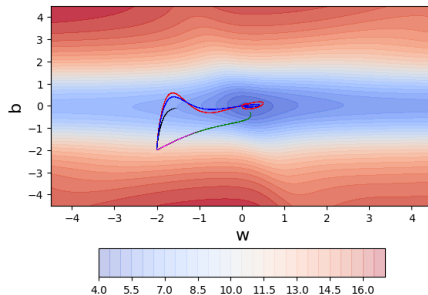
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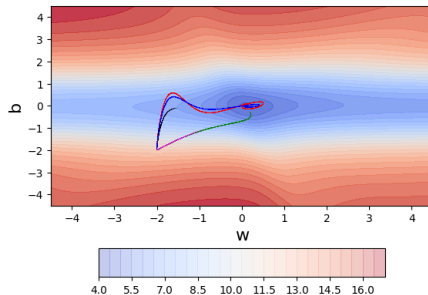
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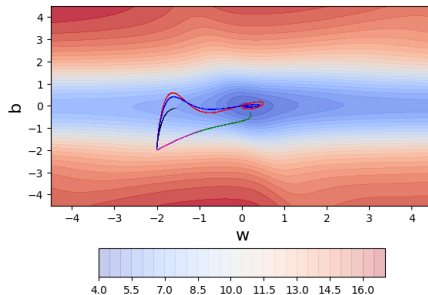
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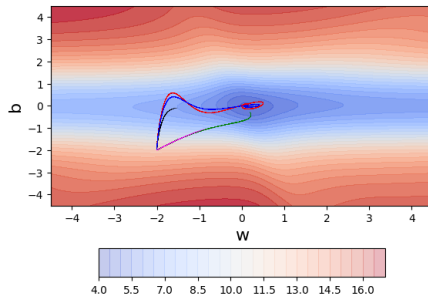
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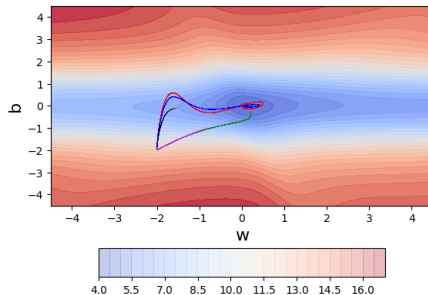
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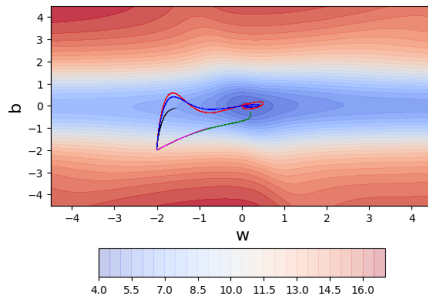
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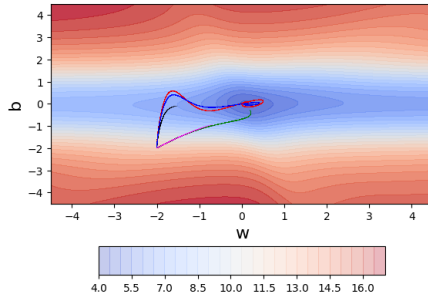

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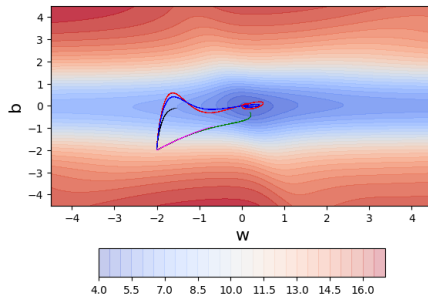
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            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



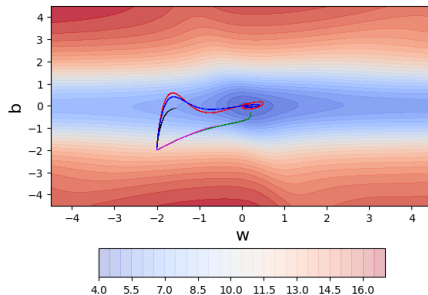
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



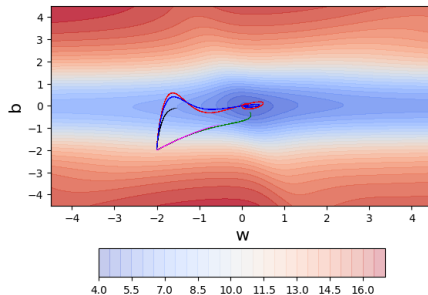
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

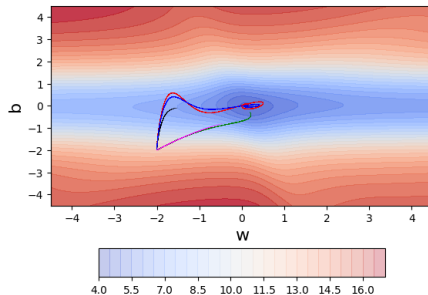
```



```
def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db
```



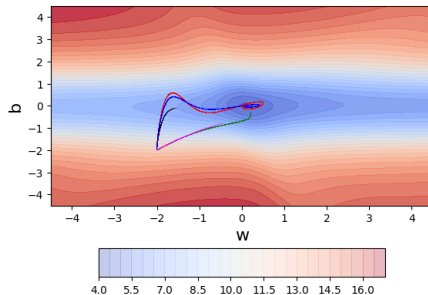
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



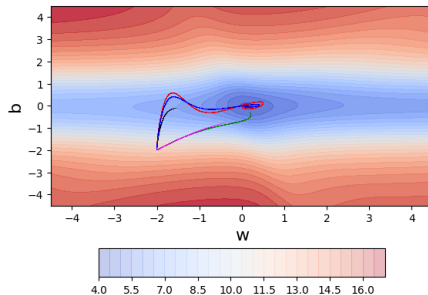
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



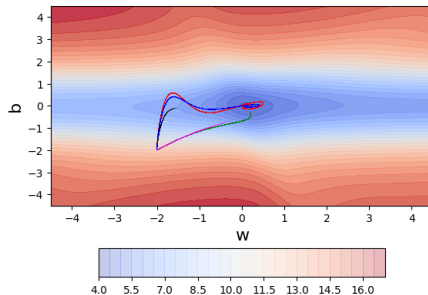
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



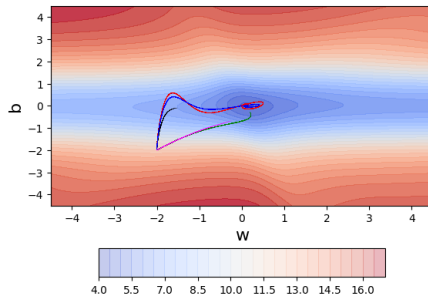

```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



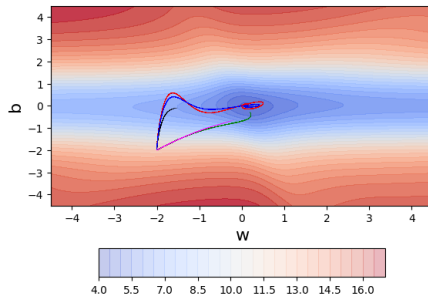
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



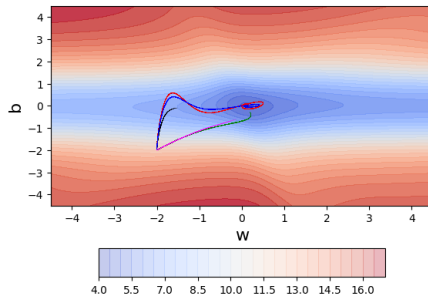
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



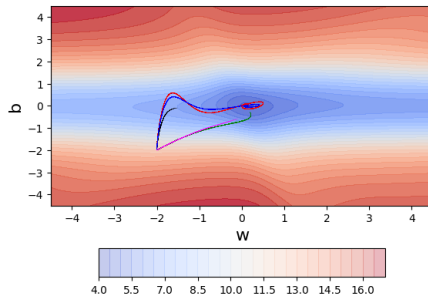
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



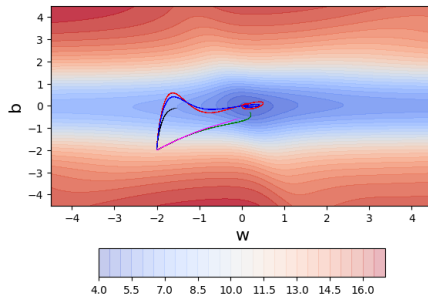
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



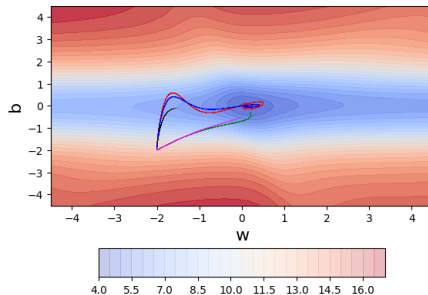
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



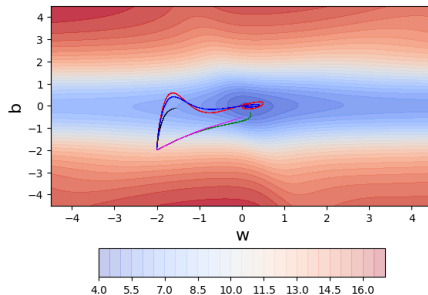
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



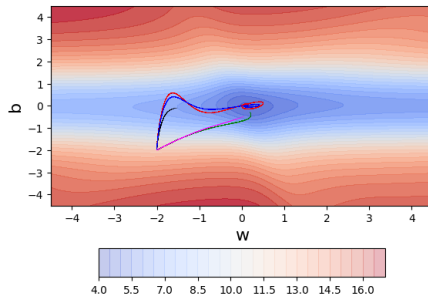
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

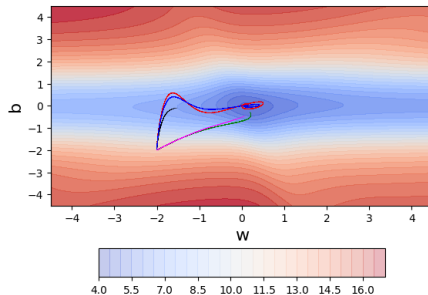
```




```
def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db
```



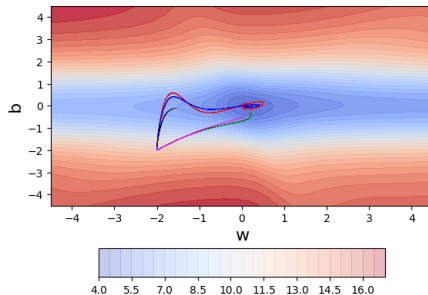
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



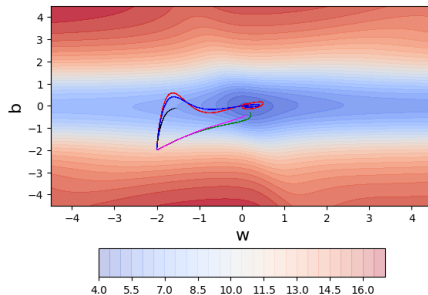
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



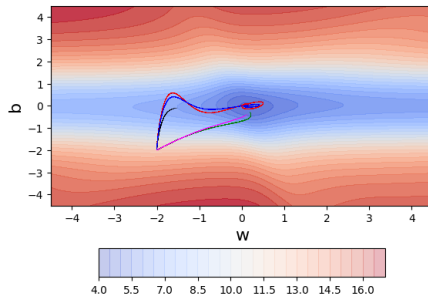
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



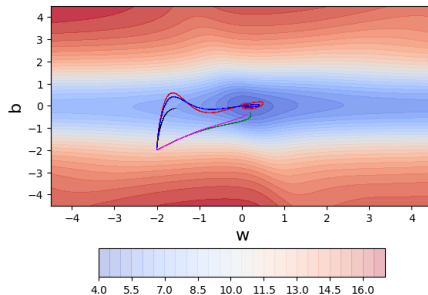
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



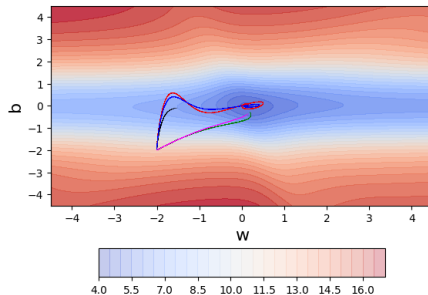
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



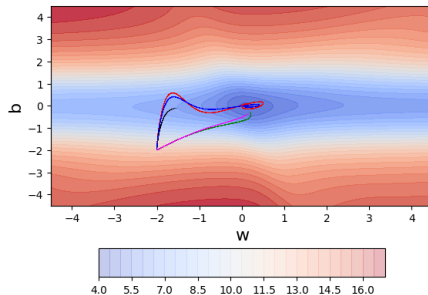
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



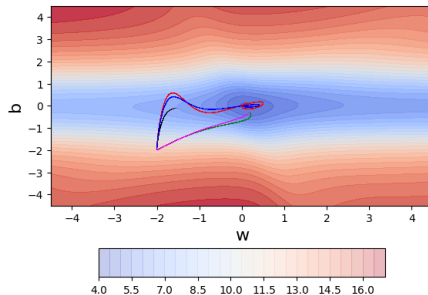
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



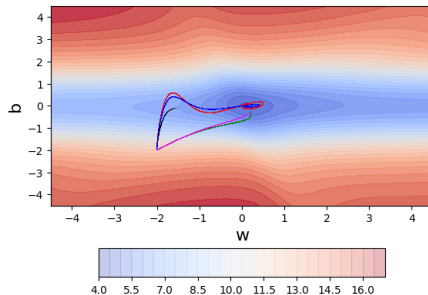

```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



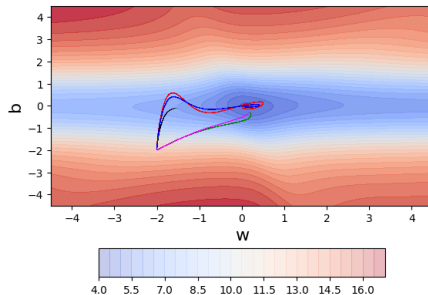
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



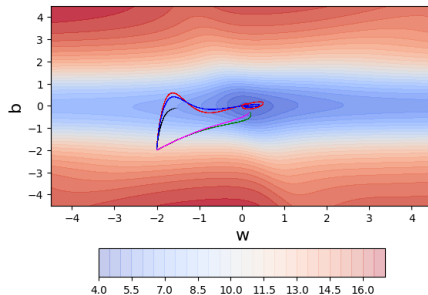
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = betal * v_w + (1 - betal) dw**2
        v_b = betal * v_b + (1 - betal) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



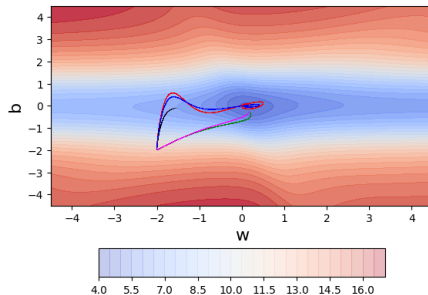
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, beta1 = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        v_w = beta1 * v_w + (1 - beta1) dw**2
        v_b = beta1 * v_b + (1 - beta1) db**2

        w = w - (eta / np.sqrt(v_w + eps)) * dw
        b = b - (eta / np.sqrt(v_b + eps)) * db

```



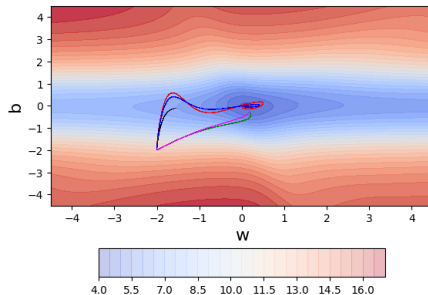
```

def do_rmsprop() :
    w, b, eta = init_w, init_b, 0.1
    v_w, b_updates, eps, betal = 0, 0, 1e-8, 0.9
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
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        w = w - (eta / np.sqrt(v_w + eps)) * dw
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```



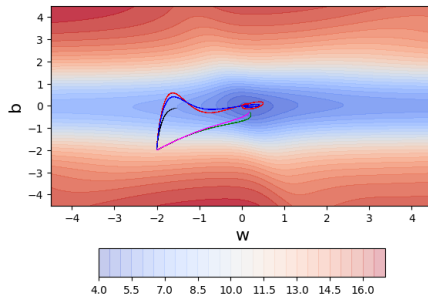
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```



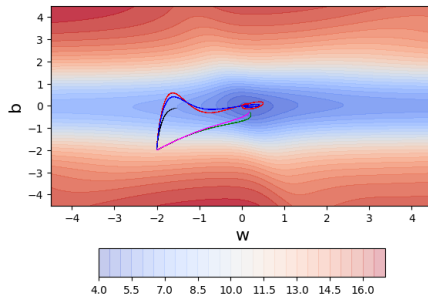
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```



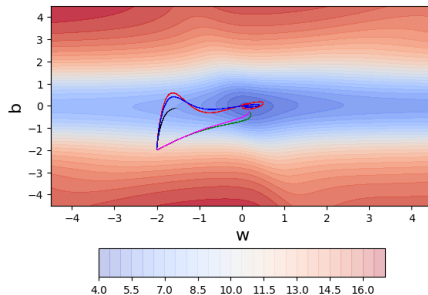
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```



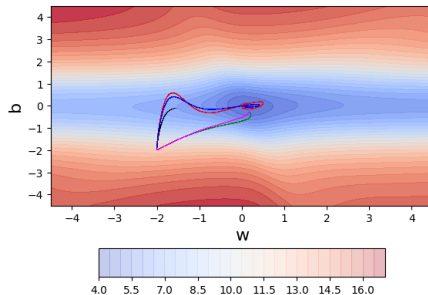

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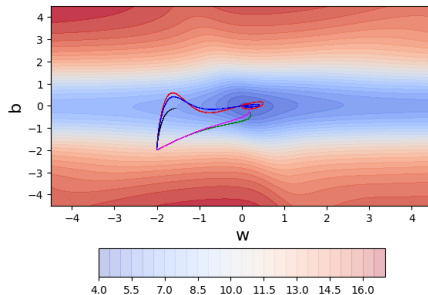
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```



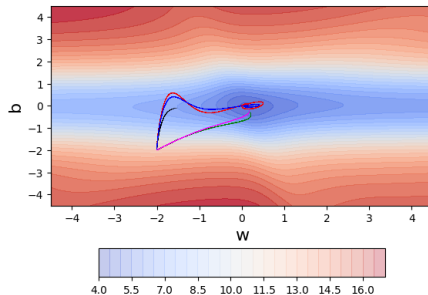
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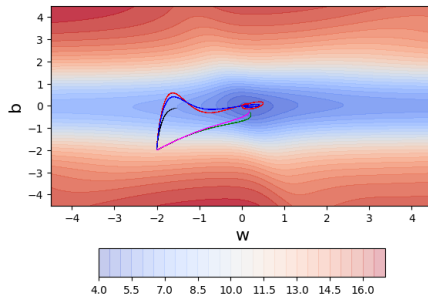
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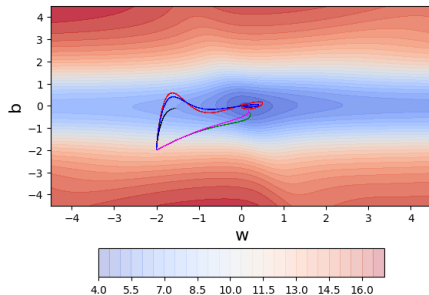
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```

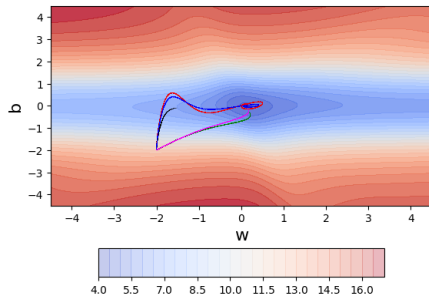


- Adagrad got stuck when it was close to convergence (it was no longer able to move in the vertical (b) direction because of the decayed learning rate)

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        w = w - (eta / np.sqrt(v_w + eps)) * dw
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```



- Adagrad got stuck when it was close to convergence (it was no longer able to move in the vertical (b) direction because of the decayed learning rate)

- RMSProp overcomes this problem by being less aggressive on the decay

Intuition

- Do everything that RMSProp does to solve the decay problem of Adagrad

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- Plus use a cumulative history of the gradients

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Update rule for Adam

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Update rule for Adam

$$m_t = \beta_1 * m_{t-1} + (1 - \beta_1) * \nabla w_t$$

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- In practice, $\beta_1 = 0.9$ and $\beta_2 = 0.999$

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$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} * \hat{m}_t$$

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$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} * \hat{m}_t$$

... and a similar set of equations for b_t

```

def do_adam() :
    w_b_dw_db = [(init_w, init_b, 0, 0)]
    w_history, b_history, error_history = [], [], [
        ], []

    w, b, eta, mini_batch_size, num_points_seen =
        init_w, init_b, 0.1, 10, 0
    m_w, m_b, v_w, v_b, m_w_hat, m_b_hat, v_w_hat,
        v_b_hat, eps, beta1, beta2 = 0, 0, 0, 0, 0,
        0, 0, 0, 1e-8, 0.9, 0.999
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        m_w = beta1 * m_w + (1-beta1)*dw
        m_b = beta1 * m_b + (1-beta1)*db

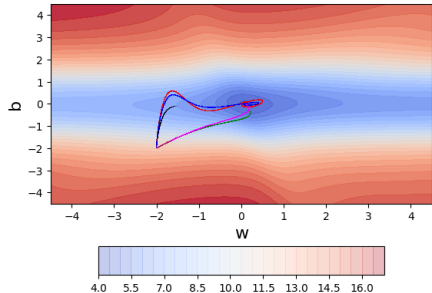
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
        m_b_hat = m_b/(1-math.pow(beta1,i+1))

        v_w_hat = v_w/(1-math.pow(beta2,i+1))
        v_b_hat = v_b/(1-math.pow(beta2,i+1))

        w = w - (eta / np.sqrt(v_w_hat + eps)) *
            m_w_hat
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```




```

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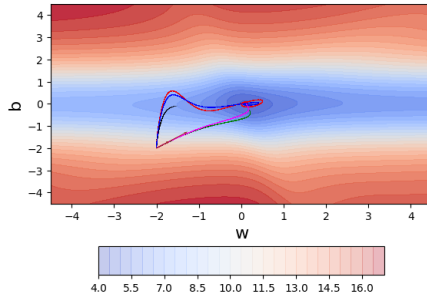
        v_w = beta2 * v_w + (1-beta2)*dw**2
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        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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        v_w_hat = v_w/(1-math.pow(beta2,i+1))
        v_b_hat = v_b/(1-math.pow(beta2,i+1))

        w = w - (eta / np.sqrt(v_w_hat + eps)) *
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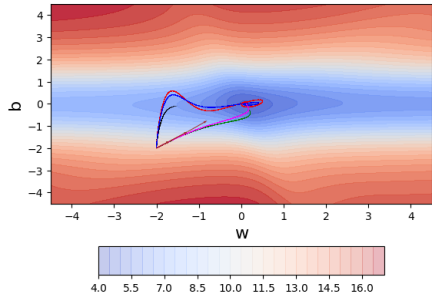
        v_w = beta2 * v_w + (1-beta2)*dw**2
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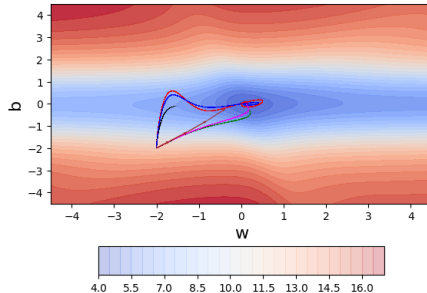
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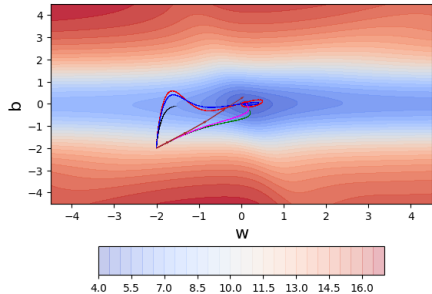
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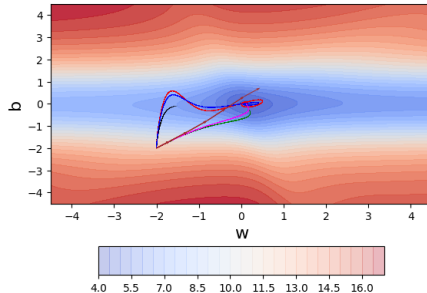
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        m_b_hat = m_b/(1-math.pow(beta1,i+1))

        v_w_hat = v_w/(1-math.pow(beta2,i+1))
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        w = w - (eta / np.sqrt(v_w_hat + eps)) *
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    w, b, eta, mini_batch_size, num_points_seen =
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    m_w, m_b, v_w, v_b, m_w_hat, m_b_hat, v_w_hat,
        v_b_hat, eps, beta1, beta2 = 0, 0, 0, 0, 0,
        0, 0, 0, 1e-8, 0.9, 0.999
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        m_w = beta1 * m_w + (1-beta1)*dw
        m_b = beta1 * m_b + (1-beta1)*db

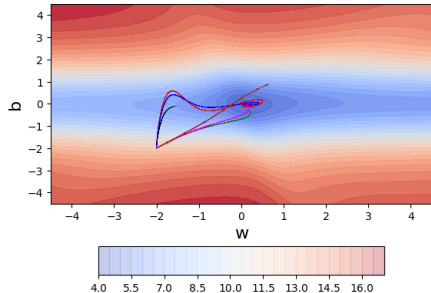
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
        m_b_hat = m_b/(1-math.pow(beta1,i+1))

        v_w_hat = v_w/(1-math.pow(beta2,i+1))
        v_b_hat = v_b/(1-math.pow(beta2,i+1))

        w = w - (eta / np.sqrt(v_w_hat + eps)) *
            m_w_hat
        b = b - (eta / np.sqrt(v_b_hat + eps)) *
            m_b_hat

```



```

def do_adam() :
    w_b_dw_db = [(init_w, init_b, 0, 0)]
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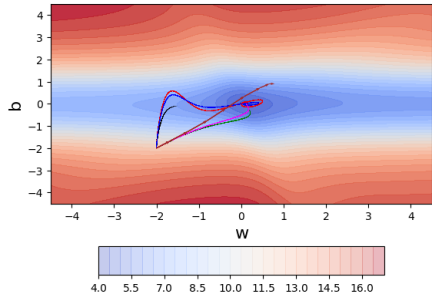
        v_w = beta2 * v_w + (1-beta2)*dw**2
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        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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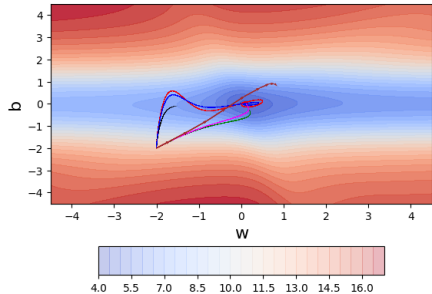
        v_w = beta2 * v_w + (1-beta2)*dw**2
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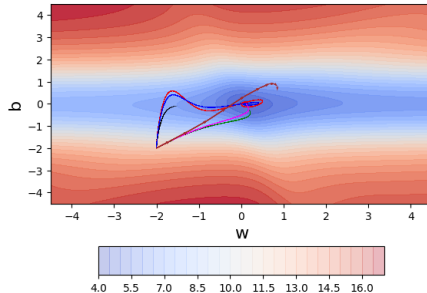
        v_w = beta2 * v_w + (1-beta2)*dw**2
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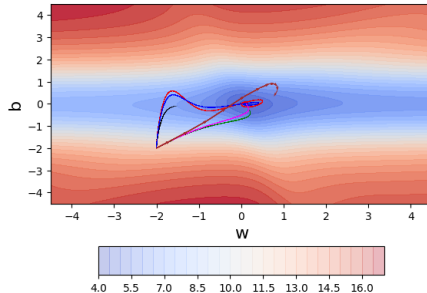
        v_w = beta2 * v_w + (1-beta2)*dw**2
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        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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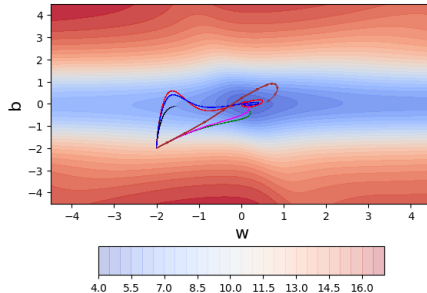
        v_w = beta2 * v_w + (1-beta2)*dw**2
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        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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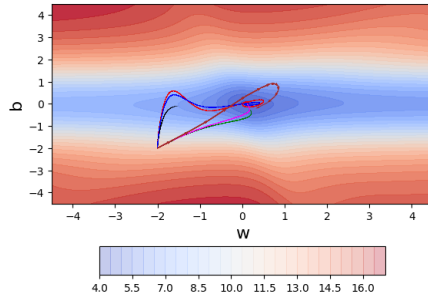
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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```



```

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        m_w = beta1 * m_w + (1-beta1)*dw
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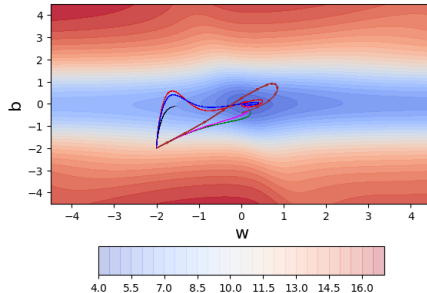
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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```



```

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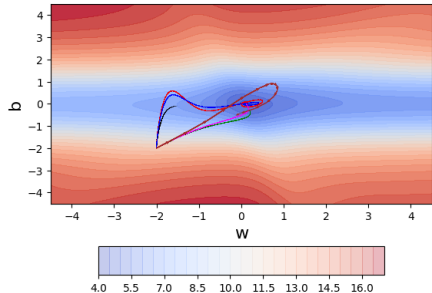
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```



```

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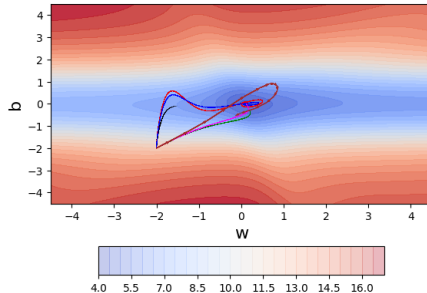
        v_w = beta2 * v_w + (1-beta2)*dw**2
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```



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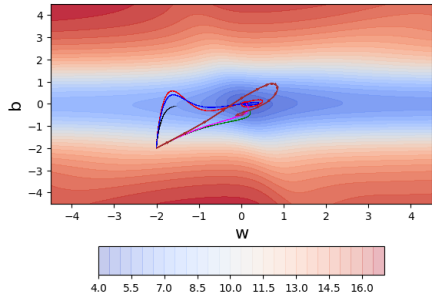
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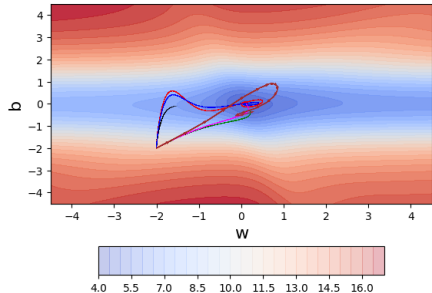
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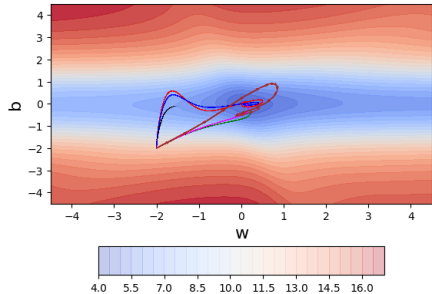
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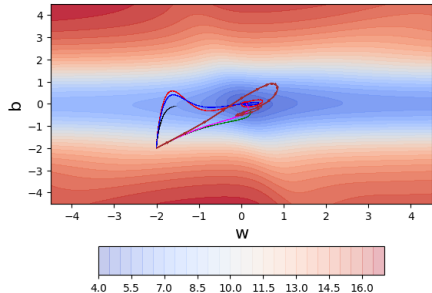
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        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
        m_b_hat = m_b/(1-math.pow(beta1,i+1))

        v_w_hat = v_w/(1-math.pow(beta2,i+1))
        v_b_hat = v_b/(1-math.pow(beta2,i+1))

        w = w - (eta / np.sqrt(v_w_hat + eps)) *
            m_w_hat
        b = b - (eta / np.sqrt(v_b_hat + eps)) *
            m_b_hat

```



```

def do_adam() :
    w_b_dw_db = [(init_w, init_b, 0, 0)]
    w_history, b_history, error_history = [], [], [
        ], []

    w, b, eta, mini_batch_size, num_points_seen =
        init_w, init_b, 0.1, 10, 0
    m_w, m_b, v_w, v_b, m_w_hat, m_b_hat, v_w_hat,
        v_b_hat, eps, beta1, beta2 = 0, 0, 0, 0, 0,
        0, 0, 0, 1e-8, 0.9, 0.999
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        m_w = beta1 * m_w + (1-beta1)*dw
        m_b = beta1 * m_b + (1-beta1)*db

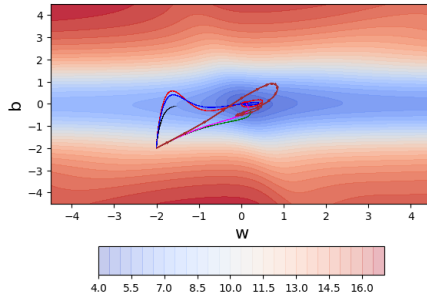
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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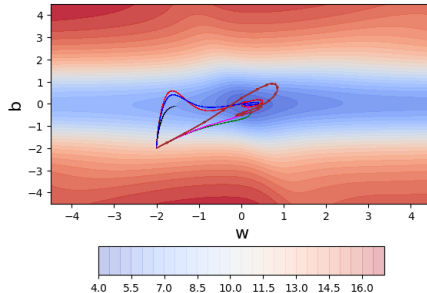
        v_w = beta2 * v_w + (1-beta2)*dw**2
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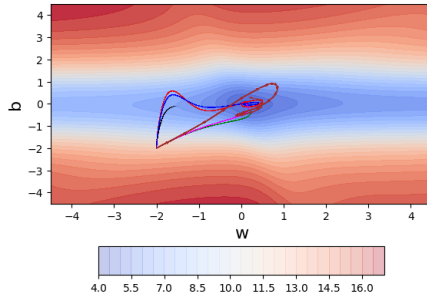
        v_w = beta2 * v_w + (1-beta2)*dw**2
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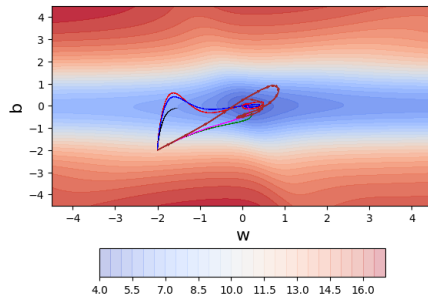
        v_w = beta2 * v_w + (1-beta2)*dw**2
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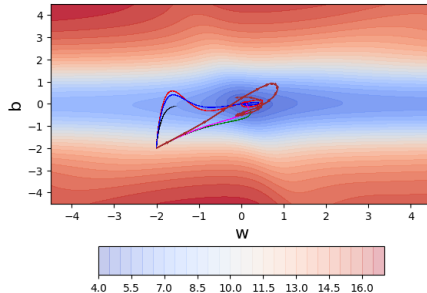
        v_w = beta2 * v_w + (1-beta2)*dw**2
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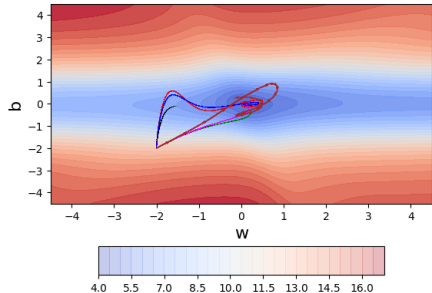
        v_w = beta2 * v_w + (1-beta2)*dw**2
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        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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```



```

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        m_b = beta1 * m_b + (1-beta1)*db

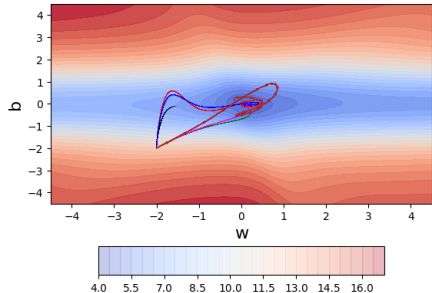
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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```



```

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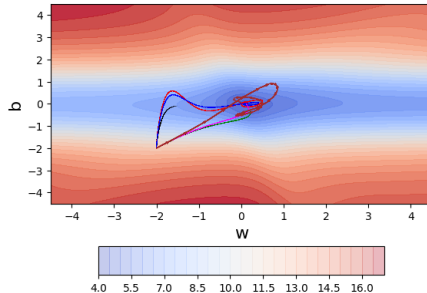
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
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```



```

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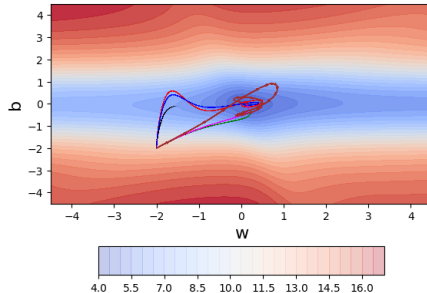
        v_w = beta2 * v_w + (1-beta2)*dw**2
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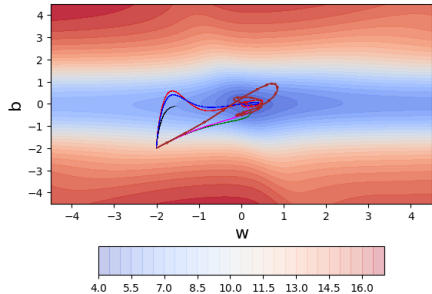
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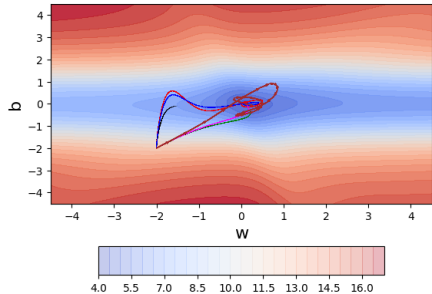
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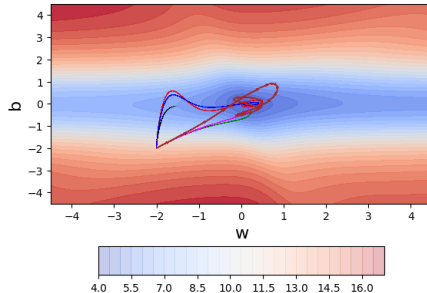
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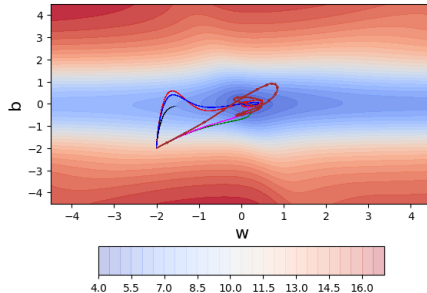
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    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
            dw += grad_w(w, b, x, y)
            db += grad_b(w, b, x, y)

        m_w = beta1 * m_w + (1-beta1)*dw
        m_b = beta1 * m_b + (1-beta1)*db

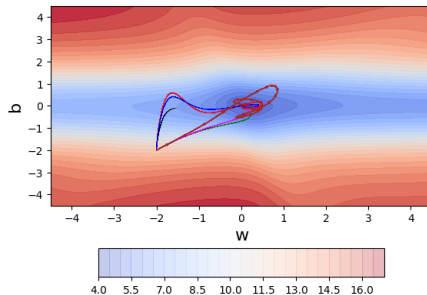
        v_w = beta2 * v_w + (1-beta2)*dw**2
        v_b = beta2 * v_b + (1-beta2)*db**2

        m_w_hat = m_w/(1-math.pow(beta1,i+1))
        m_b_hat = m_b/(1-math.pow(beta1,i+1))

        v_w_hat = v_w/(1-math.pow(beta2,i+1))
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        w = w - (eta / np.sqrt(v_w_hat + eps)) *
            m_w_hat
        b = b - (eta / np.sqrt(v_b_hat + eps)) *
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```



```

def do_adam() :
    w_b_dw_db = [(init_w, init_b, 0, 0)]
    w_history, b_history, error_history = [], [], [
    ], []

    w, b, eta, mini_batch_size, num_points_seen =
        init_w, init_b, 0.1, 10, 0
    m_w, m_b, v_w, v_b, m_w_hat, m_b_hat, v_w_hat,
        v_b_hat, eps, beta1, beta2 = 0, 0, 0, 0, 0,
        0, 0, 0, 1e-8, 0.9, 0.999
    for i in range(max_epochs) :
        dw, db = 0, 0
        for x,y in zip(X, Y) :
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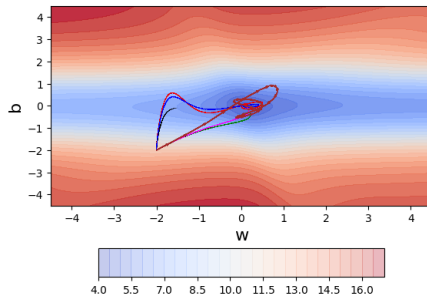
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        v_w_hat = v_w/(1-math.pow(beta2,i+1))
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```



- As expected, taking a cumulative history gives a speed up ...

Million dollar question: Which algorithm to use in practice

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- Adam might just be the best choice overall!!
- Some recent work suggest that there is a problem with Adam and it will not converge in some cases

Explanation for why we need bias correction in Adam

Update rule for Adam

$$m_t = \beta_1 * m_{t-1} + (1 - \beta_1) * \nabla w_t$$

$$v_t = \beta_2 * v_{t-1} + (1 - \beta_2) * (\nabla w_t)^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} * \hat{m}_t$$

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- In general,

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- Assumption: All g_i 's come from the same distribution i.e. $E[g_i] = E[g] \forall i$

$$E[m_t] = (1 - \beta) \sum_{i=1}^t (\beta)^{t-i} E[g_i]$$

$$\begin{aligned} &= E[g] (1 - \beta) \sum_{i=1}^t (\beta)^{t-i} \\ &= E[g] (1 - \beta) (\beta^{t-1} + \beta^{t-2} + \dots + \beta^0) \\ &= E[g] (1 - \beta) \frac{1 - \beta^t}{1 - \beta} \end{aligned}$$

the last fraction is the sum of a GP with common ratio = β

$$\begin{aligned} E[m_t] &= E[g] (1 - \beta^t) \\ E\left[\frac{m_t}{1 - \beta^t}\right] &= E[g] \end{aligned}$$

- So we have, $m_t = (1 - \beta) \sum_{i=1}^t \beta^{t-i} g_i$
- Taking Expectation on both sides

$$E[m_t] = E[(1 - \beta) \sum_{i=1}^t \beta^{t-i} g_i]$$

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the last fraction is the sum of a GP with common ratio = β

$$\begin{aligned} E[m_t] &= E[g] (1 - \beta^t) \\ E\left[\frac{m_t}{1 - \beta^t}\right] &= E[g] \\ E[\hat{m}_t] &= E[g] (\because \frac{m_t}{1 - \beta^t} = \hat{m}_t) \end{aligned}$$

- So we have, $m_t = (1 - \beta) \sum_{i=1}^t \beta^{t-i} g_i$
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$$E[m_t] = E[(1 - \beta) \sum_{i=1}^t \beta^{t-i} g_i]$$

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the last fraction is the sum of a GP with common ratio = β

$$\begin{aligned} E[m_t] &= E[g] (1 - \beta^t) \\ E\left[\frac{m_t}{1 - \beta^t}\right] &= E[g] \\ E[\hat{m}_t] &= E[g] (\because \frac{m_t}{1 - \beta^t} = \hat{m}_t) \end{aligned}$$

Hence we apply the bias correction because then the expected value of $\hat{m}_t$ is the same as the expected value of g_t